RAILWAY EXAM HUB

RRB GROUP D — COMPLETE GENERAL SCIENCE STUDY MATERIAL

33 chapters • Full concept notes • Formulae • Worked examples • 825 practice MCQs with answers and explanations

Physics: 11 chapters | Chemistry: 10 chapters | Biology: 12 chapters

Independent educational material; not an official RRB publication. Check the latest official notification for syllabus and exam details.

How to use this book

Study one chapter at a time. Learn definitions, units and principles.

For numerical questions, list known values, convert units, choose a formula and show steps.

Attempt all questions before checking answers. Record mistakes and revise the relevant concept.

These are original practice questions, not claimed to be official previous-year questions.

CHAPTER 1: Units and Measurements

Physics — Measurement is the comparison of a physical quantity with an accepted standard unit.

Concept notes and exam points

1. A physical quantity is expressed as a numerical value multiplied by a unit. A measurement is meaningful only when the unit and appropriate precision are stated.

2. The seven SI base quantities are length, mass, time, electric current, thermodynamic temperature, amount of substance and luminous intensity.

3. Their SI base units are metre (m), kilogram (kg), second (s), ampere (A), kelvin (K), mole (mol) and candela (cd).

4. Derived units are combinations of base units: velocity m/s, acceleration m/s², force newton (kg m/s²), work joule (N m) and pressure pascal (N/m²).

5. Scientific notation writes a number as a × 10■ where 1 ≤ |a| < 10. It is useful for very large and very small measurements.

6. Accuracy describes closeness to the true value; precision describes repeatability or agreement among repeated measurements. Significant figures communicate measurement precision.

7. For addition and subtraction, the result is normally rounded to the least number of decimal places among measured values. For multiplication and division, use the least number of significant figures.

8. Common prefixes: kilo = 10³, centi = 10■², milli = 10■³, micro = 10■■ and nano = 10■■. Convert units before substituting into formulae.

Formulae, units and quick facts

• density = mass/volume

• speed = distance/time

• force unit: N = kg·m/s²

• 1 km = 1000 m

• 1 hour = 3600 s

Worked examples

Problem: Convert 72 km/h into m/s.

Solution: 20 m/s. 72 × 5/18 = 20 m/s.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The SI base unit of length is.

A. metre B. kilogram C. second D. newton

Answer: A. metre. Explanation: The SI base unit of length is.

Q2. Choose the scientifically correct option: The SI base unit of length is.

A. kilogram B. second C. newton D. metre

Answer: D. metre. Explanation: The SI base unit of length is.

Q3. Which option correctly describes this fact: The SI base unit of length is?

A. second B. newton C. metre D. kilogram

Answer: C. metre. Explanation: The SI base unit of length is.

Q4. Select the best answer about this topic: The SI base unit of length is.

A. newton B. metre C. kilogram D. second

Answer: B. metre. Explanation: The SI base unit of length is.

Q5. Which of the following is true? The SI base unit of length is.

A. metre B. kilogram C. second D. newton

Answer: A. metre. Explanation: The SI base unit of length is.

Q6. Which statement is correct? The SI base unit of thermodynamic temperature is.

A. degree Celsius B. joule C. watt D. kelvin

Answer: D. kelvin. Explanation: The SI base unit of thermodynamic temperature is.

Q7. Choose the scientifically correct option: The SI base unit of thermodynamic temperature is.

A. joule B. watt C. kelvin D. degree Celsius

Answer: C. kelvin. Explanation: The SI base unit of thermodynamic temperature is.

Q8. Which option correctly describes this fact: The SI base unit of thermodynamic temperature is?

A. watt B. kelvin C. degree Celsius D. joule

Answer: B. kelvin. Explanation: The SI base unit of thermodynamic temperature is.

Q9. Select the best answer about this topic: The SI base unit of thermodynamic temperature is.

A. kelvin B. degree Celsius C. joule D. watt

Answer: A. kelvin. Explanation: The SI base unit of thermodynamic temperature is.

Q10. Which of the following is true? The SI base unit of thermodynamic temperature is.

A. degree Celsius B. joule C. watt D. kelvin

Answer: D. kelvin. Explanation: The SI base unit of thermodynamic temperature is.

Q11. Which statement is correct? The SI unit of force is.

A. pascal B. ampere C. newton D. joule

Answer: C. newton. Explanation: The SI unit of force is.

Q12. Choose the scientifically correct option: The SI unit of force is.

A. ampere B. newton C. joule D. pascal

Answer: B. newton. Explanation: The SI unit of force is.

Q13. Which option correctly describes this fact: The SI unit of force is?

A. newton B. joule C. pascal D. ampere

Answer: A. newton. Explanation: The SI unit of force is.

Q14. Select the best answer about this topic: The SI unit of force is.

A. joule B. pascal C. ampere D. newton

Answer: D. newton. Explanation: The SI unit of force is.

Q15. Which of the following is true? The SI unit of force is.

A. pascal B. ampere C. newton D. joule

Answer: C. newton. Explanation: The SI unit of force is.

Q16. Which statement is correct? The prefix milli represents.

A. 10■² B. 10■³ C. 10³ D. 10■■

Answer: B. 10■³. Explanation: The prefix milli represents.

Q17. Choose the scientifically correct option: The prefix milli represents.

A. 10■³ B. 10³ C. 10■■ D. 10■²

Answer: A. 10■³. Explanation: The prefix milli represents.

Q18. Which option correctly describes this fact: The prefix milli represents?

A. 10³ B. 10■■ C. 10■² D. 10■³

Answer: D. 10■³. Explanation: The prefix milli represents.

Q19. Select the best answer about this topic: The prefix milli represents.

A. 10■■ B. 10■² C. 10■³ D. 10³

Answer: C. 10■³. Explanation: The prefix milli represents.

Q20. Which of the following is true? The prefix milli represents.

A. 10■² B. 10■³ C. 10³ D. 10■■

Answer: B. 10■³. Explanation: The prefix milli represents.

Q21. Which statement is correct? One hour contains.

A. 3600 seconds B. 60 seconds C. 100 seconds D. 1000 seconds

Answer: A. 3600 seconds. Explanation: One hour contains.

Q22. Choose the scientifically correct option: One hour contains.

A. 60 seconds B. 100 seconds C. 1000 seconds D. 3600 seconds

Answer: D. 3600 seconds. Explanation: One hour contains.

Q23. Which option correctly describes this fact: One hour contains?

A. 100 seconds B. 1000 seconds C. 3600 seconds D. 60 seconds

Answer: C. 3600 seconds. Explanation: One hour contains.

Q24. Select the best answer about this topic: One hour contains.

A. 1000 seconds B. 3600 seconds C. 60 seconds D. 100 seconds

Answer: B. 3600 seconds. Explanation: One hour contains.

Q25. Which of the following is true? One hour contains.

A. 3600 seconds B. 60 seconds C. 100 seconds D. 1000 seconds

Answer: A. 3600 seconds. Explanation: One hour contains.

CHAPTER 2: Motion

Physics — Motion describes change in position with time relative to a chosen reference point.

Concept notes and exam points

1. Distance is the total length of the actual path and is a scalar; displacement is the shortest directed change from initial to final position and is a vector.

2. Speed is distance per unit time. Velocity is displacement per unit time and includes direction. Average speed uses total distance divided by total time.

3. Acceleration is the rate of change of velocity. It may be positive, negative (retardation in a chosen direction), or zero.

4. A distance–time graph has slope equal to speed. A velocity–time graph has slope equal to acceleration and signed area equal to displacement.

5. Uniform motion covers equal distances in equal time intervals. Non-uniform motion has changing speed or direction.

6. SI units: distance and displacement metre (m), time second (s), speed and velocity m/s, acceleration m/s².

7. For constant acceleration: $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$. These equations require straight-line motion with constant acceleration.

8. Convert km/h to m/s by multiplying by 5/18; convert m/s to km/h by multiplying by 18/5.

Formulae, units and quick facts

• speed = distance/time

• average speed = total distance/total time

• acceleration = (v-u)/t

• $v = u + at$

• $s = ut + \frac{1}{2}at^2$

• $v^2 = u^2 + 2as$

Worked examples

Problem: A vehicle travels 150 m in 30 s. Find its average speed.

Solution: 5 m/s. 150 ÷ 30 = 5 m/s.

Problem: A body changes velocity from 4 m/s to 16 m/s in 3 s. Find acceleration.

Solution: 4 m/s². (16 – 4) ÷ 3 = 4 m/s².

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Distance is a scalar quantity.

A. distance B. displacement C. velocity D. acceleration

Answer: A. distance. Explanation: Distance is a scalar quantity.

Q2. Choose the scientifically correct option: Distance is a scalar quantity.

A. displacement B. velocity C. acceleration D. distance

Answer: D. distance. Explanation: Distance is a scalar quantity.

Q3. Which option correctly describes this fact: Distance is a scalar quantity?

A. velocity B. acceleration C. distance D. displacement

Answer: C. distance. Explanation: Distance is a scalar quantity.

Q4. Select the best answer about this topic: Distance is a scalar quantity.

A. acceleration B. distance C. displacement D. velocity

Answer: B. distance. Explanation: Distance is a scalar quantity.

Q5. Which of the following is true? Distance is a scalar quantity.

A. distance B. displacement C. velocity D. acceleration

Answer: A. distance. Explanation: Distance is a scalar quantity.

Q6. Which statement is correct? The SI unit of speed is.

A. m/s² B. metre C. second D. m/s

Answer: D. m/s. Explanation: The SI unit of speed is.

Q7. Choose the scientifically correct option: The SI unit of speed is.

A. metre B. second C. m/s D. m/s²

Answer: C. m/s. Explanation: The SI unit of speed is.

Q8. Which option correctly describes this fact: The SI unit of speed is?

A. second B. m/s C. m/s² D. metre

Answer: B. m/s. Explanation: The SI unit of speed is.

Q9. Select the best answer about this topic: The SI unit of speed is.

A. m/s B. m/s² C. metre D. second

Answer: A. m/s. Explanation: The SI unit of speed is.

Q10. Which of the following is true? The SI unit of speed is.

A. m/s² B. metre C. second D. m/s

Answer: D. m/s. Explanation: The SI unit of speed is.

Q11. Which statement is correct? The slope of a distance-time graph represents.

A. force B. work C. speed D. acceleration

Answer: C. speed. Explanation: The slope of a distance-time graph represents.

Q12. Choose the scientifically correct option: The slope of a distance-time graph represents.

A. work B. speed C. acceleration D. force

Answer: B. speed. Explanation: The slope of a distance-time graph represents.

Q13. Which option correctly describes this fact: The slope of a distance-time graph represents?

A. speed B. acceleration C. force D. work

Answer: A. speed. Explanation: The slope of a distance-time graph represents.

Q14. Select the best answer about this topic: The slope of a distance-time graph represents.

A. acceleration B. force C. work D. speed

Answer: D. speed. Explanation: The slope of a distance-time graph represents.

Q15. Which of the following is true? The slope of a distance-time graph represents.

A. force B. work C. speed D. acceleration

Answer: C. speed. Explanation: The slope of a distance-time graph represents.

Q16. Which statement is correct? The area under a velocity-time graph gives.

A. mass B. displacement C. acceleration D. speed

Answer: B. displacement. Explanation: The area under a velocity-time graph gives.

Q17. Choose the scientifically correct option: The area under a velocity-time graph gives.

A. displacement B. acceleration C. speed D. mass

Answer: A. displacement. Explanation: The area under a velocity-time graph gives.

Q18. Which option correctly describes this fact: The area under a velocity-time graph gives?

A. acceleration B. speed C. mass D. displacement

Answer: D. displacement. Explanation: The area under a velocity-time graph gives.

Q19. Select the best answer about this topic: The area under a velocity-time graph gives.

A. speed B. mass C. displacement D. acceleration

Answer: C. displacement. Explanation: The area under a velocity-time graph gives.

Q20. Which of the following is true? The area under a velocity-time graph gives.

A. mass B. displacement C. acceleration D. speed

Answer: B. displacement. Explanation: The area under a velocity-time graph gives.

Q21. Which statement is correct? A body moving with constant velocity has acceleration.

A. zero B. increasing C. negative always D. infinite

Answer: A. zero. Explanation: A body moving with constant velocity has acceleration.

Q22. Choose the scientifically correct option: A body moving with constant velocity has acceleration.

A. increasing B. negative always C. infinite D. zero

Answer: D. zero. Explanation: A body moving with constant velocity has acceleration.

Q23. Which option correctly describes this fact: A body moving with constant velocity has acceleration?

A. negative always B. infinite C. zero D. increasing

Answer: C. zero. Explanation: A body moving with constant velocity has acceleration.

Q24. Select the best answer about this topic: A body moving with constant velocity has acceleration.

A. infinite B. zero C. increasing D. negative always

Answer: B. zero. Explanation: A body moving with constant velocity has acceleration.

Q25. Which of the following is true? A body moving with constant velocity has acceleration.

A. zero B. increasing C. negative always D. infinite

Answer: A. zero. Explanation: A body moving with constant velocity has acceleration.

CHAPTER 3: Force and Laws of Motion

Physics — Force is an interaction that can change an object's state of rest, uniform motion, direction or shape.

Concept notes and exam points

1. Newton's first law: an object remains at rest or in uniform straight-line motion unless acted on by a net external force. This property is inertia.

2. Inertia depends on mass. Greater mass means greater resistance to a change in motion.

3. Newton's second law states that net force equals rate of change of momentum. For constant mass, F = ma.

4. Momentum is the product of mass and velocity (p = mv). Its SI unit is kg m/s.

5. Newton's third law: every action has an equal and opposite reaction. The forces act on different bodies, so they do not cancel each other on one body.

6. Impulse equals change in momentum and is also force multiplied by the time interval when force is constant.

7. Friction opposes relative motion or its tendency. It can provide grip for walking and braking, but also causes wear and heating.

8. Seat belts and airbags increase the time over which momentum changes, reducing average impact force.

Formulae, units and quick facts

• F = ma

• momentum p = mv

• impulse = $F\Delta t = \Delta p$

• weight W = mg

Worked examples

Problem: A 5 kg object accelerates at 2 m/s². Find the net force.

Solution: 10 N. F = ma = 5 × 2 = 10 N.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The tendency of a body to resist change in motion is.

A. inertia B. pressure C. energy D. power

Answer: A. inertia. Explanation: The tendency of a body to resist change in motion is.

Q2. Choose the scientifically correct option: The tendency of a body to resist change in motion is.

A. pressure B. energy C. power D. inertia

Answer: D. inertia. Explanation: The tendency of a body to resist change in motion is.

Q3. Which option correctly describes this fact: The tendency of a body to resist change in motion is?

A. energy B. power C. inertia D. pressure

Answer: C. inertia. Explanation: The tendency of a body to resist change in motion is.

Q4. Select the best answer about this topic: The tendency of a body to resist change in motion is.

A. power B. inertia C. pressure D. energy

Answer: B. inertia. Explanation: The tendency of a body to resist change in motion is.

Q5. Which of the following is true? The tendency of a body to resist change in motion is.

A. inertia B. pressure C. energy D. power

Answer: A. inertia. Explanation: The tendency of a body to resist change in motion is.

Q6. Which statement is correct? SI unit of force is.

A. joule B. watt C. pascal D. newton

Answer: D. newton. Explanation: SI unit of force is.

Q7. Choose the scientifically correct option: SI unit of force is.

A. watt B. pascal C. newton D. joule

Answer: C. newton. Explanation: SI unit of force is.

Q8. Which option correctly describes this fact: SI unit of force is?

A. pascal B. newton C. joule D. watt

Answer: B. newton. Explanation: SI unit of force is.

Q9. Select the best answer about this topic: SI unit of force is.

A. newton B. joule C. watt D. pascal

Answer: A. newton. Explanation: SI unit of force is.

Q10. Which of the following is true? SI unit of force is.

A. joule B. watt C. pascal D. newton

Answer: D. newton. Explanation: SI unit of force is.

Q11. Which statement is correct? Momentum is equal to.

A. force × distance B. work/time C. mass × velocity D. mass × acceleration

Answer: C. mass × velocity. Explanation: Momentum is equal to.

Q12. Choose the scientifically correct option: Momentum is equal to.

A. work/time B. mass × velocity C. mass × acceleration D. force × distance

Answer: B. mass × velocity. Explanation: Momentum is equal to.

Q13. Which option correctly describes this fact: Momentum is equal to?

A. mass × velocity B. mass × acceleration C. force × distance D. work/time

Answer: A. mass × velocity. Explanation: Momentum is equal to.

Q14. Select the best answer about this topic: Momentum is equal to.

A. mass × acceleration B. force × distance C. work/time D. mass × velocity

Answer: D. mass × velocity. Explanation: Momentum is equal to.

Q15. Which of the following is true? Momentum is equal to.

A. force × distance B. work/time C. mass × velocity D. mass × acceleration

Answer: C. mass × velocity. Explanation: Momentum is equal to.

Q16. Which statement is correct? Action and reaction forces act on.

A. only stationary bodies B. different bodies C. the same body D. no bodies

Answer: B. different bodies. Explanation: Action and reaction forces act on.

Q17. Choose the scientifically correct option: Action and reaction forces act on.

A. different bodies B. the same body C. no bodies D. only stationary bodies

Answer: A. different bodies. Explanation: Action and reaction forces act on.

Q18. Which option correctly describes this fact: Action and reaction forces act on?

A. the same body B. no bodies C. only stationary bodies D. different bodies

Answer: D. different bodies. Explanation: Action and reaction forces act on.

Q19. Select the best answer about this topic: Action and reaction forces act on.

A. no bodies B. only stationary bodies C. different bodies D. the same body

Answer: C. different bodies. Explanation: Action and reaction forces act on.

Q20. Which of the following is true? Action and reaction forces act on.

A. only stationary bodies B. different bodies C. the same body D. no bodies

Answer: B. different bodies. Explanation: Action and reaction forces act on.

Q21. Which statement is correct? A seat belt reduces injury mainly by increasing.

A. time of momentum change B. body mass C. speed before collision D. gravitational acceleration

Answer: A. time of momentum change. Explanation: A seat belt reduces injury mainly by increasing.

Q22. Choose the scientifically correct option: A seat belt reduces injury mainly by increasing.

A. body mass B. speed before collision C. gravitational acceleration D. time of momentum change

Answer: D. time of momentum change. Explanation: A seat belt reduces injury mainly by increasing.

Q23. Which option correctly describes this fact: A seat belt reduces injury mainly by increasing?

A. speed before collision B. gravitational acceleration C. time of momentum change D. body mass

Answer: C. time of momentum change. Explanation: A seat belt reduces injury mainly by increasing.

Q24. Select the best answer about this topic: A seat belt reduces injury mainly by increasing.

A. gravitational acceleration B. time of momentum change C. body mass D. speed before collision

Answer: B. time of momentum change. Explanation: A seat belt reduces injury mainly by increasing.

Q25. Which of the following is true? A seat belt reduces injury mainly by increasing.

A. time of momentum change B. body mass C. speed before collision D. gravitational acceleration

Answer: A. time of momentum change. Explanation: A seat belt reduces injury mainly by increasing.

CHAPTER 4: Work, Energy and Power

Physics — Work is done when a force causes displacement in its direction or has a component along the displacement.

Concept notes and exam points

1. For constant force, work W = Fs cos θ. Work is positive when force helps displacement, negative when it opposes it, and zero when perpendicular or when there is no displacement.

2. The SI unit of work and energy is joule (J). One joule is the work done by a force of 1 N through 1 m in its direction.

3. Kinetic energy is energy due to motion: KE = ½mv². It increases with mass and with the square of speed.

4. Gravitational potential energy near Earth's surface is PE = mgh, measured relative to a chosen reference level.

5. The law of conservation of energy states that energy is not created or destroyed; it changes form. Total energy of an isolated system remains constant.

6. Power is the rate of doing work or transferring energy. Its SI unit is watt (W), where 1 W = 1 J/s.

7. Commercial electrical energy is often measured in kilowatt-hour. 1 kWh = 3.6 × 10■ J.

8. Machines may make work more convenient by changing force or direction; in ideal machines, input work equals output work.

Formulae, units and quick facts

• W = Fs cos θ

• KE = ½mv²

• PE = mgh

• P = W/t

• 1 kWh = 3.6 × 10■ J

Worked examples

Problem: A 10 N force moves an object 3 m in the force direction. Find work.

Solution: 30 J. W = Fs = 10 × 3 = 30 J.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The SI unit of work is.

A. joule B. newton C. watt D. pascal

Answer: A. joule. Explanation: The SI unit of work is.

Q2. Choose the scientifically correct option: The SI unit of work is.

A. newton B. watt C. pascal D. joule

Answer: D. joule. Explanation: The SI unit of work is.

Q3. Which option correctly describes this fact: The SI unit of work is?

A. watt B. pascal C. joule D. newton

Answer: C. joule. Explanation: The SI unit of work is.

Q4. Select the best answer about this topic: The SI unit of work is.

A. pascal B. joule C. newton D. watt

Answer: B. joule. Explanation: The SI unit of work is.

Q5. Which of the following is true? The SI unit of work is.

A. joule B. newton C. watt D. pascal

Answer: A. joule. Explanation: The SI unit of work is.

Q6. Which statement is correct? Energy possessed by a moving body is.

A. potential energy B. chemical energy only C. nuclear charge D. kinetic energy

Answer: D. kinetic energy. Explanation: Energy possessed by a moving body is.

Q7. Choose the scientifically correct option: Energy possessed by a moving body is.

A. chemical energy only B. nuclear charge C. kinetic energy D. potential energy

Answer: C. kinetic energy. Explanation: Energy possessed by a moving body is.

Q8. Which option correctly describes this fact: Energy possessed by a moving body is?

A. nuclear charge B. kinetic energy C. potential energy D. chemical energy only

Answer: B. kinetic energy. Explanation: Energy possessed by a moving body is.

Q9. Select the best answer about this topic: Energy possessed by a moving body is.

A. kinetic energy B. potential energy C. chemical energy only D. nuclear charge

Answer: A. kinetic energy. Explanation: Energy possessed by a moving body is.

Q10. Which of the following is true? Energy possessed by a moving body is.

A. potential energy B. chemical energy only C. nuclear charge D. kinetic energy

Answer: D. kinetic energy. Explanation: Energy possessed by a moving body is.

Q11. Which statement is correct? Power is the rate of doing.

A. displacement B. mass C. work D. force

Answer: C. work. Explanation: Power is the rate of doing.

Q12. Choose the scientifically correct option: Power is the rate of doing.

A. mass B. work C. force D. displacement

Answer: B. work. Explanation: Power is the rate of doing.

Q13. Which option correctly describes this fact: Power is the rate of doing?

A. work B. force C. displacement D. mass

Answer: A. work. Explanation: Power is the rate of doing.

Q14. Select the best answer about this topic: Power is the rate of doing.

A. force B. displacement C. mass D. work

Answer: D. work. Explanation: Power is the rate of doing.

Q15. Which of the following is true? Power is the rate of doing.

A. displacement B. mass C. work D. force

Answer: C. work. Explanation: Power is the rate of doing.

Q16. Which statement is correct? One kilowatt-hour equals.

A. 60 J B. 3.6 × 10■ J C. 360 J D. 1000 J

Answer: B. 3.6 × 10■ J. Explanation: One kilowatt-hour equals.

Q17. Choose the scientifically correct option: One kilowatt-hour equals.

A. 3.6 × 10■ J B. 360 J C. 1000 J D. 60 J

Answer: A. 3.6 × 10■ J. Explanation: One kilowatt-hour equals.

Q18. Which option correctly describes this fact: One kilowatt-hour equals?

A. 360 J B. 1000 J C. 60 J D. 3.6 × 10■ J

Answer: D. 3.6 × 10■ J. Explanation: One kilowatt-hour equals.

Q19. Select the best answer about this topic: One kilowatt-hour equals.

A. 1000 J B. 60 J C. 3.6 × 10■ J D. 360 J

Answer: C. 3.6 × 10■ J. Explanation: One kilowatt-hour equals.

Q20. Which of the following is true? One kilowatt-hour equals.

A. 60 J B. 3.6 × 10■ J C. 360 J D. 1000 J

Answer: B. 3.6 × 10■ J. Explanation: One kilowatt-hour equals.

Q21. Which statement is correct? Work done by a force perpendicular to displacement is.

A. zero B. maximum C. negative always D. infinite

Answer: A. zero. Explanation: Work done by a force perpendicular to displacement is.

Q22. Choose the scientifically correct option: Work done by a force perpendicular to displacement is.

A. maximum B. negative always C. infinite D. zero

Answer: D. zero. Explanation: Work done by a force perpendicular to displacement is.

Q23. Which option correctly describes this fact: Work done by a force perpendicular to displacement is?

A. negative always B. infinite C. zero D. maximum

Answer: C. zero. Explanation: Work done by a force perpendicular to displacement is.

Q24. Select the best answer about this topic: Work done by a force perpendicular to displacement is.

A. infinite B. zero C. maximum D. negative always

Answer: B. zero. Explanation: Work done by a force perpendicular to displacement is.

Q25. Which of the following is true? Work done by a force perpendicular to displacement is.

A. zero B. maximum C. negative always D. infinite

Answer: A. zero. Explanation: Work done by a force perpendicular to displacement is.

CHAPTER 5: Gravitation

Physics — Gravitation is the universal attractive force between any two masses.

Concept notes and exam points

1. Newton's law of universal gravitation: every two masses attract with force proportional to the product of their masses and inversely proportional to the square of their separation.

2. The gravitational force acts along the line joining the centres of the two bodies and is always attractive for ordinary masses.

3. Near Earth's surface, acceleration due to gravity is approximately 9.8 m/s²; it is commonly rounded to 10 m/s² in simple calculations.

4. Mass is the amount of matter and remains essentially constant with location. Weight is gravitational force and changes with local gravitational acceleration.

5. Weight W = mg. SI unit of weight is newton; SI unit of mass is kilogram.

6. An object in free fall experiences gravity as the only significant force if air resistance is neglected.

7. Pressure is force per unit area. Liquid pressure increases with depth because of the weight of liquid above.

8. Buoyant force is the upward force exerted by a fluid. An object floats when buoyancy can balance its weight.

Formulae, units and quick facts

• F = Gm■m■/r²

• weight W = mg

• pressure P = F/A

• liquid pressure p = ρgh

Worked examples

Problem: Find the weight of a 6 kg object where g = 10 m/s².

Solution: 60 N. W = mg = 6 × 10 = 60 N.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The approximate value of g near Earth's surface is.

A. 9.8 m/s² B. 98 m/s² C. 0.98 m/s² D. 6.67 m/s²

Answer: A. 9.8 m/s². Explanation: The approximate value of g near Earth's surface is.

Q2. Choose the scientifically correct option: The approximate value of g near Earth's surface is.

A. 98 m/s² B. 0.98 m/s² C. 6.67 m/s² D. 9.8 m/s²

Answer: D. 9.8 m/s². Explanation: The approximate value of g near Earth's surface is.

Q3. Which option correctly describes this fact: The approximate value of g near Earth's surface is?

A. 0.98 m/s² B. 6.67 m/s² C. 9.8 m/s² D. 98 m/s²

Answer: C. 9.8 m/s². Explanation: The approximate value of g near Earth's surface is.

Q4. Select the best answer about this topic: The approximate value of g near Earth's surface is.

A. 6.67 m/s² B. 9.8 m/s² C. 98 m/s² D. 0.98 m/s²

Answer: B. 9.8 m/s². Explanation: The approximate value of g near Earth's surface is.

Q5. Which of the following is true? The approximate value of g near Earth's surface is.

A. 9.8 m/s² B. 98 m/s² C. 0.98 m/s² D. 6.67 m/s²

Answer: A. 9.8 m/s². Explanation: The approximate value of g near Earth's surface is.

Q6. Which statement is correct? Mass is measured in.

A. newton B. pascal C. joule D. kilogram

Answer: D. kilogram. Explanation: Mass is measured in.

Q7. Choose the scientifically correct option: Mass is measured in.

A. pascal B. joule C. kilogram D. newton

Answer: C. kilogram. Explanation: Mass is measured in.

Q8. Which option correctly describes this fact: Mass is measured in?

A. joule B. kilogram C. newton D. pascal

Answer: B. kilogram. Explanation: Mass is measured in.

Q9. Select the best answer about this topic: Mass is measured in.

A. kilogram B. newton C. pascal D. joule

Answer: A. kilogram. Explanation: Mass is measured in.

Q10. Which of the following is true? Mass is measured in.

A. newton B. pascal C. joule D. kilogram

Answer: D. kilogram. Explanation: Mass is measured in.

Q11. Which statement is correct? Weight is a type of.

A. energy B. density C. force D. mass

Answer: C. force. Explanation: Weight is a type of.

Q12. Choose the scientifically correct option: Weight is a type of.

A. density B. force C. mass D. energy

Answer: B. force. Explanation: Weight is a type of.

Q13. Which option correctly describes this fact: Weight is a type of?

A. force B. mass C. energy D. density

Answer: A. force. Explanation: Weight is a type of.

Q14. Select the best answer about this topic: Weight is a type of.

A. mass B. energy C. density D. force

Answer: D. force. Explanation: Weight is a type of.

Q15. Which of the following is true? Weight is a type of.

A. energy B. density C. force D. mass

Answer: C. force. Explanation: Weight is a type of.

Q16. Which statement is correct? Gravitational force between ordinary masses is.

A. electrical B. attractive C. repulsive D. zero

Answer: B. attractive. Explanation: Gravitational force between ordinary masses is.

Q17. Choose the scientifically correct option: Gravitational force between ordinary masses is.

A. attractive B. repulsive C. zero D. electrical

Answer: A. attractive. Explanation: Gravitational force between ordinary masses is.

Q18. Which option correctly describes this fact: Gravitational force between ordinary masses is?

A. repulsive B. zero C. electrical D. attractive

Answer: D. attractive. Explanation: Gravitational force between ordinary masses is.

Q19. Select the best answer about this topic: Gravitational force between ordinary masses is.

A. zero B. electrical C. attractive D. repulsive

Answer: C. attractive. Explanation: Gravitational force between ordinary masses is.

Q20. Which of the following is true? Gravitational force between ordinary masses is.

A. electrical B. attractive C. repulsive D. zero

Answer: B. attractive. Explanation: Gravitational force between ordinary masses is.

Q21. Which statement is correct? Liquid pressure generally increases with.

A. depth B. surface colour C. container material only D. temperature only

Answer: A. depth. Explanation: Liquid pressure generally increases with.

Q22. Choose the scientifically correct option: Liquid pressure generally increases with.

A. surface colour B. container material only C. temperature only D. depth

Answer: D. depth. Explanation: Liquid pressure generally increases with.

Q23. Which option correctly describes this fact: Liquid pressure generally increases with?

A. container material only B. temperature only C. depth D. surface colour

Answer: C. depth. Explanation: Liquid pressure generally increases with.

Q24. Select the best answer about this topic: Liquid pressure generally increases with.

A. temperature only B. depth C. surface colour D. container material only

Answer: B. depth. Explanation: Liquid pressure generally increases with.

Q25. Which of the following is true? Liquid pressure generally increases with.

A. depth B. surface colour C. container material only D. temperature only

Answer: A. depth. Explanation: Liquid pressure generally increases with.

CHAPTER 6: Heat and Temperature

Physics — Heat is energy transferred because of a temperature difference; temperature indicates how hot or cold a body is.

Concept notes and exam points

1. Temperature is measured in kelvin (SI), degree Celsius and degree Fahrenheit. Heat is energy and is measured in joules.

2. Heat flows spontaneously from a higher-temperature body to a lower-temperature body until thermal equilibrium is reached.

3. Conduction transfers heat through direct interactions, especially in solids. Metals are generally good thermal conductors.

4. Convection transfers heat by bulk movement of fluids; it occurs in liquids and gases, not in rigid solids as a bulk flow.

5. Radiation transfers energy by electromagnetic waves and does not require a material medium; sunlight reaches Earth this way.

6. Specific heat capacity is the heat needed to raise the temperature of unit mass of a substance by one kelvin.

7. During a change of state, latent heat is absorbed or released while temperature remains constant for a pure substance at fixed pressure.

8. Thermal expansion is the increase in dimensions of a material when heated; gaps are left in railway tracks and bridges to accommodate expansion.

Formulae, units and quick facts

• $Q = mc\Delta T$

• $Q = mL$

• $K = °C + 273.15$

• $°F = (9/5)°C + 32$

Worked examples

Problem: How much heat raises 2 kg of a substance with c = 500 J/kg°C by 4°C?

Solution: 4000 J. $Q = mc\Delta T = 2 \times 500 \times 4 = 4000$ J.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Heat transfer through vacuum occurs by.

A. radiation B. conduction C. convection D. diffusion only

Answer: A. radiation. Explanation: Heat transfer through vacuum occurs by.

Q2. Choose the scientifically correct option: Heat transfer through vacuum occurs by.

A. conduction B. convection C. diffusion only D. radiation

Answer: D. radiation. Explanation: Heat transfer through vacuum occurs by.

Q3. Which option correctly describes this fact: Heat transfer through vacuum occurs by?

A. convection B. diffusion only C. radiation D. conduction

Answer: C. radiation. Explanation: Heat transfer through vacuum occurs by.

Q4. Select the best answer about this topic: Heat transfer through vacuum occurs by.

A. diffusion only B. radiation C. conduction D. convection

Answer: B. radiation. Explanation: Heat transfer through vacuum occurs by.

Q5. Which of the following is true? Heat transfer through vacuum occurs by.

A. radiation B. conduction C. convection D. diffusion only

Answer: A. radiation. Explanation: Heat transfer through vacuum occurs by.

Q6. Which statement is correct? SI unit of temperature is.

A. joule B. watt C. newton D. kelvin

Answer: D. kelvin. Explanation: SI unit of temperature is.

Q7. Choose the scientifically correct option: SI unit of temperature is.

A. watt B. newton C. kelvin D. joule

Answer: C. kelvin. Explanation: SI unit of temperature is.

Q8. Which option correctly describes this fact: SI unit of temperature is?

A. newton B. kelvin C. joule D. watt

Answer: B. kelvin. Explanation: SI unit of temperature is.

Q9. Select the best answer about this topic: SI unit of temperature is.

A. kelvin B. joule C. watt D. newton

Answer: A. kelvin. Explanation: SI unit of temperature is.

Q10. Which of the following is true? SI unit of temperature is.

A. joule B. watt C. newton D. kelvin

Answer: D. kelvin. Explanation: SI unit of temperature is.

Q11. Which statement is correct? Convection mainly occurs in.

A. rigid solids only B. metals only C. fluids D. vacuum

Answer: C. fluids. Explanation: Convection mainly occurs in.

Q12. Choose the scientifically correct option: Convection mainly occurs in.

A. metals only B. fluids C. vacuum D. rigid solids only

Answer: B. fluids. Explanation: Convection mainly occurs in.

Q13. Which option correctly describes this fact: Convection mainly occurs in?

A. fluids B. vacuum C. rigid solids only D. metals only

Answer: A. fluids. Explanation: Convection mainly occurs in.

Q14. Select the best answer about this topic: Convection mainly occurs in.

A. vacuum B. rigid solids only C. metals only D. fluids

Answer: D. fluids. Explanation: Convection mainly occurs in.

Q15. Which of the following is true? Convection mainly occurs in.

A. rigid solids only B. metals only C. fluids D. vacuum

Answer: C. fluids. Explanation: Convection mainly occurs in.

Q16. Which statement is correct? During melting of a pure substance, temperature generally.

A. becomes zero kelvin B. remains constant C. rises continuously D. falls continuously

Answer: B. remains constant. Explanation: During melting of a pure substance, temperature generally.

Q17. Choose the scientifically correct option: During melting of a pure substance, temperature generally.

A. remains constant B. rises continuously C. falls continuously D. becomes zero kelvin

Answer: A. remains constant. Explanation: During melting of a pure substance, temperature generally.

Q18. Which option correctly describes this fact: During melting of a pure substance, temperature generally?

A. rises continuously B. falls continuously C. becomes zero kelvin D. remains constant

Answer: D. remains constant. Explanation: During melting of a pure substance, temperature generally.

Q19. Select the best answer about this topic: During melting of a pure substance, temperature generally.

A. falls continuously B. becomes zero kelvin C. remains constant D. rises continuously

Answer: C. remains constant. Explanation: During melting of a pure substance, temperature generally.

Q20. Which of the following is true? During melting of a pure substance, temperature generally.

A. becomes zero kelvin B. remains constant C. rises continuously D. falls continuously

Answer: B. remains constant. Explanation: During melting of a pure substance, temperature generally.

Q21. Which statement is correct? Heat required for a temperature rise is calculated using.

A. Q = mcΔT B. F = ma C. P = VI D. v = u + at

Answer: A. Q = mcΔT. Explanation: Heat required for a temperature rise is calculated using.

Q22. Choose the scientifically correct option: Heat required for a temperature rise is calculated using.

A. F = ma B. P = VI C. v = u + at D. Q = mcΔT

Answer: D. Q = mcΔT. Explanation: Heat required for a temperature rise is calculated using.

Q23. Which option correctly describes this fact: Heat required for a temperature rise is calculated using?

A. P = VI B. v = u + at C. Q = mcΔT D. F = ma

Answer: C. Q = mcΔT. Explanation: Heat required for a temperature rise is calculated using.

Q24. Select the best answer about this topic: Heat required for a temperature rise is calculated using.

A. v = u + at B. Q = mcΔT C. F = ma D. P = VI

Answer: B. Q = mcΔT. Explanation: Heat required for a temperature rise is calculated using.

Q25. Which of the following is true? Heat required for a temperature rise is calculated using.

A. Q = mcΔT B. F = ma C. P = VI D. v = u + at

Answer: A. Q = mcΔT. Explanation: Heat required for a temperature rise is calculated using.

CHAPTER 7: Sound

Physics — Sound is a mechanical wave produced by vibrating sources and transmitted through a material medium.

Concept notes and exam points

1. Sound needs a medium; it cannot travel through vacuum. In air, sound is commonly transmitted as longitudinal pressure waves.

2. A longitudinal wave has compressions and rarefactions. Particles oscillate parallel to the direction of wave propagation.

3. Frequency is the number of oscillations per second and is measured in hertz (Hz). It determines pitch.

4. Amplitude is related to the energy carried by a wave and affects loudness. Loudness also depends on the ear and distance.

5. The speed of sound depends on the medium and its conditions. It is generally faster in solids than in gases.

6. The human audible range is approximately 20 Hz to 20,000 Hz. Below this is infrasound; above it is ultrasound.

7. Echo is reflected sound heard separately when the time gap is sufficient. Reflection is used in SONAR and medical ultrasound imaging.

8. Noise pollution can affect sleep, concentration and hearing. Ear protection and reducing exposure help prevent damage.

Formulae, units and quick facts

• wave speed $v = f\lambda$

• time period $T = 1/f$

• echo distance $d = vt/2$

Worked examples

Problem: A wave has frequency 200 Hz and wavelength 1.5 m. Find speed.

Solution: 300 m/s. $v = f\lambda = 200 \times 1.5 = 300$ m/s.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Sound cannot travel through.

A. vacuum B. water C. steel D. air

Answer: A. vacuum. Explanation: Sound cannot travel through.

Q2. Choose the scientifically correct option: Sound cannot travel through.

A. water B. steel C. air D. vacuum

Answer: D. vacuum. Explanation: Sound cannot travel through.

Q3. Which option correctly describes this fact: Sound cannot travel through?

A. steel B. air C. vacuum D. water

Answer: C. vacuum. Explanation: Sound cannot travel through.

Q4. Select the best answer about this topic: Sound cannot travel through.

A. air B. vacuum C. water D. steel

Answer: B. vacuum. Explanation: Sound cannot travel through.

Q5. Which of the following is true? Sound cannot travel through.

A. vacuum B. water C. steel D. air

Answer: A. vacuum. Explanation: Sound cannot travel through.

Q6. Which statement is correct? Pitch depends mainly on.

A. amplitude B. density only C. echo time D. frequency

Answer: D. frequency. Explanation: Pitch depends mainly on.

Q7. Choose the scientifically correct option: Pitch depends mainly on.

A. density only B. echo time C. frequency D. amplitude

Answer: C. frequency. Explanation: Pitch depends mainly on.

Q8. Which option correctly describes this fact: Pitch depends mainly on?

A. echo time B. frequency C. amplitude D. density only

Answer: B. frequency. Explanation: Pitch depends mainly on.

Q9. Select the best answer about this topic: Pitch depends mainly on.

A. frequency B. amplitude C. density only D. echo time

Answer: A. frequency. Explanation: Pitch depends mainly on.

Q10. Which of the following is true? Pitch depends mainly on.

A. amplitude B. density only C. echo time D. frequency

Answer: D. frequency. Explanation: Pitch depends mainly on.

Q11. Which statement is correct? The SI unit of frequency is.

A. metre B. second C. hertz D. decibel

Answer: C. hertz. Explanation: The SI unit of frequency is.

Q12. Choose the scientifically correct option: The SI unit of frequency is.

A. second B. hertz C. decibel D. metre

Answer: B. hertz. Explanation: The SI unit of frequency is.

Q13. Which option correctly describes this fact: The SI unit of frequency is?

A. hertz B. decibel C. metre D. second

Answer: A. hertz. Explanation: The SI unit of frequency is.

Q14. Select the best answer about this topic: The SI unit of frequency is.

A. decibel B. metre C. second D. hertz

Answer: D. hertz. Explanation: The SI unit of frequency is.

Q15. Which of the following is true? The SI unit of frequency is.

A. metre B. second C. hertz D. decibel

Answer: C. hertz. Explanation: The SI unit of frequency is.

Q16. Which statement is correct? Human audible frequency range is approximately.

A. 1–10 Hz B. 20 Hz–20 kHz C. 0–10 Hz D. 20 kHz–1 MHz

Answer: B. 20 Hz–20 kHz. Explanation: Human audible frequency range is approximately.

Q17. Choose the scientifically correct option: Human audible frequency range is approximately.

A. 20 Hz–20 kHz B. 0–10 Hz C. 20 kHz–1 MHz D. 1–10 Hz

Answer: A. 20 Hz–20 kHz. Explanation: Human audible frequency range is approximately.

Q18. Which option correctly describes this fact: Human audible frequency range is approximately?

A. 0–10 Hz B. 20 kHz–1 MHz C. 1–10 Hz D. 20 Hz–20 kHz

Answer: D. 20 Hz–20 kHz. Explanation: Human audible frequency range is approximately.

Q19. Select the best answer about this topic: Human audible frequency range is approximately.

A. 20 kHz–1 MHz B. 1–10 Hz C. 20 Hz–20 kHz D. 0–10 Hz

Answer: C. 20 Hz–20 kHz. Explanation: Human audible frequency range is approximately.

Q20. Which of the following is true? Human audible frequency range is approximately.

A. 1–10 Hz B. 20 Hz–20 kHz C. 0–10 Hz D. 20 kHz–1 MHz

Answer: B. 20 Hz–20 kHz. Explanation: Human audible frequency range is approximately.

Q21. Which statement is correct? SONAR uses.

A. reflected sound waves B. visible light C. radioactive decay D. static electricity

Answer: A. reflected sound waves. Explanation: SONAR uses.

Q22. Choose the scientifically correct option: SONAR uses.

A. visible light B. radioactive decay C. static electricity D. reflected sound waves

Answer: D. reflected sound waves. Explanation: SONAR uses.

Q23. Which option correctly describes this fact: SONAR uses?

A. radioactive decay B. static electricity C. reflected sound waves D. visible light

Answer: C. reflected sound waves. Explanation: SONAR uses.

Q24. Select the best answer about this topic: SONAR uses.

A. static electricity B. reflected sound waves C. visible light D. radioactive decay

Answer: B. reflected sound waves. Explanation: SONAR uses.

Q25. Which of the following is true? SONAR uses.

A. reflected sound waves B. visible light C. radioactive decay D. static electricity

Answer: A. reflected sound waves. Explanation: SONAR uses.

CHAPTER 8: Light

Physics — Light is electromagnetic radiation visible to the human eye; it can travel through vacuum.

Concept notes and exam points

1. Light travels in straight lines in a uniform transparent medium. Its speed in vacuum is approximately 3 × 10■ m/s.

2. Reflection is the bouncing back of light. The angle of incidence equals the angle of reflection, measured from the normal.

3. A plane mirror forms a virtual, erect image of the same size, laterally inverted and as far behind the mirror as the object is in front.

4. Refraction is bending of light when its speed changes as it enters another medium. It is described by Snell's law.

5. A convex lens converges parallel rays; a concave lens diverges them. Lens power is measured in dioptres.

6. A convex mirror gives a virtual, erect and diminished image and provides a wide field of view, useful in vehicle mirrors.

7. Dispersion separates white light into colours because different wavelengths refract by different amounts.

8. The human eye focuses light on the retina. The iris controls pupil size; the lens changes shape for accommodation.

Formulae, units and quick facts

• n = c/v

• P = 1/f (f in metres)

• 1/f = 1/v – 1/u (Cartesian sign convention)

Worked examples

Problem: A lens has focal length 0.5 m. Find its power.

Solution: 2 D. P = 1/f = 1/0.5 = 2 dioptres.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The speed of light in vacuum is approximately.

A. 3 × 10■ m/s B. 3 × 10■ m/s C. 340 m/s D. 1.5 × 10■ m/s

Answer: A. 3 × 10■ m/s. Explanation: The speed of light in vacuum is approximately.

Q2. Choose the scientifically correct option: The speed of light in vacuum is approximately.

A. 3 × 10■ m/s B. 340 m/s C. 1.5 × 10■ m/s D. 3 × 10■ m/s

Answer: D. 3 × 10■ m/s. Explanation: The speed of light in vacuum is approximately.

Q3. Which option correctly describes this fact: The speed of light in vacuum is approximately?

A. 340 m/s B. 1.5 × 10■ m/s C. 3 × 10■ m/s D. 3 × 10■ m/s

Answer: C. 3 × 10■ m/s. Explanation: The speed of light in vacuum is approximately.

Q4. Select the best answer about this topic: The speed of light in vacuum is approximately.

A. 1.5 × 10■ m/s B. 3 × 10■ m/s C. 3 × 10■ m/s D. 340 m/s

Answer: B. 3 × 10■ m/s. Explanation: The speed of light in vacuum is approximately.

Q5. Which of the following is true? The speed of light in vacuum is approximately.

A. 3 × 10■ m/s B. 3 × 10■ m/s C. 340 m/s D. 1.5 × 10■ m/s

Answer: A. 3 × 10■ m/s. Explanation: The speed of light in vacuum is approximately.

Q6. Which statement is correct? A plane mirror forms an image that is.

A. real and inverted B. always diminished C. always magnified D. virtual and erect

Answer: D. virtual and erect. Explanation: A plane mirror forms an image that is.

Q7. Choose the scientifically correct option: A plane mirror forms an image that is.

A. always diminished B. always magnified C. virtual and erect D. real and inverted

Answer: C. virtual and erect. Explanation: A plane mirror forms an image that is.

Q8. Which option correctly describes this fact: A plane mirror forms an image that is?

A. always magnified B. virtual and erect C. real and inverted D. always diminished

Answer: B. virtual and erect. Explanation: A plane mirror forms an image that is.

Q9. Select the best answer about this topic: A plane mirror forms an image that is.

A. virtual and erect B. real and inverted C. always diminished D. always magnified

Answer: A. virtual and erect. Explanation: A plane mirror forms an image that is.

Q10. Which of the following is true? A plane mirror forms an image that is.

A. real and inverted B. always diminished C. always magnified D. virtual and erect

Answer: D. virtual and erect. Explanation: A plane mirror forms an image that is.

Q11. Which statement is correct? A convex lens is a.

A. plane mirror B. prism only C. converging lens D. diverging lens

Answer: C. converging lens. Explanation: A convex lens is a.

Q12. Choose the scientifically correct option: A convex lens is a.

A. prism only B. converging lens C. diverging lens D. plane mirror

Answer: B. converging lens. Explanation: A convex lens is a.

Q13. Which option correctly describes this fact: A convex lens is a?

A. converging lens B. diverging lens C. plane mirror D. prism only

Answer: A. converging lens. Explanation: A convex lens is a.

Q14. Select the best answer about this topic: A convex lens is a.

A. diverging lens B. plane mirror C. prism only D. converging lens

Answer: D. converging lens. Explanation: A convex lens is a.

Q15. Which of the following is true? A convex lens is a.

A. plane mirror B. prism only C. converging lens D. diverging lens

Answer: C. converging lens. Explanation: A convex lens is a.

Q16. Which statement is correct? The splitting of white light into colours is.

A. diffusion B. dispersion C. reflection D. conduction

Answer: B. dispersion. Explanation: The splitting of white light into colours is.

Q17. Choose the scientifically correct option: The splitting of white light into colours is.

A. dispersion B. reflection C. conduction D. diffusion

Answer: A. dispersion. Explanation: The splitting of white light into colours is.

Q18. Which option correctly describes this fact: The splitting of white light into colours is?

A. reflection B. conduction C. diffusion D. dispersion

Answer: D. dispersion. Explanation: The splitting of white light into colours is.

Q19. Select the best answer about this topic: The splitting of white light into colours is.

A. conduction B. diffusion C. dispersion D. reflection

Answer: C. dispersion. Explanation: The splitting of white light into colours is.

Q20. Which of the following is true? The splitting of white light into colours is.

A. diffusion B. dispersion C. reflection D. conduction

Answer: B. dispersion. Explanation: The splitting of white light into colours is.

Q21. Which statement is correct? The image in the human eye forms on the.

A. retina B. iris C. cornea only D. pupil

Answer: A. retina. Explanation: The image in the human eye forms on the.

Q22. Choose the scientifically correct option: The image in the human eye forms on the.

A. iris B. cornea only C. pupil D. retina

Answer: D. retina. Explanation: The image in the human eye forms on the.

Q23. Which option correctly describes this fact: The image in the human eye forms on the?

A. cornea only B. pupil C. retina D. iris

Answer: C. retina. Explanation: The image in the human eye forms on the.

Q24. Select the best answer about this topic: The image in the human eye forms on the.

A. pupil B. retina C. iris D. cornea only

Answer: B. retina. Explanation: The image in the human eye forms on the.

Q25. Which of the following is true? The image in the human eye forms on the.

A. retina B. iris C. cornea only D. pupil

Answer: A. retina. Explanation: The image in the human eye forms on the.

CHAPTER 9: Electricity

Physics — Electric current is the rate of flow of electric charge through a cross-section of a conductor.

Concept notes and exam points

1. Current I = Q/t and is measured in ampere (A). Conventional current direction is the direction of positive charge flow.

2. Potential difference is work done per unit charge and is measured in volt (V). Resistance opposes current and is measured in ohm (Ω).

3. Ohm's law states that current through a conductor is directly proportional to potential difference when physical conditions such as temperature remain constant.

4. In a series circuit, current is the same through all components and resistances add. In parallel, potential difference is common and reciprocal resistances add.

5. Electrical power is the rate of electrical energy transfer. For a resistor, $P = VI = I^2R = V^2/R$.

6. A fuse is connected in series and melts when excessive current flows, protecting wiring. Earthing provides a low-resistance safety path.

7. Good conductors have many mobile charge carriers; insulators have very few. Copper and aluminium are common conductors.

8. Domestic electricity is dangerous; never touch exposed live wires. Safety devices and qualified electricians are essential.

Formulae, units and quick facts

• $I = Q/t$

• $V = IR$

• $P = VI = I^2R = V^2/R$

• $E = Pt$

Worked examples

Problem: A 12 V supply is connected to a 4 Ω resistor. Find current.

Solution: 3 A. $I = V/R = 12/4 = 3$ A.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? SI unit of electric current is.

A. ampere B. volt C. ohm D. watt

Answer: A. ampere. Explanation: SI unit of electric current is.

Q2. Choose the scientifically correct option: SI unit of electric current is.

A. volt B. ohm C. watt D. ampere

Answer: D. ampere. Explanation: SI unit of electric current is.

Q3. Which option correctly describes this fact: SI unit of electric current is?

A. ohm B. watt C. ampere D. volt

Answer: C. ampere. Explanation: SI unit of electric current is.

Q4. Select the best answer about this topic: SI unit of electric current is.

A. watt B. ampere C. volt D. ohm

Answer: B. ampere. Explanation: SI unit of electric current is.

Q5. Which of the following is true? SI unit of electric current is.

A. ampere B. volt C. ohm D. watt

Answer: A. ampere. Explanation: SI unit of electric current is.

Q6. Which statement is correct? Ohm's law is expressed as.

A. P = VI B. Q = It² C. R = V I D. V = IR

Answer: D. V = IR. Explanation: Ohm's law is expressed as.

Q7. Choose the scientifically correct option: Ohm's law is expressed as.

A. Q = It² B. R = V I C. V = IR D. P = VI

Answer: C. V = IR. Explanation: Ohm's law is expressed as.

Q8. Which option correctly describes this fact: Ohm's law is expressed as?

A. R = V I B. V = IR C. P = VI D. Q = It²

Answer: B. V = IR. Explanation: Ohm's law is expressed as.

Q9. Select the best answer about this topic: Ohm's law is expressed as.

A. V = IR B. P = VI C. Q = It² D. R = V I

Answer: A. V = IR. Explanation: Ohm's law is expressed as.

Q10. Which of the following is true? Ohm's law is expressed as.

A. P = VI B. Q = It² C. R = V I D. V = IR

Answer: D. V = IR. Explanation: Ohm's law is expressed as.

Q11. Which statement is correct? In a series circuit, the same quantity through each resistor is.

A. resistance B. power C. current D. voltage

Answer: C. current. Explanation: In a series circuit, the same quantity through each resistor is.

Q12. Choose the scientifically correct option: In a series circuit, the same quantity through each resistor is.

A. power B. current C. voltage D. resistance

Answer: B. current. Explanation: In a series circuit, the same quantity through each resistor is.

Q13. Which option correctly describes this fact: In a series circuit, the same quantity through each resistor is?

A. current B. voltage C. resistance D. power

Answer: A. current. Explanation: In a series circuit, the same quantity through each resistor is.

Q14. Select the best answer about this topic: In a series circuit, the same quantity through each resistor is.

A. voltage B. resistance C. power D. current

Answer: D. current. Explanation: In a series circuit, the same quantity through each resistor is.

Q15. Which of the following is true? In a series circuit, the same quantity through each resistor is.

A. resistance B. power C. current D. voltage

Answer: C. current. Explanation: In a series circuit, the same quantity through each resistor is.

Q16. Which statement is correct? A fuse is connected in.

A. open circuit B. series C. parallel only D. earth wire only

Answer: B. series. Explanation: A fuse is connected in.

Q17. Choose the scientifically correct option: A fuse is connected in.

A. series B. parallel only C. earth wire only D. open circuit

Answer: A. series. Explanation: A fuse is connected in.

Q18. Which option correctly describes this fact: A fuse is connected in?

A. parallel only B. earth wire only C. open circuit D. series

Answer: D. series. Explanation: A fuse is connected in.

Q19. Select the best answer about this topic: A fuse is connected in.

A. earth wire only B. open circuit C. series D. parallel only

Answer: C. series. Explanation: A fuse is connected in.

Q20. Which of the following is true? A fuse is connected in.

A. open circuit B. series C. parallel only D. earth wire only

Answer: B. series. Explanation: A fuse is connected in.

Q21. Which statement is correct? Electrical power is measured in.

A. watt B. coulomb C. ohm D. tesla

Answer: A. watt. Explanation: Electrical power is measured in.

Q22. Choose the scientifically correct option: Electrical power is measured in.

A. coulomb B. ohm C. tesla D. watt

Answer: D. watt. Explanation: Electrical power is measured in.

Q23. Which option correctly describes this fact: Electrical power is measured in?

A. ohm B. tesla C. watt D. coulomb

Answer: C. watt. Explanation: Electrical power is measured in.

Q24. Select the best answer about this topic: Electrical power is measured in.

A. tesla B. watt C. coulomb D. ohm

Answer: B. watt. Explanation: Electrical power is measured in.

Q25. Which of the following is true? Electrical power is measured in.

A. watt B. coulomb C. ohm D. tesla

Answer: A. watt. Explanation: Electrical power is measured in.

CHAPTER 10: Magnetism and Electromagnetism

Physics — Magnetism is associated with magnetic fields produced by magnets and moving electric charges.

Concept notes and exam points

1. A magnet has north and south poles. Like poles repel and unlike poles attract; magnetic poles occur in pairs in ordinary magnets.

2. Magnetic field is the region where a magnetic force can be experienced. Field lines outside a bar magnet run from north to south.

3. A compass needle aligns approximately with Earth's local magnetic field and is used to find direction.

4. Electric current produces a magnetic field around a conductor. The right-hand thumb rule gives the field direction around a straight wire.

5. A solenoid produces a field similar to a bar magnet; inserting a soft iron core makes an electromagnet stronger.

6. Electromagnets are used in electric bells, relays, cranes and motors because their magnetism can be switched or varied.

7. An electric motor converts electrical energy into mechanical energy. A generator converts mechanical energy into electrical energy.

8. Electromagnetic induction is production of an emf when magnetic flux linked with a circuit changes.

Formulae, units and quick facts

• $F = BIL \sin\theta$

• Faraday induction: induced emf $\propto$ rate of change of magnetic flux

• motor: electrical $\rightarrow$ mechanical

• generator: mechanical $\rightarrow$ electrical

Worked examples

Problem: A 2 m wire carries 3 A perpendicular to a 0.5 T field. Find force.

Solution: 3 N. $F = BIL = 0.5 \times 3 \times 2 = 3$ N.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Like magnetic poles.

A. repel B. attract C. vanish D. produce gravity

Answer: A. repel. Explanation: Like magnetic poles.

Q2. Choose the scientifically correct option: Like magnetic poles.

A. attract B. vanish C. produce gravity D. repel

Answer: D. repel. Explanation: Like magnetic poles.

Q3. Which option correctly describes this fact: Like magnetic poles?

A. vanish B. produce gravity C. repel D. attract

Answer: C. repel. Explanation: Like magnetic poles.

Q4. Select the best answer about this topic: Like magnetic poles.

A. produce gravity B. repel C. attract D. vanish

Answer: B. repel. Explanation: Like magnetic poles.

Q5. Which of the following is true? Like magnetic poles.

A. repel B. attract C. vanish D. produce gravity

Answer: A. repel. Explanation: Like magnetic poles.

Q6. Which statement is correct? Magnetic field around a straight current-carrying wire is.

A. straight radial B. always zero C. random D. circular

Answer: D. circular. Explanation: Magnetic field around a straight current-carrying wire is.

Q7. Choose the scientifically correct option: Magnetic field around a straight current-carrying wire is.

A. always zero B. random C. circular D. straight radial

Answer: C. circular. Explanation: Magnetic field around a straight current-carrying wire is.

Q8. Which option correctly describes this fact: Magnetic field around a straight current-carrying wire is?

A. random B. circular C. straight radial D. always zero

Answer: B. circular. Explanation: Magnetic field around a straight current-carrying wire is.

Q9. Select the best answer about this topic: Magnetic field around a straight current-carrying wire is.

A. circular B. straight radial C. always zero D. random

Answer: A. circular. Explanation: Magnetic field around a straight current-carrying wire is.

Q10. Which of the following is true? Magnetic field around a straight current-carrying wire is.

A. straight radial B. always zero C. random D. circular

Answer: D. circular. Explanation: Magnetic field around a straight current-carrying wire is.

Q11. Which statement is correct? A soft iron core is used in an electromagnet to.

A. make it permanent only B. remove the field C. increase magnetic strength D. stop all current

Answer: C. increase magnetic strength. Explanation: A soft iron core is used in an electromagnet to.

Q12. Choose the scientifically correct option: A soft iron core is used in an electromagnet to.

A. remove the field B. increase magnetic strength C. stop all current D. make it permanent only

Answer: B. increase magnetic strength. Explanation: A soft iron core is used in an electromagnet to.

Q13. Which option correctly describes this fact: A soft iron core is used in an electromagnet to?

A. increase magnetic strength B. stop all current C. make it permanent only D. remove the field

Answer: A. increase magnetic strength. Explanation: A soft iron core is used in an electromagnet to.

Q14. Select the best answer about this topic: A soft iron core is used in an electromagnet to.

A. stop all current B. make it permanent only C. remove the field D. increase magnetic strength

Answer: D. increase magnetic strength. Explanation: A soft iron core is used in an electromagnet to.

Q15. Which of the following is true? A soft iron core is used in an electromagnet to.

A. make it permanent only B. remove the field C. increase magnetic strength D. stop all current

Answer: C. increase magnetic strength. Explanation: A soft iron core is used in an electromagnet to.

Q16. Which statement is correct? An electric motor converts electrical energy into.

A. sound only B. mechanical energy C. chemical energy D. nuclear energy

Answer: B. mechanical energy. Explanation: An electric motor converts electrical energy into.

Q17. Choose the scientifically correct option: An electric motor converts electrical energy into.

A. mechanical energy B. chemical energy C. nuclear energy D. sound only

Answer: A. mechanical energy. Explanation: An electric motor converts electrical energy into.

Q18. Which option correctly describes this fact: An electric motor converts electrical energy into?

A. chemical energy B. nuclear energy C. sound only D. mechanical energy

Answer: D. mechanical energy. Explanation: An electric motor converts electrical energy into.

Q19. Select the best answer about this topic: An electric motor converts electrical energy into.

A. nuclear energy B. sound only C. mechanical energy D. chemical energy

Answer: C. mechanical energy. Explanation: An electric motor converts electrical energy into.

Q20. Which of the following is true? An electric motor converts electrical energy into.

A. sound only B. mechanical energy C. chemical energy D. nuclear energy

Answer: B. mechanical energy. Explanation: An electric motor converts electrical energy into.

Q21. Which statement is correct? A generator works on.

A. electromagnetic induction B. evaporation C. electrolysis only D. friction only

Answer: A. electromagnetic induction. Explanation: A generator works on.

Q22. Choose the scientifically correct option: A generator works on.

A. evaporation B. electrolysis only C. friction only D. electromagnetic induction

Answer: D. electromagnetic induction. Explanation: A generator works on.

Q23. Which option correctly describes this fact: A generator works on?

A. electrolysis only B. friction only C. electromagnetic induction D. evaporation

Answer: C. electromagnetic induction. Explanation: A generator works on.

Q24. Select the best answer about this topic: A generator works on.

A. friction only B. electromagnetic induction C. evaporation D. electrolysis only

Answer: B. electromagnetic induction. Explanation: A generator works on.

Q25. Which of the following is true? A generator works on.

A. electromagnetic induction B. evaporation C. electrolysis only D. friction only

Answer: A. electromagnetic induction. Explanation: A generator works on.

CHAPTER 11: Sources of Energy

Physics — Energy sources are resources from which useful energy is obtained for domestic, industrial and transport needs.

Concept notes and exam points

1. Renewable sources are replenished naturally on a human timescale, such as solar, wind, hydro and sustainably managed biomass.

2. Non-renewable sources such as coal, petroleum, natural gas and uranium are finite or form too slowly for rapid replacement.

3. Fossil fuels release energy by combustion but emit carbon dioxide and pollutants; efficient use and pollution controls are important.

4. Solar photovoltaic cells convert sunlight directly into electricity; solar thermal systems use solar heat.

5. Wind turbines convert kinetic energy of moving air into electrical energy. Output depends on wind speed and site conditions.

6. Hydroelectricity uses falling or flowing water to turn turbines. Dams can affect ecosystems and communities.

7. Biogas is produced by anaerobic decomposition of organic matter and contains mainly methane and carbon dioxide.

8. Energy conservation reduces demand through efficient appliances, insulation, public transport and avoiding unnecessary consumption.

Formulae, units and quick facts

• efficiency = useful output energy/input energy × 100%

• 1 kWh = 3.6 × 10■ J

Worked examples

Problem: A device receives 500 J and gives 350 J useful output. Find efficiency.

Solution: 70%. Efficiency = 350/500 × 100 = 70%.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Which is a renewable energy source?.

A. solar B. coal C. petroleum D. natural gas

Answer: A. solar. Explanation: Which is a renewable energy source?.

Q2. Choose the scientifically correct option: Which is a renewable energy source?.

A. coal B. petroleum C. natural gas D. solar

Answer: D. solar. Explanation: Which is a renewable energy source?.

Q3. Which option correctly describes this fact: Which is a renewable energy source??

A. petroleum B. natural gas C. solar D. coal

Answer: C. solar. Explanation: Which is a renewable energy source?.

Q4. Select the best answer about this topic: Which is a renewable energy source?.

A. natural gas B. solar C. coal D. petroleum

Answer: B. solar. Explanation: Which is a renewable energy source?.

Q5. Which of the following is true? Which is a renewable energy source?.

A. solar B. coal C. petroleum D. natural gas

Answer: A. solar. Explanation: Which is a renewable energy source?.

Q6. Which statement is correct? Photovoltaic cells convert solar energy into.

A. sound energy B. coal C. nuclear fuel D. electrical energy

Answer: D. electrical energy. Explanation: Photovoltaic cells convert solar energy into.

Q7. Choose the scientifically correct option: Photovoltaic cells convert solar energy into.

A. coal B. nuclear fuel C. electrical energy D. sound energy

Answer: C. electrical energy. Explanation: Photovoltaic cells convert solar energy into.

Q8. Which option correctly describes this fact: Photovoltaic cells convert solar energy into?

A. nuclear fuel B. electrical energy C. sound energy D. coal

Answer: B. electrical energy. Explanation: Photovoltaic cells convert solar energy into.

Q9. Select the best answer about this topic: Photovoltaic cells convert solar energy into.

A. electrical energy B. sound energy C. coal D. nuclear fuel

Answer: A. electrical energy. Explanation: Photovoltaic cells convert solar energy into.

Q10. Which of the following is true? Photovoltaic cells convert solar energy into.

A. sound energy B. coal C. nuclear fuel D. electrical energy

Answer: D. electrical energy. Explanation: Photovoltaic cells convert solar energy into.

Q11. Which statement is correct? Biogas is rich in.

A. nitrogen only B. hydrogen chloride C. methane D. oxygen

Answer: C. methane. Explanation: Biogas is rich in.

Q12. Choose the scientifically correct option: Biogas is rich in.

A. hydrogen chloride B. methane C. oxygen D. nitrogen only

Answer: B. methane. Explanation: Biogas is rich in.

Q13. Which option correctly describes this fact: Biogas is rich in?

A. methane B. oxygen C. nitrogen only D. hydrogen chloride

Answer: A. methane. Explanation: Biogas is rich in.

Q14. Select the best answer about this topic: Biogas is rich in.

A. oxygen B. nitrogen only C. hydrogen chloride D. methane

Answer: D. methane. Explanation: Biogas is rich in.

Q15. Which of the following is true? Biogas is rich in.

A. nitrogen only B. hydrogen chloride C. methane D. oxygen

Answer: C. methane. Explanation: Biogas is rich in.

Q16. Which statement is correct? Fossil fuel combustion releases significant.

A. neon B. carbon dioxide C. helium D. argon

Answer: B. carbon dioxide. Explanation: Fossil fuel combustion releases significant.

Q17. Choose the scientifically correct option: Fossil fuel combustion releases significant.

A. carbon dioxide B. helium C. argon D. neon

Answer: A. carbon dioxide. Explanation: Fossil fuel combustion releases significant.

Q18. Which option correctly describes this fact: Fossil fuel combustion releases significant?

A. helium B. argon C. neon D. carbon dioxide

Answer: D. carbon dioxide. Explanation: Fossil fuel combustion releases significant.

Q19. Select the best answer about this topic: Fossil fuel combustion releases significant.

A. argon B. neon C. carbon dioxide D. helium

Answer: C. carbon dioxide. Explanation: Fossil fuel combustion releases significant.

Q20. Which of the following is true? Fossil fuel combustion releases significant.

A. neon B. carbon dioxide C. helium D. argon

Answer: B. carbon dioxide. Explanation: Fossil fuel combustion releases significant.

Q21. Which statement is correct? Energy efficiency is useful output divided by.

A. input B. mass C. temperature D. time only

Answer: A. input. Explanation: Energy efficiency is useful output divided by.

Q22. Choose the scientifically correct option: Energy efficiency is useful output divided by.

A. mass B. temperature C. time only D. input

Answer: D. input. Explanation: Energy efficiency is useful output divided by.

Q23. Which option correctly describes this fact: Energy efficiency is useful output divided by?

A. temperature B. time only C. input D. mass

Answer: C. input. Explanation: Energy efficiency is useful output divided by.

Q24. Select the best answer about this topic: Energy efficiency is useful output divided by.

A. time only B. input C. mass D. temperature

Answer: B. input. Explanation: Energy efficiency is useful output divided by.

Q25. Which of the following is true? Energy efficiency is useful output divided by.

A. input B. mass C. temperature D. time only

Answer: A. input. Explanation: Energy efficiency is useful output divided by.

CHAPTER 12: Matter and Its Nature

Chemistry — Matter has mass and occupies space. Its particles are in constant motion and attract one another.

Concept notes and exam points

1. Solids have a definite shape and volume; liquids have definite volume but take the shape of their container; gases have neither fixed shape nor fixed volume.

2. Diffusion is mixing caused by random particle motion. It is generally faster in gases than liquids and faster at higher temperature.

3. Melting changes solid to liquid; freezing is the reverse. Vaporisation changes liquid to gas; condensation changes gas to liquid; sublimation is direct solid-to-gas change.

4. During a pure substance's change of state at fixed pressure, temperature remains constant while latent heat is absorbed or released.

5. Evaporation occurs at the surface at temperatures below boiling and causes cooling because higher-energy particles escape.

6. An element contains one type of atom; a compound contains two or more elements chemically combined in a fixed ratio; a mixture has variable composition.

7. Filtration separates insoluble solids from liquids; evaporation recovers dissolved solids; distillation separates liquids by boiling points; chromatography separates components by different movement.

8. Physical changes generally do not form new substances; chemical changes produce substances with different compositions and properties.

Formulae, units and quick facts

• density = mass/volume

• Q = mL (latent heat)

• temperature conversion: K = °C + 273.15

Worked examples

Problem: A 200 g sample occupies 50 cm³. Find density.

Solution: 4 g/cm³. Density = mass/volume = 200/50 = 4 g/cm³.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Matter has.

A. mass and occupies space B. only colour C. no particles D. no volume

Answer: A. mass and occupies space. Explanation: Matter has.

Q2. Choose the scientifically correct option: Matter has.

A. only colour B. no particles C. no volume D. mass and occupies space

Answer: D. mass and occupies space. Explanation: Matter has.

Q3. Which option correctly describes this fact: Matter has?

A. no particles B. no volume C. mass and occupies space D. only colour

Answer: C. mass and occupies space. Explanation: Matter has.

Q4. Select the best answer about this topic: Matter has.

A. no volume B. mass and occupies space C. only colour D. no particles

Answer: B. mass and occupies space. Explanation: Matter has.

Q5. Which of the following is true? Matter has.

A. mass and occupies space B. only colour C. no particles D. no volume

Answer: A. mass and occupies space. Explanation: Matter has.

Q6. Which statement is correct? The state with fixed shape and fixed volume is.

A. liquid B. gas C. vapour D. solid

Answer: D. solid. Explanation: The state with fixed shape and fixed volume is.

Q7. Choose the scientifically correct option: The state with fixed shape and fixed volume is.

A. gas B. vapour C. solid D. liquid

Answer: C. solid. Explanation: The state with fixed shape and fixed volume is.

Q8. Which option correctly describes this fact: The state with fixed shape and fixed volume is?

A. vapour B. solid C. liquid D. gas

Answer: B. solid. Explanation: The state with fixed shape and fixed volume is.

Q9. Select the best answer about this topic: The state with fixed shape and fixed volume is.

A. solid B. liquid C. gas D. vapour

Answer: A. solid. Explanation: The state with fixed shape and fixed volume is.

Q10. Which of the following is true? The state with fixed shape and fixed volume is.

A. liquid B. gas C. vapour D. solid

Answer: D. solid. Explanation: The state with fixed shape and fixed volume is.

Q11. Which statement is correct? Direct change from solid to gas is.

A. freezing B. melting C. sublimation D. condensation

Answer: C. sublimation. Explanation: Direct change from solid to gas is.

Q12. Choose the scientifically correct option: Direct change from solid to gas is.

A. melting B. sublimation C. condensation D. freezing

Answer: B. sublimation. Explanation: Direct change from solid to gas is.

Q13. Which option correctly describes this fact: Direct change from solid to gas is?

A. sublimation B. condensation C. freezing D. melting

Answer: A. sublimation. Explanation: Direct change from solid to gas is.

Q14. Select the best answer about this topic: Direct change from solid to gas is.

A. condensation B. freezing C. melting D. sublimation

Answer: D. sublimation. Explanation: Direct change from solid to gas is.

Q15. Which of the following is true? Direct change from solid to gas is.

A. freezing B. melting C. sublimation D. condensation

Answer: C. sublimation. Explanation: Direct change from solid to gas is.

Q16. Which statement is correct? A method to separate insoluble sand from water is.

A. crystallisation only B. filtration C. chromatography D. magnetisation

Answer: B. filtration. Explanation: A method to separate insoluble sand from water is.

Q17. Choose the scientifically correct option: A method to separate insoluble sand from water is.

A. filtration B. chromatography C. magnetisation D. crystallisation only

Answer: A. filtration. Explanation: A method to separate insoluble sand from water is.

Q18. Which option correctly describes this fact: A method to separate insoluble sand from water is?

A. chromatography B. magnetisation C. crystallisation only D. filtration

Answer: D. filtration. Explanation: A method to separate insoluble sand from water is.

Q19. Select the best answer about this topic: A method to separate insoluble sand from water is.

A. magnetisation B. crystallisation only C. filtration D. chromatography

Answer: C. filtration. Explanation: A method to separate insoluble sand from water is.

Q20. Which of the following is true? A method to separate insoluble sand from water is.

A. crystallisation only B. filtration C. chromatography D. magnetisation

Answer: B. filtration. Explanation: A method to separate insoluble sand from water is.

Q21. Which statement is correct? Evaporation causes cooling because.

A. higher-energy particles escape B. all particles stop moving C. mass becomes zero D. pressure becomes infinite

Answer: A. higher-energy particles escape. Explanation: Evaporation causes cooling because.

Q22. Choose the scientifically correct option: Evaporation causes cooling because.

A. all particles stop moving B. mass becomes zero C. pressure becomes infinite D. higher-energy particles escape

Answer: D. higher-energy particles escape. Explanation: Evaporation causes cooling because.

Q23. Which option correctly describes this fact: Evaporation causes cooling because?

A. mass becomes zero B. pressure becomes infinite C. higher-energy particles escape D. all particles stop moving

Answer: C. higher-energy particles escape. Explanation: Evaporation causes cooling because.

Q24. Select the best answer about this topic: Evaporation causes cooling because.

A. pressure becomes infinite B. higher-energy particles escape C. all particles stop moving D. mass becomes zero

Answer: B. higher-energy particles escape. Explanation: Evaporation causes cooling because.

Q25. Which of the following is true? Evaporation causes cooling because.

A. higher-energy particles escape B. all particles stop moving C. mass becomes zero D. pressure becomes infinite

Answer: A. higher-energy particles escape. Explanation: Evaporation causes cooling because.

CHAPTER 13: Atoms and Molecules

Chemistry — Atoms are the basic units of elements that take part in chemical changes; molecules are groups of atoms bonded together.

Concept notes and exam points

1. An atom contains a nucleus of protons and neutrons, surrounded by electrons. Protons are positive, electrons negative and neutrons neutral.

2. Atomic number Z equals the number of protons. In a neutral atom, electrons equal protons.

3. Mass number A is the total number of protons and neutrons. Neutrons = A – Z.

4. Isotopes have the same atomic number but different mass numbers. They have similar chemical properties but different nuclear properties.

5. A molecule may contain atoms of the same element (O■) or different elements (H■O). Ionic compounds such as NaCl are represented by formula units.

6. Valency describes combining capacity and helps write chemical formulae. The total positive and negative charge in an ionic formula must balance.

7. The mole is the SI unit for amount of substance. One mole contains approximately 6.022×10^{23} specified entities (Avogadro constant).

8. Relative atomic and molecular masses are compared with one-twelfth the mass of a carbon-12 atom.

Formulae, units and quick facts

• neutrons = mass number – atomic number

• moles n = mass/molar mass

• number of particles = $n \times 6.022 \times 10^{23}$

Worked examples

Problem: An atom has atomic number 11 and mass number 23. How many neutrons?

Solution: 12. Neutrons = 23 – 11 = 12.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Atomic number equals number of.

A. protons B. neutrons C. protons + neutrons D. molecules

Answer: A. protons. Explanation: Atomic number equals number of.

Q2. Choose the scientifically correct option: Atomic number equals number of.

A. neutrons B. protons + neutrons C. molecules D. protons

Answer: D. protons. Explanation: Atomic number equals number of.

Q3. Which option correctly describes this fact: Atomic number equals number of?

A. protons + neutrons B. molecules C. protons D. neutrons

Answer: C. protons. Explanation: Atomic number equals number of.

Q4. Select the best answer about this topic: Atomic number equals number of.

A. molecules B. protons C. neutrons D. protons + neutrons

Answer: B. protons. Explanation: Atomic number equals number of.

Q5. Which of the following is true? Atomic number equals number of.

A. protons B. neutrons C. protons + neutrons D. molecules

Answer: A. protons. Explanation: Atomic number equals number of.

Q6. Which statement is correct? A neutral atom has equal numbers of.

A. protons and neutrons B. electrons and neutrons C. atoms and molecules D. protons and electrons

Answer: D. protons and electrons. Explanation: A neutral atom has equal numbers of.

Q7. Choose the scientifically correct option: A neutral atom has equal numbers of.

A. electrons and neutrons B. atoms and molecules C. protons and electrons D. protons and neutrons

Answer: C. protons and electrons. Explanation: A neutral atom has equal numbers of.

Q8. Which option correctly describes this fact: A neutral atom has equal numbers of?

A. atoms and molecules B. protons and electrons C. protons and neutrons D. electrons and neutrons

Answer: B. protons and electrons. Explanation: A neutral atom has equal numbers of.

Q9. Select the best answer about this topic: A neutral atom has equal numbers of.

A. protons and electrons B. protons and neutrons C. electrons and neutrons D. atoms and molecules

Answer: A. protons and electrons. Explanation: A neutral atom has equal numbers of.

Q10. Which of the following is true? A neutral atom has equal numbers of.

A. protons and neutrons B. electrons and neutrons C. atoms and molecules D. protons and electrons

Answer: D. protons and electrons. Explanation: A neutral atom has equal numbers of.

Q11. Which statement is correct? Isotopes have same.

A. number of neutrons B. physical mass C. atomic number D. mass number

Answer: C. atomic number. Explanation: Isotopes have same.

Q12. Choose the scientifically correct option: Isotopes have same.

A. physical mass B. atomic number C. mass number D. number of neutrons

Answer: B. atomic number. Explanation: Isotopes have same.

Q13. Which option correctly describes this fact: Isotopes have same?

A. atomic number B. mass number C. number of neutrons D. physical mass

Answer: A. atomic number. Explanation: Isotopes have same.

Q14. Select the best answer about this topic: Isotopes have same.

A. mass number B. number of neutrons C. physical mass D. atomic number

Answer: D. atomic number. Explanation: Isotopes have same.

Q15. Which of the following is true? Isotopes have same.

A. number of neutrons B. physical mass C. atomic number D. mass number

Answer: C. atomic number. Explanation: Isotopes have same.

Q16. Which statement is correct? One mole contains approximately.

A. 3.14 entities B. 6.022×10^{23} entities C. 100 entities D. 10^{3} entities

Answer: B. 6.022×10^{23} entities. Explanation: One mole contains approximately.

Q17. Choose the scientifically correct option: One mole contains approximately.

A. 6.022×10^{23} entities B. 100 entities C. 10^{3} entities D. 3.14 entities

Answer: A. 6.022×10^{23} entities. Explanation: One mole contains approximately.

Q18. Which option correctly describes this fact: One mole contains approximately?

A. 100 entities B. 10^{3} entities C. 3.14 entities D. 6.022×10^{23} entities

Answer: D. 6.022×10^{23} entities. Explanation: One mole contains approximately.

Q19. Select the best answer about this topic: One mole contains approximately.

A. 10^{3} entities B. 3.14 entities C. 6.022×10^{23} entities D. 100 entities

Answer: C. 6.022×10^{23} entities. Explanation: One mole contains approximately.

Q20. Which of the following is true? One mole contains approximately.

A. 3.14 entities B. 6.022×10^{23} entities C. 100 entities D. 10^{3} entities

Answer: B. 6.022×10^{23} entities. Explanation: One mole contains approximately.

Q21. Which statement is correct? The neutron has.

A. no electric charge B. positive charge C. negative charge D. double positive charge

Answer: A. no electric charge. Explanation: The neutron has.

Q22. Choose the scientifically correct option: The neutron has.

A. positive charge B. negative charge C. double positive charge D. no electric charge

Answer: D. no electric charge. Explanation: The neutron has.

Q23. Which option correctly describes this fact: The neutron has?

A. negative charge B. double positive charge C. no electric charge D. positive charge

Answer: C. no electric charge. Explanation: The neutron has.

Q24. Select the best answer about this topic: The neutron has.

A. double positive charge B. no electric charge C. positive charge D. negative charge

Answer: B. no electric charge. Explanation: The neutron has.

Q25. Which of the following is true? The neutron has.

A. no electric charge B. positive charge C. negative charge D. double positive charge

Answer: A. no electric charge. Explanation: The neutron has.

CHAPTER 14: Chemical Reactions and Equations

Chemistry — A chemical reaction converts reactants into products by rearranging atoms and changing chemical bonds.

Concept notes and exam points

1. A chemical equation represents a reaction using symbols and formulae. It must be balanced because atoms are conserved.

2. Combination reaction: two or more reactants form one product. Decomposition: one compound breaks into simpler substances.

3. Displacement reaction: a more reactive element displaces a less reactive element from its compound. Double displacement involves exchange of ions.

4. Oxidation can mean addition of oxygen or removal of hydrogen; reduction can mean removal of oxygen or addition of hydrogen. Modernly, oxidation is electron loss and reduction electron gain.

5. Redox reactions involve oxidation and reduction together. Corrosion of iron is an oxidation process requiring oxygen and moisture.

6. Exothermic reactions release heat; endothermic reactions absorb heat. Combustion is commonly exothermic.

7. A precipitate is an insoluble solid formed when solutions react. Gas evolution, colour change and temperature change can indicate a reaction.

8. Catalysts alter reaction rate without being consumed overall; they do not change the equilibrium position in a reversible reaction.

Formulae, units and quick facts

• mass of reactants = mass of products in a closed system

• oxidation = loss of electrons

• reduction = gain of electrons

Worked examples

Problem: Balance: H■ + O■ → H■O.

Solution: 2H■ + O■ → 2H■O. There are 4 hydrogen and 2 oxygen atoms on both sides.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? A balanced equation obeys conservation of.

A. atoms/mass B. colour C. temperature only D. volume always

Answer: A. atoms/mass. Explanation: A balanced equation obeys conservation of.

Q2. Choose the scientifically correct option: A balanced equation obeys conservation of.

A. colour B. temperature only C. volume always D. atoms/mass

Answer: D. atoms/mass. Explanation: A balanced equation obeys conservation of.

Q3. Which option correctly describes this fact: A balanced equation obeys conservation of?

A. temperature only B. volume always C. atoms/mass D. colour

Answer: C. atoms/mass. Explanation: A balanced equation obeys conservation of.

Q4. Select the best answer about this topic: A balanced equation obeys conservation of.

A. volume always B. atoms/mass C. colour D. temperature only

Answer: B. atoms/mass. Explanation: A balanced equation obeys conservation of.

Q5. Which of the following is true? A balanced equation obeys conservation of.

A. atoms/mass B. colour C. temperature only D. volume always

Answer: A. atoms/mass. Explanation: A balanced equation obeys conservation of.

Q6. Which statement is correct? A reaction that releases heat is.

A. endothermic B. photolytic only C. neutral D. exothermic

Answer: D. exothermic. Explanation: A reaction that releases heat is.

Q7. Choose the scientifically correct option: A reaction that releases heat is.

A. photolytic only B. neutral C. exothermic D. endothermic

Answer: C. exothermic. Explanation: A reaction that releases heat is.

Q8. Which option correctly describes this fact: A reaction that releases heat is?

A. neutral B. exothermic C. endothermic D. photolytic only

Answer: B. exothermic. Explanation: A reaction that releases heat is.

Q9. Select the best answer about this topic: A reaction that releases heat is.

A. exothermic B. endothermic C. photolytic only D. neutral

Answer: A. exothermic. Explanation: A reaction that releases heat is.

Q10. Which of the following is true? A reaction that releases heat is.

A. endothermic B. photolytic only C. neutral D. exothermic

Answer: D. exothermic. Explanation: A reaction that releases heat is.

Q11. Which statement is correct? Oxidation in electron terms is.

A. loss of neutrons B. gain of protons C. loss of electrons D. gain of electrons

Answer: C. loss of electrons. Explanation: Oxidation in electron terms is.

Q12. Choose the scientifically correct option: Oxidation in electron terms is.

A. gain of protons B. loss of electrons C. gain of electrons D. loss of neutrons

Answer: B. loss of electrons. Explanation: Oxidation in electron terms is.

Q13. Which option correctly describes this fact: Oxidation in electron terms is?

A. loss of electrons B. gain of electrons C. loss of neutrons D. gain of protons

Answer: A. loss of electrons. Explanation: Oxidation in electron terms is.

Q14. Select the best answer about this topic: Oxidation in electron terms is.

A. gain of electrons B. loss of neutrons C. gain of protons D. loss of electrons

Answer: D. loss of electrons. Explanation: Oxidation in electron terms is.

Q15. Which of the following is true? Oxidation in electron terms is.

A. loss of neutrons B. gain of protons C. loss of electrons D. gain of electrons

Answer: C. loss of electrons. Explanation: Oxidation in electron terms is.

Q16. Which statement is correct? A catalyst changes the.

A. atomic number B. rate of reaction C. identity of products always D. total mass

Answer: B. rate of reaction. Explanation: A catalyst changes the.

Q17. Choose the scientifically correct option: A catalyst changes the.

A. rate of reaction B. identity of products always C. total mass D. atomic number

Answer: A. rate of reaction. Explanation: A catalyst changes the.

Q18. Which option correctly describes this fact: A catalyst changes the?

A. identity of products always B. total mass C. atomic number D. rate of reaction

Answer: D. rate of reaction. Explanation: A catalyst changes the.

Q19. Select the best answer about this topic: A catalyst changes the.

A. total mass B. atomic number C. rate of reaction D. identity of products always

Answer: C. rate of reaction. Explanation: A catalyst changes the.

Q20. Which of the following is true? A catalyst changes the.

A. atomic number B. rate of reaction C. identity of products always D. total mass

Answer: B. rate of reaction. Explanation: A catalyst changes the.

Q21. Which statement is correct? Rusting of iron is a form of.

A. corrosion B. distillation C. sublimation D. neutralisation

Answer: A. corrosion. Explanation: Rusting of iron is a form of.

Q22. Choose the scientifically correct option: Rusting of iron is a form of.

A. distillation B. sublimation C. neutralisation D. corrosion

Answer: D. corrosion. Explanation: Rusting of iron is a form of.

Q23. Which option correctly describes this fact: Rusting of iron is a form of?

A. sublimation B. neutralisation C. corrosion D. distillation

Answer: C. corrosion. Explanation: Rusting of iron is a form of.

Q24. Select the best answer about this topic: Rusting of iron is a form of.

A. neutralisation B. corrosion C. distillation D. sublimation

Answer: B. corrosion. Explanation: Rusting of iron is a form of.

Q25. Which of the following is true? Rusting of iron is a form of.

A. corrosion B. distillation C. sublimation D. neutralisation

Answer: A. corrosion. Explanation: Rusting of iron is a form of.

CHAPTER 15: Acids, Bases and Salts

Chemistry — Acids and bases are important classes of substances identified by their behaviour in water and with indicators.

Concept notes and exam points

1. Arrhenius acids produce H■ (more accurately H■O■) in aqueous solution; bases produce OH■ ions in aqueous solution.

2. Acids turn blue litmus red; bases turn red litmus blue. Universal indicator gives a range of colours related to pH.

3. The pH scale is commonly 0–14 at room temperature: below 7 acidic, 7 neutral and above 7 alkaline. pH is logarithmic.

4. Neutralisation is reaction of an acid with a base to form salt and water; it is generally exothermic.

5. Strong acids ionise extensively in water; weak acids ionise partially. Strength is not the same as concentration.

6. Common acids: hydrochloric acid in gastric juice, sulfuric acid in batteries, acetic acid in vinegar and citric acid in citrus fruits.

7. Common bases: sodium hydroxide, potassium hydroxide, calcium hydroxide and magnesium hydroxide. Alkalis are bases soluble in water.

8. Salts are ionic compounds. Baking soda is sodium hydrogen carbonate (NaHCO■); washing soda is sodium carbonate decahydrate (Na■CO■·10H■O); bleaching powder is commonly represented as CaOCl■.

Formulae, units and quick facts

• pH = –log■■[H■]

• acid + base → salt + water

• HCl + NaOH → NaCl + H■O

Worked examples

Problem: A solution has pH 3. Is it acidic or basic?

Solution: Acidic. A pH below 7 is acidic.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? An acid turns blue litmus.

A. red B. blue C. green D. colourless always

Answer: A. red. Explanation: An acid turns blue litmus.

Q2. Choose the scientifically correct option: An acid turns blue litmus.

A. blue B. green C. colourless always D. red

Answer: D. red. Explanation: An acid turns blue litmus.

Q3. Which option correctly describes this fact: An acid turns blue litmus?

A. green B. colourless always C. red D. blue

Answer: C. red. Explanation: An acid turns blue litmus.

Q4. Select the best answer about this topic: An acid turns blue litmus.

A. colourless always B. red C. blue D. green

Answer: B. red. Explanation: An acid turns blue litmus.

Q5. Which of the following is true? An acid turns blue litmus.

A. red B. blue C. green D. colourless always

Answer: A. red. Explanation: An acid turns blue litmus.

Q6. Which statement is correct? A neutral solution at 25°C has pH.

A. 1 B. 14 C. 10 D. 7

Answer: D. 7. Explanation: A neutral solution at 25°C has pH.

Q7. Choose the scientifically correct option: A neutral solution at 25°C has pH.

A. 14 B. 10 C. 7 D. 1

Answer: C. 7. Explanation: A neutral solution at 25°C has pH.

Q8. Which option correctly describes this fact: A neutral solution at 25°C has pH?

A. 10 B. 7 C. 1 D. 14

Answer: B. 7. Explanation: A neutral solution at 25°C has pH.

Q9. Select the best answer about this topic: A neutral solution at 25°C has pH.

A. 7 B. 1 C. 14 D. 10

Answer: A. 7. Explanation: A neutral solution at 25°C has pH.

Q10. Which of the following is true? A neutral solution at 25°C has pH.

A. 1 B. 14 C. 10 D. 7

Answer: D. 7. Explanation: A neutral solution at 25°C has pH.

Q11. Which statement is correct? Vinegar contains.

A. nitric acid B. hydrochloric acid C. acetic acid D. sulfuric acid

Answer: C. acetic acid. Explanation: Vinegar contains.

Q12. Choose the scientifically correct option: Vinegar contains.

A. hydrochloric acid B. acetic acid C. sulfuric acid D. nitric acid

Answer: B. acetic acid. Explanation: Vinegar contains.

Q13. Which option correctly describes this fact: Vinegar contains?

A. acetic acid B. sulfuric acid C. nitric acid D. hydrochloric acid

Answer: A. acetic acid. Explanation: Vinegar contains.

Q14. Select the best answer about this topic: Vinegar contains.

A. sulfuric acid B. nitric acid C. hydrochloric acid D. acetic acid

Answer: D. acetic acid. Explanation: Vinegar contains.

Q15. Which of the following is true? Vinegar contains.

A. nitric acid B. hydrochloric acid C. acetic acid D. sulfuric acid

Answer: C. acetic acid. Explanation: Vinegar contains.

Q16. Which statement is correct? Baking soda is.

A. sodium hydroxide B. sodium hydrogen carbonate C. sodium chloride D. calcium carbonate

Answer: B. sodium hydrogen carbonate. Explanation: Baking soda is.

Q17. Choose the scientifically correct option: Baking soda is.

A. sodium hydrogen carbonate B. sodium chloride C. calcium carbonate D. sodium hydroxide

Answer: A. sodium hydrogen carbonate. Explanation: Baking soda is.

Q18. Which option correctly describes this fact: Baking soda is?

A. sodium chloride B. calcium carbonate C. sodium hydroxide D. sodium hydrogen carbonate

Answer: D. sodium hydrogen carbonate. Explanation: Baking soda is.

Q19. Select the best answer about this topic: Baking soda is.

A. calcium carbonate B. sodium hydroxide C. sodium hydrogen carbonate D. sodium chloride

Answer: C. sodium hydrogen carbonate. Explanation: Baking soda is.

Q20. Which of the following is true? Baking soda is.

A. sodium hydroxide B. sodium hydrogen carbonate C. sodium chloride D. calcium carbonate

Answer: B. sodium hydrogen carbonate. Explanation: Baking soda is.

Q21. Which statement is correct? Neutralisation produces salt and.

A. water B. oxygen C. hydrogen only D. carbon dioxide always

Answer: A. water. Explanation: Neutralisation produces salt and.

Q22. Choose the scientifically correct option: Neutralisation produces salt and.

A. oxygen B. hydrogen only C. carbon dioxide always D. water

Answer: D. water. Explanation: Neutralisation produces salt and.

Q23. Which option correctly describes this fact: Neutralisation produces salt and?

A. hydrogen only B. carbon dioxide always C. water D. oxygen

Answer: C. water. Explanation: Neutralisation produces salt and.

Q24. Select the best answer about this topic: Neutralisation produces salt and.

A. carbon dioxide always B. water C. oxygen D. hydrogen only

Answer: B. water. Explanation: Neutralisation produces salt and.

Q25. Which of the following is true? Neutralisation produces salt and.

A. water B. oxygen C. hydrogen only D. carbon dioxide always

Answer: A. water. Explanation: Neutralisation produces salt and.

CHAPTER 16: Metals and Non-metals

Chemistry — Elements are classified by characteristic physical and chemical properties; metals generally conduct heat and electricity and are malleable.

Concept notes and exam points

1. Metals are generally lustrous, malleable, ductile and good conductors. Exceptions exist: mercury is liquid at room temperature and some metals are relatively soft.

2. Non-metals are generally poor conductors, brittle as solids and may be gases or solids; graphite is a notable conducting non-metal.

3. Metals react with oxygen to form metal oxides, often basic; some oxides are amphoteric, reacting with both acids and bases.

4. Many metals react with acids to produce salt and hydrogen, but less reactive metals such as copper do not liberate hydrogen from dilute non-oxidising acids.

5. The reactivity series arranges metals by tendency to lose electrons. A more reactive metal can displace a less reactive metal from its salt solution.

6. Ionic compounds form by electron transfer, commonly from metals to non-metals. They have high melting points and conduct electricity when molten or dissolved.

7. Corrosion is gradual deterioration by chemical reaction with the environment. Painting, oiling, galvanising and alloying can protect iron.

8. Alloys are mixtures designed for improved properties: brass is copper-zinc; bronze is copper-tin; stainless steel contains iron, chromium and often nickel.

Formulae, units and quick facts

• metal + acid → salt + hydrogen (for suitable reactive metals)

• iron rust: hydrated iron(III) oxide

• galvanising = zinc coating

Worked examples

Problem: Which metal is commonly used to galvanise iron?

Solution: Zinc. Zinc coating protects iron from corrosion.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? A common property of metals is.

A. good electrical conductivity B. brittleness in all cases C. poor heat conduction D. gaseous state

Answer: A. good electrical conductivity. Explanation: A common property of metals is.

Q2. Choose the scientifically correct option: A common property of metals is.

A. brittleness in all cases B. poor heat conduction C. gaseous state D. good electrical conductivity

Answer: D. good electrical conductivity. Explanation: A common property of metals is.

Q3. Which option correctly describes this fact: A common property of metals is?

A. poor heat conduction B. gaseous state C. good electrical conductivity D. brittleness in all cases

Answer: C. good electrical conductivity. Explanation: A common property of metals is.

Q4. Select the best answer about this topic: A common property of metals is.

A. gaseous state B. good electrical conductivity C. brittleness in all cases D. poor heat conduction

Answer: B. good electrical conductivity. Explanation: A common property of metals is.

Q5. Which of the following is true? A common property of metals is.

A. good electrical conductivity B. brittleness in all cases C. poor heat conduction D. gaseous state

Answer: A. good electrical conductivity. Explanation: A common property of metals is.

Q6. Which statement is correct? The liquid metal at room temperature is.

A. iron B. copper C. aluminium D. mercury

Answer: D. mercury. Explanation: The liquid metal at room temperature is.

Q7. Choose the scientifically correct option: The liquid metal at room temperature is.

A. copper B. aluminium C. mercury D. iron

Answer: C. mercury. Explanation: The liquid metal at room temperature is.

Q8. Which option correctly describes this fact: The liquid metal at room temperature is?

A. aluminium B. mercury C. iron D. copper

Answer: B. mercury. Explanation: The liquid metal at room temperature is.

Q9. Select the best answer about this topic: The liquid metal at room temperature is.

A. mercury B. iron C. copper D. aluminium

Answer: A. mercury. Explanation: The liquid metal at room temperature is.

Q10. Which of the following is true? The liquid metal at room temperature is.

A. iron B. copper C. aluminium D. mercury

Answer: D. mercury. Explanation: The liquid metal at room temperature is.

Q11. Which statement is correct? Brass is an alloy of.

A. iron and carbon B. lead and mercury C. copper and zinc D. copper and tin

Answer: C. copper and zinc. Explanation: Brass is an alloy of.

Q12. Choose the scientifically correct option: Brass is an alloy of.

A. lead and mercury B. copper and zinc C. copper and tin D. iron and carbon

Answer: B. copper and zinc. Explanation: Brass is an alloy of.

Q13. Which option correctly describes this fact: Brass is an alloy of?

A. copper and zinc B. copper and tin C. iron and carbon D. lead and mercury

Answer: A. copper and zinc. Explanation: Brass is an alloy of.

Q14. Select the best answer about this topic: Brass is an alloy of.

A. copper and tin B. iron and carbon C. lead and mercury D. copper and zinc

Answer: D. copper and zinc. Explanation: Brass is an alloy of.

Q15. Which of the following is true? Brass is an alloy of.

A. iron and carbon B. lead and mercury C. copper and zinc D. copper and tin

Answer: C. copper and zinc. Explanation: Brass is an alloy of.

Q16. Which statement is correct? Galvanising uses a coating of.

A. silver B. zinc C. sodium D. carbon

Answer: B. zinc. Explanation: Galvanising uses a coating of.

Q17. Choose the scientifically correct option: Galvanising uses a coating of.

A. zinc B. sodium C. carbon D. silver

Answer: A. zinc. Explanation: Galvanising uses a coating of.

Q18. Which option correctly describes this fact: Galvanising uses a coating of?

A. sodium B. carbon C. silver D. zinc

Answer: D. zinc. Explanation: Galvanising uses a coating of.

Q19. Select the best answer about this topic: Galvanising uses a coating of.

A. carbon B. silver C. zinc D. sodium

Answer: C. zinc. Explanation: Galvanising uses a coating of.

Q20. Which of the following is true? Galvanising uses a coating of.

A. silver B. zinc C. sodium D. carbon

Answer: B. zinc. Explanation: Galvanising uses a coating of.

Q21. Which statement is correct? Graphite conducts electricity because it has.

A. delocalised electrons B. no electrons C. only ionic bonds D. no carbon atoms

Answer: A. delocalised electrons. Explanation: Graphite conducts electricity because it has.

Q22. Choose the scientifically correct option: Graphite conducts electricity because it has.

A. no electrons B. only ionic bonds C. no carbon atoms D. delocalised electrons

Answer: D. delocalised electrons. Explanation: Graphite conducts electricity because it has.

Q23. Which option correctly describes this fact: Graphite conducts electricity because it has?

A. only ionic bonds B. no carbon atoms C. delocalised electrons D. no electrons

Answer: C. delocalised electrons. Explanation: Graphite conducts electricity because it has.

Q24. Select the best answer about this topic: Graphite conducts electricity because it has.

A. no carbon atoms B. delocalised electrons C. no electrons D. only ionic bonds

Answer: B. delocalised electrons. Explanation: Graphite conducts electricity because it has.

Q25. Which of the following is true? Graphite conducts electricity because it has.

A. delocalised electrons B. no electrons C. only ionic bonds D. no carbon atoms

Answer: A. delocalised electrons. Explanation: Graphite conducts electricity because it has.

CHAPTER 17: Periodic Classification of Elements

Chemistry — The periodic table arranges elements so that properties show repeating patterns with increasing atomic number.

Concept notes and exam points

1. Modern periodic law: properties of elements are periodic functions of their atomic numbers.

2. The modern periodic table has 7 periods (rows) and 18 groups (columns). Elements in a group often have similar valence-electron configurations and chemical properties.

3. Across a period, atomic radius generally decreases while effective nuclear attraction increases; down a group, atomic radius generally increases.

4. Metallic character generally decreases across a period and increases down a group. Non-metallic character shows the opposite broad trend.

5. Group 1 elements are alkali metals; Group 2 are alkaline earth metals; Group 17 are halogens; Group 18 are noble gases.

6. Noble gases have filled outer shells (helium has a duplet) and are generally unreactive. Halogens are reactive non-metals.

7. The period number is related to the number of occupied electron shells for main-group elements. Group position helps indicate valence electrons.

8. Hydrogen has a unique position because it shows similarities with both alkali metals and halogens.

Formulae, units and quick facts

• periods = 7

• groups = 18

• atomic number = proton number

Worked examples

Problem: How many groups are in the modern periodic table?

Solution: 18. The modern periodic table contains 18 vertical groups.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Modern periodic law is based on.

A. atomic number B. atomic mass only C. density D. melting point

Answer: A. atomic number. Explanation: Modern periodic law is based on.

Q2. Choose the scientifically correct option: Modern periodic law is based on.

A. atomic mass only B. density C. melting point D. atomic number

Answer: D. atomic number. Explanation: Modern periodic law is based on.

Q3. Which option correctly describes this fact: Modern periodic law is based on?

A. density B. melting point C. atomic number D. atomic mass only

Answer: C. atomic number. Explanation: Modern periodic law is based on.

Q4. Select the best answer about this topic: Modern periodic law is based on.

A. melting point B. atomic number C. atomic mass only D. density

Answer: B. atomic number. Explanation: Modern periodic law is based on.

Q5. Which of the following is true? Modern periodic law is based on.

A. atomic number B. atomic mass only C. density D. melting point

Answer: A. atomic number. Explanation: Modern periodic law is based on.

Q6. Which statement is correct? The modern periodic table has how many periods?.

A. 8 B. 18 C. 9 D. 7

Answer: D. 7. Explanation: The modern periodic table has how many periods?.

Q7. Choose the scientifically correct option: The modern periodic table has how many periods?.

A. 18 B. 9 C. 7 D. 8

Answer: C. 7. Explanation: The modern periodic table has how many periods?.

Q8. Which option correctly describes this fact: The modern periodic table has how many periods??

A. 9 B. 7 C. 8 D. 18

Answer: B. 7. Explanation: The modern periodic table has how many periods?.

Q9. Select the best answer about this topic: The modern periodic table has how many periods?.

A. 7 B. 8 C. 18 D. 9

Answer: A. 7. Explanation: The modern periodic table has how many periods?.

Q10. Which of the following is true? The modern periodic table has how many periods?.

A. 8 B. 18 C. 9 D. 7

Answer: D. 7. Explanation: The modern periodic table has how many periods?.

Q11. Which statement is correct? Halogens belong to group.

A. 2 B. 18 C. 17 D. 1

Answer: C. 17. Explanation: Halogens belong to group.

Q12. Choose the scientifically correct option: Halogens belong to group.

A. 18 B. 17 C. 1 D. 2

Answer: B. 17. Explanation: Halogens belong to group.

Q13. Which option correctly describes this fact: Halogens belong to group?

A. 17 B. 1 C. 2 D. 18

Answer: A. 17. Explanation: Halogens belong to group.

Q14. Select the best answer about this topic: Halogens belong to group.

A. 1 B. 2 C. 18 D. 17

Answer: D. 17. Explanation: Halogens belong to group.

Q15. Which of the following is true? Halogens belong to group.

A. 2 B. 18 C. 17 D. 1

Answer: C. 17. Explanation: Halogens belong to group.

Q16. Which statement is correct? Noble gases are generally.

A. strong acids B. unreactive C. highly reactive metals D. liquids only

Answer: B. unreactive. Explanation: Noble gases are generally.

Q17. Choose the scientifically correct option: Noble gases are generally.

A. unreactive B. highly reactive metals C. liquids only D. strong acids

Answer: A. unreactive. Explanation: Noble gases are generally.

Q18. Which option correctly describes this fact: Noble gases are generally?

A. highly reactive metals B. liquids only C. strong acids D. unreactive

Answer: D. unreactive. Explanation: Noble gases are generally.

Q19. Select the best answer about this topic: Noble gases are generally.

A. liquids only B. strong acids C. unreactive D. highly reactive metals

Answer: C. unreactive. Explanation: Noble gases are generally.

Q20. Which of the following is true? Noble gases are generally.

A. strong acids B. unreactive C. highly reactive metals D. liquids only

Answer: B. unreactive. Explanation: Noble gases are generally.

Q21. Which statement is correct? Atomic radius generally increases.

A. down a group B. across a period left to right C. with proton number only in all cases D. when electrons are removed

Answer: A. down a group. Explanation: Atomic radius generally increases.

Q22. Choose the scientifically correct option: Atomic radius generally increases.

A. across a period left to right B. with proton number only in all cases C. when electrons are removed D. down a group

Answer: D. down a group. Explanation: Atomic radius generally increases.

Q23. Which option correctly describes this fact: Atomic radius generally increases?

A. with proton number only in all cases B. when electrons are removed C. down a group D. across a period left to right

Answer: C. down a group. Explanation: Atomic radius generally increases.

Q24. Select the best answer about this topic: Atomic radius generally increases.

A. when electrons are removed B. down a group C. across a period left to right D. with proton number only in all cases

Answer: B. down a group. Explanation: Atomic radius generally increases.

Q25. Which of the following is true? Atomic radius generally increases.

A. down a group B. across a period left to right C. with proton number only in all cases D. when electrons are removed

Answer: A. down a group. Explanation: Atomic radius generally increases.

CHAPTER 18: Carbon and Its Compounds

Chemistry — Carbon forms a vast range of compounds because it is tetravalent and can bond to itself in chains and rings.

Concept notes and exam points

1. Carbon has atomic number 6 and four valence electrons; it commonly forms four covalent bonds.

2. Catenation is carbon's ability to form bonds with other carbon atoms, producing chains, branches and rings.

3. Saturated hydrocarbons contain only single carbon-carbon bonds (alkanes); unsaturated hydrocarbons contain double or triple bonds (alkenes and alkynes).

4. Homologous series members share a functional group and general formula; successive members differ by –CH■– and have gradually changing physical properties.

5. Functional groups determine characteristic reactions: alcohol –OH, aldehyde –CHO, ketone >C=O, carboxylic acid –COOH and halogen –X.

6. Combustion of hydrocarbons in sufficient oxygen produces carbon dioxide and water; incomplete combustion may produce carbon monoxide or soot.

7. Ethanol is an alcohol; ethanoic acid is a carboxylic acid. Esterification of an alcohol with a carboxylic acid forms an ester and water.

8. Soaps form scum in hard water; detergents generally work better in hard water. Soap molecules have hydrophilic and hydrophobic parts.

Formulae, units and quick facts

• alkane: C■H■■■■

• alkene: C■H■■

• alkyne: C■H■■■■

Worked examples

Problem: What is the molecular formula of propane?

Solution: C■H■. For alkanes C■H■■■■; n = 3 gives C■H■.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Carbon commonly forms how many covalent bonds?.

A. 4 B. 2 C. 1 D. 6

Answer: A. 4. Explanation: Carbon commonly forms how many covalent bonds?.

Q2. Choose the scientifically correct option: Carbon commonly forms how many covalent bonds?.

A. 2 B. 1 C. 6 D. 4

Answer: D. 4. Explanation: Carbon commonly forms how many covalent bonds?.

Q3. Which option correctly describes this fact: Carbon commonly forms how many covalent bonds??

A. 1 B. 6 C. 4 D. 2

Answer: C. 4. Explanation: Carbon commonly forms how many covalent bonds?.

Q4. Select the best answer about this topic: Carbon commonly forms how many covalent bonds?.

A. 6 B. 4 C. 2 D. 1

Answer: B. 4. Explanation: Carbon commonly forms how many covalent bonds?.

Q5. Which of the following is true? Carbon commonly forms how many covalent bonds?.

A. 4 B. 2 C. 1 D. 6

Answer: A. 4. Explanation: Carbon commonly forms how many covalent bonds?.

Q6. Which statement is correct? The ability of carbon to bond with itself is.

A. neutralisation B. ionisation C. sublimation D. catenation

Answer: D. catenation. Explanation: The ability of carbon to bond with itself is.

Q7. Choose the scientifically correct option: The ability of carbon to bond with itself is.

A. ionisation B. sublimation C. catenation D. neutralisation

Answer: C. catenation. Explanation: The ability of carbon to bond with itself is.

Q8. Which option correctly describes this fact: The ability of carbon to bond with itself is?

A. sublimation B. catenation C. neutralisation D. ionisation

Answer: B. catenation. Explanation: The ability of carbon to bond with itself is.

Q9. Select the best answer about this topic: The ability of carbon to bond with itself is.

A. catenation B. neutralisation C. ionisation D. sublimation

Answer: A. catenation. Explanation: The ability of carbon to bond with itself is.

Q10. Which of the following is true? The ability of carbon to bond with itself is.

A. neutralisation B. ionisation C. sublimation D. catenation

Answer: D. catenation. Explanation: The ability of carbon to bond with itself is.

Q11. Which statement is correct? Alkenes contain at least one.

A. ionic bond B. metallic bond C. C=C double bond D. C–C single bond only

Answer: C. C=C double bond. Explanation: Alkenes contain at least one.

Q12. Choose the scientifically correct option: Alkenes contain at least one.

A. metallic bond B. C=C double bond C. C–C single bond only D. ionic bond

Answer: B. C=C double bond. Explanation: Alkenes contain at least one.

Q13. Which option correctly describes this fact: Alkenes contain at least one?

A. C=C double bond B. C–C single bond only C. ionic bond D. metallic bond

Answer: A. C=C double bond. Explanation: Alkenes contain at least one.

Q14. Select the best answer about this topic: Alkenes contain at least one.

A. C–C single bond only B. ionic bond C. metallic bond D. C=C double bond

Answer: D. C=C double bond. Explanation: Alkenes contain at least one.

Q15. Which of the following is true? Alkenes contain at least one.

A. ionic bond B. metallic bond C. C=C double bond D. C–C single bond only

Answer: C. C=C double bond. Explanation: Alkenes contain at least one.

Q16. Which statement is correct? The functional group of alcohols is.

A. –NH■ B. –OH C. –COOH D. –CHO

Answer: B. –OH. Explanation: The functional group of alcohols is.

Q17. Choose the scientifically correct option: The functional group of alcohols is.

A. –OH B. –COOH C. –CHO D. –NH■

Answer: A. –OH. Explanation: The functional group of alcohols is.

Q18. Which option correctly describes this fact: The functional group of alcohols is?

A. –COOH B. –CHO C. –NH■ D. –OH

Answer: D. –OH. Explanation: The functional group of alcohols is.

Q19. Select the best answer about this topic: The functional group of alcohols is.

A. –CHO B. –NH■ C. –OH D. –COOH

Answer: C. –OH. Explanation: The functional group of alcohols is.

Q20. Which of the following is true? The functional group of alcohols is.

A. –NH■ B. –OH C. –COOH D. –CHO

Answer: B. –OH. Explanation: The functional group of alcohols is.

Q21. Which statement is correct? Incomplete combustion may produce poisonous.

A. carbon monoxide B. helium C. oxygen D. argon

Answer: A. carbon monoxide. Explanation: Incomplete combustion may produce poisonous.

Q22. Choose the scientifically correct option: Incomplete combustion may produce poisonous.

A. helium B. oxygen C. argon D. carbon monoxide

Answer: D. carbon monoxide. Explanation: Incomplete combustion may produce poisonous.

Q23. Which option correctly describes this fact: Incomplete combustion may produce poisonous?

A. oxygen B. argon C. carbon monoxide D. helium

Answer: C. carbon monoxide. Explanation: Incomplete combustion may produce poisonous.

Q24. Select the best answer about this topic: Incomplete combustion may produce poisonous.

A. argon B. carbon monoxide C. helium D. oxygen

Answer: B. carbon monoxide. Explanation: Incomplete combustion may produce poisonous.

Q25. Which of the following is true? Incomplete combustion may produce poisonous.

A. carbon monoxide B. helium C. oxygen D. argon

Answer: A. carbon monoxide. Explanation: Incomplete combustion may produce poisonous.

CHAPTER 19: Fuels and Combustion

Chemistry — A fuel releases usable energy, usually as heat, through chemical or nuclear processes.

Concept notes and exam points

1. Combustion is a chemical reaction with an oxidant, commonly oxygen, releasing heat and often light.

2. Ignition temperature is the minimum temperature at which a substance catches fire and continues burning under specified conditions.

3. A fuel's calorific value is the heat released by complete combustion of a unit mass (or volume) of fuel.

4. Ideal fuels are readily available, economical, have high calorific value, burn at a controllable rate and produce minimal pollution or residue.

5. Fossil fuels include coal, petroleum and natural gas. They are non-renewable and combustion contributes to greenhouse gas emissions.

6. LPG mainly contains propane and butane; CNG mainly contains methane. Both burn relatively cleanly compared with many solid fuels, but are still fossil-derived in common use.

7. Fire requires fuel, oxygen and sufficient heat. Removing any one component can extinguish a fire. Different fire classes require suitable extinguishers.

8. Carbon monoxide forms during incomplete combustion and binds strongly to haemoglobin, reducing oxygen transport; adequate ventilation is essential.

Formulae, units and quick facts

• calorific value = heat released/mass of fuel

• complete hydrocarbon combustion → CO■ + H■O

Worked examples

Problem: A fuel releases 80,000 kJ from 2 kg. Find calorific value.

Solution: 40,000 kJ/kg. 80,000 ÷ 2 = 40,000 kJ/kg.

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The three components of the fire triangle are fuel, heat and.

A. oxygen B. nitrogen C. carbon dioxide D. water vapour

Answer: A. oxygen. Explanation: The three components of the fire triangle are fuel, heat and.

Q2. Choose the scientifically correct option: The three components of the fire triangle are fuel, heat and.

A. nitrogen B. carbon dioxide C. water vapour D. oxygen

Answer: D. oxygen. Explanation: The three components of the fire triangle are fuel, heat and.

Q3. Which option correctly describes this fact: The three components of the fire triangle are fuel, heat and?

A. carbon dioxide B. water vapour C. oxygen D. nitrogen

Answer: C. oxygen. Explanation: The three components of the fire triangle are fuel, heat and.

Q4. Select the best answer about this topic: The three components of the fire triangle are fuel, heat and.

A. water vapour B. oxygen C. nitrogen D. carbon dioxide

Answer: B. oxygen. Explanation: The three components of the fire triangle are fuel, heat and.

Q5. Which of the following is true? The three components of the fire triangle are fuel, heat and.

A. oxygen B. nitrogen C. carbon dioxide D. water vapour

Answer: A. oxygen. Explanation: The three components of the fire triangle are fuel, heat and.

Q6. Which statement is correct? CNG mainly contains.

A. butane only B. hydrogen chloride C. carbon monoxide D. methane

Answer: D. methane. Explanation: CNG mainly contains.

Q7. Choose the scientifically correct option: CNG mainly contains.

A. hydrogen chloride B. carbon monoxide C. methane D. butane only

Answer: C. methane. Explanation: CNG mainly contains.

Q8. Which option correctly describes this fact: CNG mainly contains?

A. carbon monoxide B. methane C. butane only D. hydrogen chloride

Answer: B. methane. Explanation: CNG mainly contains.

Q9. Select the best answer about this topic: CNG mainly contains.

A. methane B. butane only C. hydrogen chloride D. carbon monoxide

Answer: A. methane. Explanation: CNG mainly contains.

Q10. Which of the following is true? CNG mainly contains.

A. butane only B. hydrogen chloride C. carbon monoxide D. methane

Answer: D. methane. Explanation: CNG mainly contains.

Q11. Which statement is correct? Incomplete combustion can produce.

A. helium B. ozone C. carbon monoxide D. only oxygen

Answer: C. carbon monoxide. Explanation: Incomplete combustion can produce.

Q12. Choose the scientifically correct option: Incomplete combustion can produce.

A. ozone B. carbon monoxide C. only oxygen D. helium

Answer: B. carbon monoxide. Explanation: Incomplete combustion can produce.

Q13. Which option correctly describes this fact: Incomplete combustion can produce?

A. carbon monoxide B. only oxygen C. helium D. ozone

Answer: A. carbon monoxide. Explanation: Incomplete combustion can produce.

Q14. Select the best answer about this topic: Incomplete combustion can produce.

A. only oxygen B. helium C. ozone D. carbon monoxide

Answer: D. carbon monoxide. Explanation: Incomplete combustion can produce.

Q15. Which of the following is true? Incomplete combustion can produce.

A. helium B. ozone C. carbon monoxide D. only oxygen

Answer: C. carbon monoxide. Explanation: Incomplete combustion can produce.

Q16. Which statement is correct? Calorific value is usually expressed per unit.

A. pressure only B. mass or volume C. colour D. temperature

Answer: B. mass or volume. Explanation: Calorific value is usually expressed per unit.

Q17. Choose the scientifically correct option: Calorific value is usually expressed per unit.

A. mass or volume B. colour C. temperature D. pressure only

Answer: A. mass or volume. Explanation: Calorific value is usually expressed per unit.

Q18. Which option correctly describes this fact: Calorific value is usually expressed per unit?

A. colour B. temperature C. pressure only D. mass or volume

Answer: D. mass or volume. Explanation: Calorific value is usually expressed per unit.

Q19. Select the best answer about this topic: Calorific value is usually expressed per unit.

A. temperature B. pressure only C. mass or volume D. colour

Answer: C. mass or volume. Explanation: Calorific value is usually expressed per unit.

Q20. Which of the following is true? Calorific value is usually expressed per unit.

A. pressure only B. mass or volume C. colour D. temperature

Answer: B. mass or volume. Explanation: Calorific value is usually expressed per unit.

Q21. Which statement is correct? Ignition temperature is the minimum temperature for.

A. sustained ignition B. freezing C. condensation D. melting

Answer: A. sustained ignition. Explanation: Ignition temperature is the minimum temperature for.

Q22. Choose the scientifically correct option: Ignition temperature is the minimum temperature for.

A. freezing B. condensation C. melting D. sustained ignition

Answer: D. sustained ignition. Explanation: Ignition temperature is the minimum temperature for.

Q23. Which option correctly describes this fact: Ignition temperature is the minimum temperature for?

A. condensation B. melting C. sustained ignition D. freezing

Answer: C. sustained ignition. Explanation: Ignition temperature is the minimum temperature for.

Q24. Select the best answer about this topic: Ignition temperature is the minimum temperature for.

A. melting B. sustained ignition C. freezing D. condensation

Answer: B. sustained ignition. Explanation: Ignition temperature is the minimum temperature for.

Q25. Which of the following is true? Ignition temperature is the minimum temperature for.

A. sustained ignition B. freezing C. condensation D. melting

Answer: A. sustained ignition. Explanation: Ignition temperature is the minimum temperature for.

CHAPTER 20: Chemistry in Everyday Life

Chemistry — Chemistry explains materials and reactions used in food, cleaning, medicines, agriculture and household products.

Concept notes and exam points

1. Hard water contains dissolved calcium and magnesium ions. Temporary hardness is commonly associated with bicarbonates; permanent hardness with sulfates and chlorides.

2. Soap reacts with calcium and magnesium ions to form insoluble scum. Detergents are formulated to remain effective in hard water.

3. Bleaching agents are used for whitening and disinfection; they must be handled according to label directions and never mixed casually with acids or ammonia.

4. Antacids neutralise excess stomach acid. Examples include magnesium hydroxide and calcium carbonate; medicines should be used as directed.

5. Fertilisers supply essential plant nutrients, especially nitrogen, phosphorus and potassium (NPK). Excess runoff can cause eutrophication.

6. Food preservatives slow spoilage by microorganisms or oxidation. Safe storage, hygiene and correct use limits are important.

7. Polymers are large molecules made of repeating units. Examples include polyethylene, PVC, nylon and natural cellulose.

8. Pesticides control pests but can harm non-target organisms; integrated pest management and safe handling reduce risks.

Formulae, units and quick facts

• hardness ions: Ca^{2}■, Mg^{2}■

• NPK = nitrogen, phosphorus, potassium

Worked examples

Problem: Which ions are mainly responsible for hard water?

Solution: Calcium and magnesium. Sodium and potassium

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Hard water mainly contains.

A. Ca^{2}■ and Mg^{2}■ B. only Na■ C. only H■ D. only Cl■

Answer: A. Ca^{2}■ and Mg^{2}■. Explanation: Hard water mainly contains.

Q2. Choose the scientifically correct option: Hard water mainly contains.

A. only Na■ B. only H■ C. only Cl■ D. Ca^{2}■ and Mg^{2}■

Answer: D. Ca^{2}■ and Mg^{2}■. Explanation: Hard water mainly contains.

Q3. Which option correctly describes this fact: Hard water mainly contains?

A. only H■ B. only Cl■ C. Ca^{2}■ and Mg^{2}■ D. only Na■

Answer: C. Ca^{2}■ and Mg^{2}■. Explanation: Hard water mainly contains.

Q4. Select the best answer about this topic: Hard water mainly contains.

A. only Cl■ B. Ca^{2}■ and Mg^{2}■ C. only Na■ D. only H■

Answer: B. Ca^{2}■ and Mg^{2}■. Explanation: Hard water mainly contains.

Q5. Which of the following is true? Hard water mainly contains.

A. Ca^{2}■ and Mg^{2}■ B. only Na■ C. only H■ D. only Cl■

Answer: A. Ca^{2}■ and Mg^{2}■. Explanation: Hard water mainly contains.

Q6. Which statement is correct? The three primary fertiliser nutrients are.

A. C, H and O B. Na, Cl and Ag C. Fe, Cu and Zn only D. N, P and K

Answer: D. N, P and K. Explanation: The three primary fertiliser nutrients are.

Q7. Choose the scientifically correct option: The three primary fertiliser nutrients are.

A. Na, Cl and Ag B. Fe, Cu and Zn only C. N, P and K D. C, H and O

Answer: C. N, P and K. Explanation: The three primary fertiliser nutrients are.

Q8. Which option correctly describes this fact: The three primary fertiliser nutrients are?

A. Fe, Cu and Zn only B. N, P and K C. C, H and O D. Na, Cl and Ag

Answer: B. N, P and K. Explanation: The three primary fertiliser nutrients are.

Q9. Select the best answer about this topic: The three primary fertiliser nutrients are.

A. N, P and K B. C, H and O C. Na, Cl and Ag D. Fe, Cu and Zn only

Answer: A. N, P and K. Explanation: The three primary fertiliser nutrients are.

Q10. Which of the following is true? The three primary fertiliser nutrients are.

A. C, H and O B. Na, Cl and Ag C. Fe, Cu and Zn only D. N, P and K

Answer: D. N, P and K. Explanation: The three primary fertiliser nutrients are.

Q11. Which statement is correct? Antacids are used to.

A. digest cellulose B. sterilise drinking water C. neutralise excess stomach acid D. increase acidity

Answer: C. neutralise excess stomach acid. Explanation: Antacids are used to.

Q12. Choose the scientifically correct option: Antacids are used to.

A. sterilise drinking water B. neutralise excess stomach acid C. increase acidity D. digest cellulose

Answer: B. neutralise excess stomach acid. Explanation: Antacids are used to.

Q13. Which option correctly describes this fact: Antacids are used to?

A. neutralise excess stomach acid B. increase acidity C. digest cellulose D. sterilise drinking water

Answer: A. neutralise excess stomach acid. Explanation: Antacids are used to.

Q14. Select the best answer about this topic: Antacids are used to.

A. increase acidity B. digest cellulose C. sterilise drinking water D. neutralise excess stomach acid

Answer: D. neutralise excess stomach acid. Explanation: Antacids are used to.

Q15. Which of the following is true? Antacids are used to.

A. digest cellulose B. sterilise drinking water C. neutralise excess stomach acid D. increase acidity

Answer: C. neutralise excess stomach acid. Explanation: Antacids are used to.

Q16. Which statement is correct? A polymer is made of repeating.

A. water droplets B. monomer-derived units C. metal atoms only D. ions without bonds

Answer: B. monomer-derived units. Explanation: A polymer is made of repeating.

Q17. Choose the scientifically correct option: A polymer is made of repeating.

A. monomer-derived units B. metal atoms only C. ions without bonds D. water droplets

Answer: A. monomer-derived units. Explanation: A polymer is made of repeating.

Q18. Which option correctly describes this fact: A polymer is made of repeating?

A. metal atoms only B. ions without bonds C. water droplets D. monomer-derived units

Answer: D. monomer-derived units. Explanation: A polymer is made of repeating.

Q19. Select the best answer about this topic: A polymer is made of repeating.

A. ions without bonds B. water droplets C. monomer-derived units D. metal atoms only

Answer: C. monomer-derived units. Explanation: A polymer is made of repeating.

Q20. Which of the following is true? A polymer is made of repeating.

A. water droplets B. monomer-derived units C. metal atoms only D. ions without bonds

Answer: B. monomer-derived units. Explanation: A polymer is made of repeating.

Q21. Which statement is correct? Soap forms scum with.

A. calcium and magnesium ions B. helium C. oxygen gas D. glucose

Answer: A. calcium and magnesium ions. Explanation: Soap forms scum with.

Q22. Choose the scientifically correct option: Soap forms scum with.

A. helium B. oxygen gas C. glucose D. calcium and magnesium ions

Answer: D. calcium and magnesium ions. Explanation: Soap forms scum with.

Q23. Which option correctly describes this fact: Soap forms scum with?

A. oxygen gas B. glucose C. calcium and magnesium ions D. helium

Answer: C. calcium and magnesium ions. Explanation: Soap forms scum with.

Q24. Select the best answer about this topic: Soap forms scum with.

A. glucose B. calcium and magnesium ions C. helium D. oxygen gas

Answer: B. calcium and magnesium ions. Explanation: Soap forms scum with.

Q25. Which of the following is true? Soap forms scum with.

A. calcium and magnesium ions B. helium C. oxygen gas D. glucose

Answer: A. calcium and magnesium ions. Explanation: Soap forms scum with.

CHAPTER 21: Environmental Chemistry

Chemistry — Environmental chemistry studies chemical processes in air, water and soil and how pollutants affect ecosystems and health.

Concept notes and exam points

1. Air pollution includes particulate matter, sulfur dioxide, nitrogen oxides, carbon monoxide, ground-level ozone and volatile organic compounds.

2. Sulfur dioxide and nitrogen oxides can form acids in the atmosphere, contributing to acid rain that damages aquatic ecosystems, vegetation and buildings.

3. The enhanced greenhouse effect results from increased concentrations of greenhouse gases such as carbon dioxide, methane and nitrous oxide.

4. Ozone in the stratosphere absorbs harmful ultraviolet radiation; ground-level ozone is a pollutant that irritates the respiratory system.

5. Water pollution can arise from sewage, industrial effluents, fertilisers, pesticides, oil and plastics. Treatment and source control are essential.

6. Eutrophication occurs when excess nutrients stimulate algal growth; decomposition can reduce dissolved oxygen and harm aquatic life.

7. Biodegradable materials can be decomposed by microorganisms under suitable conditions; non-biodegradable waste can persist and accumulate.

8. The 3Rs are reduce, reuse and recycle. Waste segregation, safe disposal, cleaner production and conservation protect environmental quality.

Formulae, units and quick facts

• greenhouse gases: CO■, CH■, N■O, water vapour

• acid rain precursors: SO■ and NO■

Worked examples

Problem: Which gases are major contributors to acid rain?

Solution: SO■ and nitrogen oxides. oxygen and helium

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Stratospheric ozone protects life by absorbing.

A. ultraviolet radiation B. radio waves C. sound waves D. infrared only

Answer: A. ultraviolet radiation. Explanation: Stratospheric ozone protects life by absorbing.

Q2. Choose the scientifically correct option: Stratospheric ozone protects life by absorbing.

A. radio waves B. sound waves C. infrared only D. ultraviolet radiation

Answer: D. ultraviolet radiation. Explanation: Stratospheric ozone protects life by absorbing.

Q3. Which option correctly describes this fact: Stratospheric ozone protects life by absorbing?

A. sound waves B. infrared only C. ultraviolet radiation D. radio waves

Answer: C. ultraviolet radiation. Explanation: Stratospheric ozone protects life by absorbing.

Q4. Select the best answer about this topic: Stratospheric ozone protects life by absorbing.

A. infrared only B. ultraviolet radiation C. radio waves D. sound waves

Answer: B. ultraviolet radiation. Explanation: Stratospheric ozone protects life by absorbing.

Q5. Which of the following is true? Stratospheric ozone protects life by absorbing.

A. ultraviolet radiation B. radio waves C. sound waves D. infrared only

Answer: A. ultraviolet radiation. Explanation: Stratospheric ozone protects life by absorbing.

Q6. Which statement is correct? Acid rain is associated with emissions of.

A. oxygen and nitrogen only B. helium C. hydrogen D. SO■ and NO■

Answer: D. SO■ and NO■. Explanation: Acid rain is associated with emissions of.

Q7. Choose the scientifically correct option: Acid rain is associated with emissions of.

A. helium B. hydrogen C. SO■ and NO■ D. oxygen and nitrogen only

Answer: C. SO■ and NO■. Explanation: Acid rain is associated with emissions of.

Q8. Which option correctly describes this fact: Acid rain is associated with emissions of?

A. hydrogen B. SO■ and NO■ C. oxygen and nitrogen only D. helium

Answer: B. SO■ and NO■. Explanation: Acid rain is associated with emissions of.

Q9. Select the best answer about this topic: Acid rain is associated with emissions of.

A. SO■ and NO■ B. oxygen and nitrogen only C. helium D. hydrogen

Answer: A. SO■ and NO■. Explanation: Acid rain is associated with emissions of.

Q10. Which of the following is true? Acid rain is associated with emissions of.

A. oxygen and nitrogen only B. helium C. hydrogen D. SO■ and NO■

Answer: D. SO■ and NO■. Explanation: Acid rain is associated with emissions of.

Q11. Which statement is correct? Eutrophication is commonly caused by excess.

A. salt crystals only B. oxygen C. nutrients D. sand only

Answer: C. nutrients. Explanation: Eutrophication is commonly caused by excess.

Q12. Choose the scientifically correct option: Eutrophication is commonly caused by excess.

A. oxygen B. nutrients C. sand only D. salt crystals only

Answer: B. nutrients. Explanation: Eutrophication is commonly caused by excess.

Q13. Which option correctly describes this fact: Eutrophication is commonly caused by excess?

A. nutrients B. sand only C. salt crystals only D. oxygen

Answer: A. nutrients. Explanation: Eutrophication is commonly caused by excess.

Q14. Select the best answer about this topic: Eutrophication is commonly caused by excess.

A. sand only B. salt crystals only C. oxygen D. nutrients

Answer: D. nutrients. Explanation: Eutrophication is commonly caused by excess.

Q15. Which of the following is true? Eutrophication is commonly caused by excess.

A. salt crystals only B. oxygen C. nutrients D. sand only

Answer: C. nutrients. Explanation: Eutrophication is commonly caused by excess.

Q16. Which statement is correct? A major greenhouse gas is.

A. helium B. carbon dioxide C. argon D. neon

Answer: B. carbon dioxide. Explanation: A major greenhouse gas is.

Q17. Choose the scientifically correct option: A major greenhouse gas is.

A. carbon dioxide B. argon C. neon D. helium

Answer: A. carbon dioxide. Explanation: A major greenhouse gas is.

Q18. Which option correctly describes this fact: A major greenhouse gas is?

A. argon B. neon C. helium D. carbon dioxide

Answer: D. carbon dioxide. Explanation: A major greenhouse gas is.

Q19. Select the best answer about this topic: A major greenhouse gas is.

A. neon B. helium C. carbon dioxide D. argon

Answer: C. carbon dioxide. Explanation: A major greenhouse gas is.

Q20. Which of the following is true? A major greenhouse gas is.

A. helium B. carbon dioxide C. argon D. neon

Answer: B. carbon dioxide. Explanation: A major greenhouse gas is.

Q21. Which statement is correct? The 3Rs are reduce, reuse and.

A. recycle B. refuel C. react D. radiate

Answer: A. recycle. Explanation: The 3Rs are reduce, reuse and.

Q22. Choose the scientifically correct option: The 3Rs are reduce, reuse and.

A. refuel B. react C. radiate D. recycle

Answer: D. recycle. Explanation: The 3Rs are reduce, reuse and.

Q23. Which option correctly describes this fact: The 3Rs are reduce, reuse and?

A. react B. radiate C. recycle D. refuel

Answer: C. recycle. Explanation: The 3Rs are reduce, reuse and.

Q24. Select the best answer about this topic: The 3Rs are reduce, reuse and.

A. radiate B. recycle C. refuel D. react

Answer: B. recycle. Explanation: The 3Rs are reduce, reuse and.

Q25. Which of the following is true? The 3Rs are reduce, reuse and.

A. recycle B. refuel C. react D. radiate

Answer: A. recycle. Explanation: The 3Rs are reduce, reuse and.

CHAPTER 22: Cell Biology

Biology — The cell is the basic structural and functional unit of life; organisms may be unicellular or multicellular.

Concept notes and exam points

1. Cell theory states that living organisms are made of cells, the cell is the basic unit of life, and cells arise from pre-existing cells.

2. Prokaryotic cells lack a membrane-bound nucleus and membrane-bound organelles; bacteria are prokaryotes. Eukaryotic cells have a true nucleus.

3. The plasma membrane is selectively permeable and regulates movement of substances. Diffusion and osmosis are important transport processes.

4. The nucleus contains DNA and controls many cell activities. Chromosomes carry genes, which are units of heredity.

5. Mitochondria are major sites of aerobic respiration and ATP production. Ribosomes synthesise proteins.

6. Chloroplasts contain chlorophyll and carry out photosynthesis in plants and algae. The cell wall provides support in plants.

7. The endoplasmic reticulum helps synthesise and transport proteins or lipids; the Golgi apparatus modifies, sorts and packages materials.

8. Plant cells commonly have a cellulose cell wall, chloroplasts in photosynthetic tissues and a large central vacuole; animal cells lack a cell wall.

Formulae, units and quick facts

• photosynthesis: 6CO■ + 6H■O → C■H■■O■ + 6O■ (light, chlorophyll)

• ATP = cellular energy carrier

Worked examples

Problem: Which organelle is the major site of aerobic respiration?

Solution: Mitochondrion. Ribosome

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The basic unit of life is the.

A. cell B. organ C. tissue D. organ system

Answer: A. cell. Explanation: The basic unit of life is the.

Q2. Choose the scientifically correct option: The basic unit of life is the.

A. organ B. tissue C. organ system D. cell

Answer: D. cell. Explanation: The basic unit of life is the.

Q3. Which option correctly describes this fact: The basic unit of life is the?

A. tissue B. organ system C. cell D. organ

Answer: C. cell. Explanation: The basic unit of life is the.

Q4. Select the best answer about this topic: The basic unit of life is the.

A. organ system B. cell C. organ D. tissue

Answer: B. cell. Explanation: The basic unit of life is the.

Q5. Which of the following is true? The basic unit of life is the.

A. cell B. organ C. tissue D. organ system

Answer: A. cell. Explanation: The basic unit of life is the.

Q6. Which statement is correct? Prokaryotes lack a membrane-bound.

A. plasma membrane B. cytoplasm C. ribosome D. nucleus

Answer: D. nucleus. Explanation: Prokaryotes lack a membrane-bound.

Q7. Choose the scientifically correct option: Prokaryotes lack a membrane-bound.

A. cytoplasm B. ribosome C. nucleus D. plasma membrane

Answer: C. nucleus. Explanation: Prokaryotes lack a membrane-bound.

Q8. Which option correctly describes this fact: Prokaryotes lack a membrane-bound?

A. ribosome B. nucleus C. plasma membrane D. cytoplasm

Answer: B. nucleus. Explanation: Prokaryotes lack a membrane-bound.

Q9. Select the best answer about this topic: Prokaryotes lack a membrane-bound.

A. nucleus B. plasma membrane C. cytoplasm D. ribosome

Answer: A. nucleus. Explanation: Prokaryotes lack a membrane-bound.

Q10. Which of the following is true? Prokaryotes lack a membrane-bound.

A. plasma membrane B. cytoplasm C. ribosome D. nucleus

Answer: D. nucleus. Explanation: Prokaryotes lack a membrane-bound.

Q11. Which statement is correct? Protein synthesis occurs at.

A. vacuoles B. chloroplast grana only C. ribosomes D. lysosomes

Answer: C. ribosomes. Explanation: Protein synthesis occurs at.

Q12. Choose the scientifically correct option: Protein synthesis occurs at.

A. chloroplast grana only B. ribosomes C. lysosomes D. vacuoles

Answer: B. ribosomes. Explanation: Protein synthesis occurs at.

Q13. Which option correctly describes this fact: Protein synthesis occurs at?

A. ribosomes B. lysosomes C. vacuoles D. chloroplast grana only

Answer: A. ribosomes. Explanation: Protein synthesis occurs at.

Q14. Select the best answer about this topic: Protein synthesis occurs at.

A. lysosomes B. vacuoles C. chloroplast grana only D. ribosomes

Answer: D. ribosomes. Explanation: Protein synthesis occurs at.

Q15. Which of the following is true? Protein synthesis occurs at.

A. vacuoles B. chloroplast grana only C. ribosomes D. lysosomes

Answer: C. ribosomes. Explanation: Protein synthesis occurs at.

Q16. Which statement is correct? Photosynthesis takes place in.

A. Golgi apparatus B. chloroplasts C. mitochondria D. nucleus

Answer: B. chloroplasts. Explanation: Photosynthesis takes place in.

Q17. Choose the scientifically correct option: Photosynthesis takes place in.

A. chloroplasts B. mitochondria C. nucleus D. Golgi apparatus

Answer: A. chloroplasts. Explanation: Photosynthesis takes place in.

Q18. Which option correctly describes this fact: Photosynthesis takes place in?

A. mitochondria B. nucleus C. Golgi apparatus D. chloroplasts

Answer: D. chloroplasts. Explanation: Photosynthesis takes place in.

Q19. Select the best answer about this topic: Photosynthesis takes place in.

A. nucleus B. Golgi apparatus C. chloroplasts D. mitochondria

Answer: C. chloroplasts. Explanation: Photosynthesis takes place in.

Q20. Which of the following is true? Photosynthesis takes place in.

A. Golgi apparatus B. chloroplasts C. mitochondria D. nucleus

Answer: B. chloroplasts. Explanation: Photosynthesis takes place in.

Q21. Which statement is correct? The plasma membrane is described as.

A. selectively permeable B. completely impermeable C. made of cellulose only D. absent in animal cells

Answer: A. selectively permeable. Explanation: The plasma membrane is described as.

Q22. Choose the scientifically correct option: The plasma membrane is described as.

A. completely impermeable B. made of cellulose only C. absent in animal cells D. selectively permeable

Answer: D. selectively permeable. Explanation: The plasma membrane is described as.

Q23. Which option correctly describes this fact: The plasma membrane is described as?

A. made of cellulose only B. absent in animal cells C. selectively permeable D. completely impermeable

Answer: C. selectively permeable. Explanation: The plasma membrane is described as.

Q24. Select the best answer about this topic: The plasma membrane is described as.

A. absent in animal cells B. selectively permeable C. completely impermeable D. made of cellulose only

Answer: B. selectively permeable. Explanation: The plasma membrane is described as.

Q25. Which of the following is true? The plasma membrane is described as.

A. selectively permeable B. completely impermeable C. made of cellulose only D. absent in animal cells

Answer: A. selectively permeable. Explanation: The plasma membrane is described as.

CHAPTER 23: Human Body and Tissues

Biology — A tissue is a group of similar cells working together to perform a particular function.

Concept notes and exam points

1. The four basic animal tissue types are epithelial, connective, muscular and nervous tissue.

2. Epithelial tissue covers body surfaces and lines cavities; it protects, absorbs and secretes. It may be simple or stratified.

3. Connective tissue supports and connects structures. Examples include bone, cartilage, blood, adipose tissue, tendons and ligaments.

4. Muscle tissue contracts to produce movement. Skeletal muscle is usually voluntary and striated; smooth muscle is involuntary; cardiac muscle forms the heart.

5. Nervous tissue contains neurons that transmit impulses and supporting glial cells. Neurons have a cell body, dendrites and an axon.

6. Organs contain different tissues working together. Organ systems coordinate major functions such as digestion, respiration and circulation.

7. Bone provides support, protection and mineral storage; cartilage provides flexible support and reduces friction at joints.

8. Tendons connect muscle to bone; ligaments connect bone to bone. These are common exam distinctions.

Formulae, units and quick facts

• tendon: muscle-to-bone

• ligament: bone-to-bone

Worked examples

Problem: Which tissue connects muscle to bone?

Solution: Tendon. Ligament

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The four basic animal tissue types include.

A. epithelial, connective, muscular and nervous B. xylem, phloem, epidermis and cork C. bone, leaf, root and stem D. blood, air, water and soil

Answer: A. epithelial, connective, muscular and nervous. Explanation: The four basic animal tissue types include.

Q2. Choose the scientifically correct option: The four basic animal tissue types include.

A. xylem, phloem, epidermis and cork B. bone, leaf, root and stem C. blood, air, water and soil D. epithelial, connective, muscular and nervous

Answer: D. epithelial, connective, muscular and nervous. Explanation: The four basic animal tissue types include.

Q3. Which option correctly describes this fact: The four basic animal tissue types include?

A. bone, leaf, root and stem B. blood, air, water and soil C. epithelial, connective, muscular and nervous D. xylem, phloem, epidermis and cork

Answer: C. epithelial, connective, muscular and nervous. Explanation: The four basic animal tissue types include.

Q4. Select the best answer about this topic: The four basic animal tissue types include.

A. blood, air, water and soil B. epithelial, connective, muscular and nervous C. xylem, phloem, epidermis and cork D. bone, leaf, root and stem

Answer: B. epithelial, connective, muscular and nervous. Explanation: The four basic animal tissue types include.

Q5. Which of the following is true? The four basic animal tissue types include.

A. epithelial, connective, muscular and nervous B. xylem, phloem, epidermis and cork C. bone, leaf, root and stem D. blood, air, water and soil

Answer: A. epithelial, connective, muscular and nervous. Explanation: The four basic animal tissue types include.

Q6. Which statement is correct? The heart is mainly made of.

A. smooth muscle only B. skeletal muscle C. epithelial tissue D. cardiac muscle

Answer: D. cardiac muscle. Explanation: The heart is mainly made of.

Q7. Choose the scientifically correct option: The heart is mainly made of.

A. skeletal muscle B. epithelial tissue C. cardiac muscle D. smooth muscle only

Answer: C. cardiac muscle. Explanation: The heart is mainly made of.

Q8. Which option correctly describes this fact: The heart is mainly made of?

A. epithelial tissue B. cardiac muscle C. smooth muscle only D. skeletal muscle

Answer: B. cardiac muscle. Explanation: The heart is mainly made of.

Q9. Select the best answer about this topic: The heart is mainly made of.

A. cardiac muscle B. smooth muscle only C. skeletal muscle D. epithelial tissue

Answer: A. cardiac muscle. Explanation: The heart is mainly made of.

Q10. Which of the following is true? The heart is mainly made of.

A. smooth muscle only B. skeletal muscle C. epithelial tissue D. cardiac muscle

Answer: D. cardiac muscle. Explanation: The heart is mainly made of.

Q11. Which statement is correct? Ligaments connect.

A. nerve to muscle B. skin to hair C. bone to bone D. muscle to bone

Answer: C. bone to bone. Explanation: Ligaments connect.

Q12. Choose the scientifically correct option: Ligaments connect.

A. skin to hair B. bone to bone C. muscle to bone D. nerve to muscle

Answer: B. bone to bone. Explanation: Ligaments connect.

Q13. Which option correctly describes this fact: Ligaments connect?

A. bone to bone B. muscle to bone C. nerve to muscle D. skin to hair

Answer: A. bone to bone. Explanation: Ligaments connect.

Q14. Select the best answer about this topic: Ligaments connect.

A. muscle to bone B. nerve to muscle C. skin to hair D. bone to bone

Answer: D. bone to bone. Explanation: Ligaments connect.

Q15. Which of the following is true? Ligaments connect.

A. nerve to muscle B. skin to hair C. bone to bone D. muscle to bone

Answer: C. bone to bone. Explanation: Ligaments connect.

Q16. Which statement is correct? Blood is classified as.

A. skeletal muscle B. connective tissue C. epithelial tissue D. nervous tissue

Answer: B. connective tissue. Explanation: Blood is classified as.

Q17. Choose the scientifically correct option: Blood is classified as.

A. connective tissue B. epithelial tissue C. nervous tissue D. skeletal muscle

Answer: A. connective tissue. Explanation: Blood is classified as.

Q18. Which option correctly describes this fact: Blood is classified as?

A. epithelial tissue B. nervous tissue C. skeletal muscle D. connective tissue

Answer: D. connective tissue. Explanation: Blood is classified as.

Q19. Select the best answer about this topic: Blood is classified as.

A. nervous tissue B. skeletal muscle C. connective tissue D. epithelial tissue

Answer: C. connective tissue. Explanation: Blood is classified as.

Q20. Which of the following is true? Blood is classified as.

A. skeletal muscle B. connective tissue C. epithelial tissue D. nervous tissue

Answer: B. connective tissue. Explanation: Blood is classified as.

Q21. Which statement is correct? The functional cell of nervous tissue is the.

A. neuron B. nephron C. alveolus D. platelet

Answer: A. neuron. Explanation: The functional cell of nervous tissue is the.

Q22. Choose the scientifically correct option: The functional cell of nervous tissue is the.

A. nephron B. alveolus C. platelet D. neuron

Answer: D. neuron. Explanation: The functional cell of nervous tissue is the.

Q23. Which option correctly describes this fact: The functional cell of nervous tissue is the?

A. alveolus B. platelet C. neuron D. nephron

Answer: C. neuron. Explanation: The functional cell of nervous tissue is the.

Q24. Select the best answer about this topic: The functional cell of nervous tissue is the.

A. platelet B. neuron C. nephron D. alveolus

Answer: B. neuron. Explanation: The functional cell of nervous tissue is the.

Q25. Which of the following is true? The functional cell of nervous tissue is the.

A. neuron B. nephron C. alveolus D. platelet

Answer: A. neuron. Explanation: The functional cell of nervous tissue is the.

CHAPTER 24: Digestive System

Biology — Digestion breaks large food molecules into smaller soluble molecules that can be absorbed and used by cells.

Concept notes and exam points

1. The alimentary canal includes mouth, oesophagus, stomach, small intestine, large intestine, rectum and anus.

2. Mechanical digestion begins with chewing. Salivary amylase starts starch digestion in the mouth.

3. The stomach churns food and secretes hydrochloric acid and enzymes; acid helps provide a suitable environment and kills many microbes.

4. Bile is produced by the liver and stored in the gallbladder. It emulsifies fats but is not itself a digestive enzyme.

5. The pancreas releases digestive enzymes and bicarbonate into the small intestine. The small intestine is the main site of digestion and nutrient absorption.

6. Villi and microvilli increase the surface area of the small intestine for absorption. Blood capillaries absorb many nutrients; lacteals absorb much dietary fat.

7. The large intestine absorbs water and salts and houses gut microbes; faeces are stored in the rectum before elimination.

8. A balanced diet contains carbohydrates, proteins, fats, vitamins, minerals, fibre and water in appropriate amounts.

Formulae, units and quick facts

• starch digestion begins: salivary amylase

• protein digestion begins mainly in stomach: pepsin

• bile: fat emulsification

Worked examples

Problem: Which organ produces bile?

Solution: Liver. Pancreas

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Starch digestion begins in the.

A. mouth B. large intestine C. rectum D. oesophagus

Answer: A. mouth. Explanation: Starch digestion begins in the.

Q2. Choose the scientifically correct option: Starch digestion begins in the.

A. large intestine B. rectum C. oesophagus D. mouth

Answer: D. mouth. Explanation: Starch digestion begins in the.

Q3. Which option correctly describes this fact: Starch digestion begins in the?

A. rectum B. oesophagus C. mouth D. large intestine

Answer: C. mouth. Explanation: Starch digestion begins in the.

Q4. Select the best answer about this topic: Starch digestion begins in the.

A. oesophagus B. mouth C. large intestine D. rectum

Answer: B. mouth. Explanation: Starch digestion begins in the.

Q5. Which of the following is true? Starch digestion begins in the.

A. mouth B. large intestine C. rectum D. oesophagus

Answer: A. mouth. Explanation: Starch digestion begins in the.

Q6. Which statement is correct? Bile is stored in the.

A. pancreas B. kidney C. spleen D. gallbladder

Answer: D. gallbladder. Explanation: Bile is stored in the.

Q7. Choose the scientifically correct option: Bile is stored in the.

A. kidney B. spleen C. gallbladder D. pancreas

Answer: C. gallbladder. Explanation: Bile is stored in the.

Q8. Which option correctly describes this fact: Bile is stored in the?

A. spleen B. gallbladder C. pancreas D. kidney

Answer: B. gallbladder. Explanation: Bile is stored in the.

Q9. Select the best answer about this topic: Bile is stored in the.

A. gallbladder B. pancreas C. kidney D. spleen

Answer: A. gallbladder. Explanation: Bile is stored in the.

Q10. Which of the following is true? Bile is stored in the.

A. pancreas B. kidney C. spleen D. gallbladder

Answer: D. gallbladder. Explanation: Bile is stored in the.

Q11. Which statement is correct? Most nutrient absorption occurs in the.

A. large intestine B. oesophagus C. small intestine D. stomach

Answer: C. small intestine. Explanation: Most nutrient absorption occurs in the.

Q12. Choose the scientifically correct option: Most nutrient absorption occurs in the.

A. oesophagus B. small intestine C. stomach D. large intestine

Answer: B. small intestine. Explanation: Most nutrient absorption occurs in the.

Q13. Which option correctly describes this fact: Most nutrient absorption occurs in the?

A. small intestine B. stomach C. large intestine D. oesophagus

Answer: A. small intestine. Explanation: Most nutrient absorption occurs in the.

Q14. Select the best answer about this topic: Most nutrient absorption occurs in the.

A. stomach B. large intestine C. oesophagus D. small intestine

Answer: D. small intestine. Explanation: Most nutrient absorption occurs in the.

Q15. Which of the following is true? Most nutrient absorption occurs in the.

A. large intestine B. oesophagus C. small intestine D. stomach

Answer: C. small intestine. Explanation: Most nutrient absorption occurs in the.

Q16. Which statement is correct? Villi mainly increase.

A. tooth strength B. absorptive surface area C. stomach acidity D. bile production

Answer: B. absorptive surface area. Explanation: Villi mainly increase.

Q17. Choose the scientifically correct option: Villi mainly increase.

A. absorptive surface area B. stomach acidity C. bile production D. tooth strength

Answer: A. absorptive surface area. Explanation: Villi mainly increase.

Q18. Which option correctly describes this fact: Villi mainly increase?

A. stomach acidity B. bile production C. tooth strength D. absorptive surface area

Answer: D. absorptive surface area. Explanation: Villi mainly increase.

Q19. Select the best answer about this topic: Villi mainly increase.

A. bile production B. tooth strength C. absorptive surface area D. stomach acidity

Answer: C. absorptive surface area. Explanation: Villi mainly increase.

Q20. Which of the following is true? Villi mainly increase.

A. tooth strength B. absorptive surface area C. stomach acidity D. bile production

Answer: B. absorptive surface area. Explanation: Villi mainly increase.

Q21. Which statement is correct? The enzyme in saliva that begins starch digestion is.

A. amylase B. pepsin C. trypsin D. lipase

Answer: A. amylase. Explanation: The enzyme in saliva that begins starch digestion is.

Q22. Choose the scientifically correct option: The enzyme in saliva that begins starch digestion is.

A. pepsin B. trypsin C. lipase D. amylase

Answer: D. amylase. Explanation: The enzyme in saliva that begins starch digestion is.

Q23. Which option correctly describes this fact: The enzyme in saliva that begins starch digestion is?

A. trypsin B. lipase C. amylase D. pepsin

Answer: C. amylase. Explanation: The enzyme in saliva that begins starch digestion is.

Q24. Select the best answer about this topic: The enzyme in saliva that begins starch digestion is.

A. lipase B. amylase C. pepsin D. trypsin

Answer: B. amylase. Explanation: The enzyme in saliva that begins starch digestion is.

Q25. Which of the following is true? The enzyme in saliva that begins starch digestion is.

A. amylase B. pepsin C. trypsin D. lipase

Answer: A. amylase. Explanation: The enzyme in saliva that begins starch digestion is.

CHAPTER 25: Respiratory System

Biology — Respiration is the process by which cells release usable energy from food; breathing moves air into and out of the lungs.

Concept notes and exam points

1. The human respiratory pathway includes nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles and alveoli.

2. Alveoli are tiny air sacs with thin moist walls and a rich capillary supply, providing a large surface area for gas exchange.

3. Oxygen diffuses from alveolar air into blood; carbon dioxide diffuses from blood into alveoli down partial-pressure gradients.

4. The diaphragm contracts and flattens during inhalation, increasing thoracic volume and drawing air into the lungs.

5. During exhalation at rest, the diaphragm relaxes and thoracic volume decreases, helping air move out.

6. Most oxygen is transported bound to haemoglobin in red blood cells. Much carbon dioxide is transported as bicarbonate ions in plasma.

7. Aerobic cellular respiration uses oxygen to release energy from glucose, producing carbon dioxide and water; ATP captures usable energy.

8. Smoking damages airways and alveoli and increases risk of chronic obstructive pulmonary disease, lung cancer and cardiovascular disease.

Formulae, units and quick facts

• aerobic respiration: glucose + oxygen → carbon dioxide + water + energy

• alveoli: gas exchange

Worked examples

Problem: Where does gas exchange occur in human lungs?

Solution: Alveoli. Trachea

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The main site of gas exchange is the.

A. alveolus B. bronchus C. nasal cavity D. larynx

Answer: A. alveolus. Explanation: The main site of gas exchange is the.

Q2. Choose the scientifically correct option: The main site of gas exchange is the.

A. bronchus B. nasal cavity C. larynx D. alveolus

Answer: D. alveolus. Explanation: The main site of gas exchange is the.

Q3. Which option correctly describes this fact: The main site of gas exchange is the?

A. nasal cavity B. larynx C. alveolus D. bronchus

Answer: C. alveolus. Explanation: The main site of gas exchange is the.

Q4. Select the best answer about this topic: The main site of gas exchange is the.

A. larynx B. alveolus C. bronchus D. nasal cavity

Answer: B. alveolus. Explanation: The main site of gas exchange is the.

Q5. Which of the following is true? The main site of gas exchange is the.

A. alveolus B. bronchus C. nasal cavity D. larynx

Answer: A. alveolus. Explanation: The main site of gas exchange is the.

Q6. Which statement is correct? During inhalation the diaphragm.

A. relaxes and domes upward B. stops moving C. closes the trachea D. contracts and flattens

Answer: D. contracts and flattens. Explanation: During inhalation the diaphragm.

Q7. Choose the scientifically correct option: During inhalation the diaphragm.

A. stops moving B. closes the trachea C. contracts and flattens D. relaxes and domes upward

Answer: C. contracts and flattens. Explanation: During inhalation the diaphragm.

Q8. Which option correctly describes this fact: During inhalation the diaphragm?

A. closes the trachea B. contracts and flattens C. relaxes and domes upward D. stops moving

Answer: B. contracts and flattens. Explanation: During inhalation the diaphragm.

Q9. Select the best answer about this topic: During inhalation the diaphragm.

A. contracts and flattens B. relaxes and domes upward C. stops moving D. closes the trachea

Answer: A. contracts and flattens. Explanation: During inhalation the diaphragm.

Q10. Which of the following is true? During inhalation the diaphragm.

A. relaxes and domes upward B. stops moving C. closes the trachea D. contracts and flattens

Answer: D. contracts and flattens. Explanation: During inhalation the diaphragm.

Q11. Which statement is correct? Oxygen is mainly carried by.

A. plasma proteins only B. bile C. haemoglobin D. platelets

Answer: C. haemoglobin. Explanation: Oxygen is mainly carried by.

Q12. Choose the scientifically correct option: Oxygen is mainly carried by.

A. bile B. haemoglobin C. platelets D. plasma proteins only

Answer: B. haemoglobin. Explanation: Oxygen is mainly carried by.

Q13. Which option correctly describes this fact: Oxygen is mainly carried by?

A. haemoglobin B. platelets C. plasma proteins only D. bile

Answer: A. haemoglobin. Explanation: Oxygen is mainly carried by.

Q14. Select the best answer about this topic: Oxygen is mainly carried by.

A. platelets B. plasma proteins only C. bile D. haemoglobin

Answer: D. haemoglobin. Explanation: Oxygen is mainly carried by.

Q15. Which of the following is true? Oxygen is mainly carried by.

A. plasma proteins only B. bile C. haemoglobin D. platelets

Answer: C. haemoglobin. Explanation: Oxygen is mainly carried by.

Q16. Which statement is correct? The windpipe is the.

A. pharynx B. trachea C. oesophagus D. bronchiole

Answer: B. trachea. Explanation: The windpipe is the.

Q17. Choose the scientifically correct option: The windpipe is the.

A. trachea B. oesophagus C. bronchiole D. pharynx

Answer: A. trachea. Explanation: The windpipe is the.

Q18. Which option correctly describes this fact: The windpipe is the?

A. oesophagus B. bronchiole C. pharynx D. trachea

Answer: D. trachea. Explanation: The windpipe is the.

Q19. Select the best answer about this topic: The windpipe is the.

A. bronchiole B. pharynx C. trachea D. oesophagus

Answer: C. trachea. Explanation: The windpipe is the.

Q20. Which of the following is true? The windpipe is the.

A. pharynx B. trachea C. oesophagus D. bronchiole

Answer: B. trachea. Explanation: The windpipe is the.

Q21. Which statement is correct? Aerobic respiration requires.

A. oxygen B. nitrogen gas as reactant C. sunlight directly D. chlorophyll

Answer: A. oxygen. Explanation: Aerobic respiration requires.

Q22. Choose the scientifically correct option: Aerobic respiration requires.

A. nitrogen gas as reactant B. sunlight directly C. chlorophyll D. oxygen

Answer: D. oxygen. Explanation: Aerobic respiration requires.

Q23. Which option correctly describes this fact: Aerobic respiration requires?

A. sunlight directly B. chlorophyll C. oxygen D. nitrogen gas as reactant

Answer: C. oxygen. Explanation: Aerobic respiration requires.

Q24. Select the best answer about this topic: Aerobic respiration requires.

A. chlorophyll B. oxygen C. nitrogen gas as reactant D. sunlight directly

Answer: B. oxygen. Explanation: Aerobic respiration requires.

Q25. Which of the following is true? Aerobic respiration requires.

A. oxygen B. nitrogen gas as reactant C. sunlight directly D. chlorophyll

Answer: A. oxygen. Explanation: Aerobic respiration requires.

CHAPTER 26: Circulatory System and Blood

Biology — The circulatory system transports gases, nutrients, hormones and wastes around the body.

Concept notes and exam points

1. The human heart has four chambers: right and left atria and right and left ventricles. Valves help prevent backflow.

2. The right side pumps deoxygenated blood to the lungs through the pulmonary artery; the left side pumps oxygenated blood to the body through the aorta.

3. Pulmonary circulation is between heart and lungs; systemic circulation is between heart and the rest of the body.

4. Arteries carry blood away from the heart and generally have thick elastic walls. Veins return blood to the heart and often contain valves.

5. Capillaries have very thin walls and allow exchange of gases, nutrients and wastes with tissues.

6. Red blood cells contain haemoglobin and transport oxygen. White blood cells defend against pathogens. Platelets help blood clot.

7. Plasma is the liquid component of blood and transports nutrients, hormones, proteins, salts and waste products.

8. The pulse is the pressure wave produced by each heartbeat in arteries. Blood pressure is commonly recorded as systolic over diastolic pressure.

Formulae, units and quick facts

• heart → lungs: pulmonary circulation

• heart → body: systemic circulation

Worked examples

Problem: Which blood cells transport most oxygen?

Solution: Red blood cells. White blood cells

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The human heart has.

A. four chambers B. two chambers C. three chambers D. five chambers

Answer: A. four chambers. Explanation: The human heart has.

Q2. Choose the scientifically correct option: The human heart has.

A. two chambers B. three chambers C. five chambers D. four chambers

Answer: D. four chambers. Explanation: The human heart has.

Q3. Which option correctly describes this fact: The human heart has?

A. three chambers B. five chambers C. four chambers D. two chambers

Answer: C. four chambers. Explanation: The human heart has.

Q4. Select the best answer about this topic: The human heart has.

A. five chambers B. four chambers C. two chambers D. three chambers

Answer: B. four chambers. Explanation: The human heart has.

Q5. Which of the following is true? The human heart has.

A. four chambers B. two chambers C. three chambers D. five chambers

Answer: A. four chambers. Explanation: The human heart has.

Q6. Which statement is correct? The pulmonary artery carries blood from heart to.

A. body tissues B. kidneys C. liver D. lungs

Answer: D. lungs. Explanation: The pulmonary artery carries blood from heart to.

Q7. Choose the scientifically correct option: The pulmonary artery carries blood from heart to.

A. kidneys B. liver C. lungs D. body tissues

Answer: C. lungs. Explanation: The pulmonary artery carries blood from heart to.

Q8. Which option correctly describes this fact: The pulmonary artery carries blood from heart to?

A. liver B. lungs C. body tissues D. kidneys

Answer: B. lungs. Explanation: The pulmonary artery carries blood from heart to.

Q9. Select the best answer about this topic: The pulmonary artery carries blood from heart to.

A. lungs B. body tissues C. kidneys D. liver

Answer: A. lungs. Explanation: The pulmonary artery carries blood from heart to.

Q10. Which of the following is true? The pulmonary artery carries blood from heart to.

A. body tissues B. kidneys C. liver D. lungs

Answer: D. lungs. Explanation: The pulmonary artery carries blood from heart to.

Q11. Which statement is correct? Blood clotting mainly involves.

A. neurons B. alveoli C. platelets D. red cells only

Answer: C. platelets. Explanation: Blood clotting mainly involves.

Q12. Choose the scientifically correct option: Blood clotting mainly involves.

A. alveoli B. platelets C. red cells only D. neurons

Answer: B. platelets. Explanation: Blood clotting mainly involves.

Q13. Which option correctly describes this fact: Blood clotting mainly involves?

A. platelets B. red cells only C. neurons D. alveoli

Answer: A. platelets. Explanation: Blood clotting mainly involves.

Q14. Select the best answer about this topic: Blood clotting mainly involves.

A. red cells only B. neurons C. alveoli D. platelets

Answer: D. platelets. Explanation: Blood clotting mainly involves.

Q15. Which of the following is true? Blood clotting mainly involves.

A. neurons B. alveoli C. platelets D. red cells only

Answer: C. platelets. Explanation: Blood clotting mainly involves.

Q16. Which statement is correct? Arteries carry blood.

A. only to lungs B. away from the heart C. towards the heart D. only without oxygen

Answer: B. away from the heart. Explanation: Arteries carry blood.

Q17. Choose the scientifically correct option: Arteries carry blood.

A. away from the heart B. towards the heart C. only without oxygen D. only to lungs

Answer: A. away from the heart. Explanation: Arteries carry blood.

Q18. Which option correctly describes this fact: Arteries carry blood?

A. towards the heart B. only without oxygen C. only to lungs D. away from the heart

Answer: D. away from the heart. Explanation: Arteries carry blood.

Q19. Select the best answer about this topic: Arteries carry blood.

A. only without oxygen B. only to lungs C. away from the heart D. towards the heart

Answer: C. away from the heart. Explanation: Arteries carry blood.

Q20. Which of the following is true? Arteries carry blood.

A. only to lungs B. away from the heart C. towards the heart D. only without oxygen

Answer: B. away from the heart. Explanation: Arteries carry blood.

Q21. Which statement is correct? The oxygen-carrying pigment is.

A. haemoglobin B. chlorophyll C. melanin D. insulin

Answer: A. haemoglobin. Explanation: The oxygen-carrying pigment is.

Q22. Choose the scientifically correct option: The oxygen-carrying pigment is.

A. chlorophyll B. melanin C. insulin D. haemoglobin

Answer: D. haemoglobin. Explanation: The oxygen-carrying pigment is.

Q23. Which option correctly describes this fact: The oxygen-carrying pigment is?

A. melanin B. insulin C. haemoglobin D. chlorophyll

Answer: C. haemoglobin. Explanation: The oxygen-carrying pigment is.

Q24. Select the best answer about this topic: The oxygen-carrying pigment is.

A. insulin B. haemoglobin C. chlorophyll D. melanin

Answer: B. haemoglobin. Explanation: The oxygen-carrying pigment is.

Q25. Which of the following is true? The oxygen-carrying pigment is.

A. haemoglobin B. chlorophyll C. melanin D. insulin

Answer: A. haemoglobin. Explanation: The oxygen-carrying pigment is.

CHAPTER 27: Excretory System

Biology — Excretion removes metabolic wastes and helps regulate water, salts and acid-base balance.

Concept notes and exam points

1. The urinary system includes two kidneys, two ureters, urinary bladder and urethra.

2. The nephron is the functional unit of the kidney. It includes a glomerulus, Bowman's capsule and renal tubule.

3. Ultrafiltration occurs at the glomerulus; useful substances and much water are reabsorbed along the tubule; secretion adds selected substances to the filtrate.

4. Urine commonly contains water, urea, salts and other dissolved wastes. Healthy urine normally contains little or no glucose or protein.

5. Urea is produced in the liver from ammonia generated during amino-acid metabolism and is carried by blood to the kidneys.

6. The kidneys help maintain blood volume, electrolyte balance and pH. Antidiuretic hormone (ADH) influences water reabsorption.

7. The lungs excrete carbon dioxide and water vapour; skin excretes small amounts of water, salts and urea in sweat.

8. Dialysis can remove wastes and excess fluid from blood when kidneys fail, but it does not replace every kidney function.

Formulae, units and quick facts

• nephron = functional unit of kidney

• urea formed in liver

• urine pathway: kidney → ureter → bladder → urethra

Worked examples

Problem: The functional unit of the kidney is the

Solution: nephron. neuron

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The functional unit of kidney is.

A. nephron B. neuron C. alveolus D. villus

Answer: A. nephron. Explanation: The functional unit of kidney is.

Q2. Choose the scientifically correct option: The functional unit of kidney is.

A. neuron B. alveolus C. villus D. nephron

Answer: D. nephron. Explanation: The functional unit of kidney is.

Q3. Which option correctly describes this fact: The functional unit of kidney is?

A. alveolus B. villus C. nephron D. neuron

Answer: C. nephron. Explanation: The functional unit of kidney is.

Q4. Select the best answer about this topic: The functional unit of kidney is.

A. villus B. nephron C. neuron D. alveolus

Answer: B. nephron. Explanation: The functional unit of kidney is.

Q5. Which of the following is true? The functional unit of kidney is.

A. nephron B. neuron C. alveolus D. villus

Answer: A. nephron. Explanation: The functional unit of kidney is.

Q6. Which statement is correct? Urea is mainly formed in the.

A. kidney B. urinary bladder C. lungs D. liver

Answer: D. liver. Explanation: Urea is mainly formed in the.

Q7. Choose the scientifically correct option: Urea is mainly formed in the.

A. urinary bladder B. lungs C. liver D. kidney

Answer: C. liver. Explanation: Urea is mainly formed in the.

Q8. Which option correctly describes this fact: Urea is mainly formed in the?

A. lungs B. liver C. kidney D. urinary bladder

Answer: B. liver. Explanation: Urea is mainly formed in the.

Q9. Select the best answer about this topic: Urea is mainly formed in the.

A. liver B. kidney C. urinary bladder D. lungs

Answer: A. liver. Explanation: Urea is mainly formed in the.

Q10. Which of the following is true? Urea is mainly formed in the.

A. kidney B. urinary bladder C. lungs D. liver

Answer: D. liver. Explanation: Urea is mainly formed in the.

Q11. Which statement is correct? Urine travels from kidney to bladder through the.

A. aorta B. nephron C. ureter D. urethra

Answer: C. ureter. Explanation: Urine travels from kidney to bladder through the.

Q12. Choose the scientifically correct option: Urine travels from kidney to bladder through the.

A. nephron B. ureter C. urethra D. aorta

Answer: B. ureter. Explanation: Urine travels from kidney to bladder through the.

Q13. Which option correctly describes this fact: Urine travels from kidney to bladder through the?

A. ureter B. urethra C. aorta D. nephron

Answer: A. ureter. Explanation: Urine travels from kidney to bladder through the.

Q14. Select the best answer about this topic: Urine travels from kidney to bladder through the.

A. urethra B. aorta C. nephron D. ureter

Answer: D. ureter. Explanation: Urine travels from kidney to bladder through the.

Q15. Which of the following is true? Urine travels from kidney to bladder through the.

A. aorta B. nephron C. ureter D. urethra

Answer: C. ureter. Explanation: Urine travels from kidney to bladder through the.

Q16. Which statement is correct? Ultrafiltration occurs in the.

A. urethra B. glomerulus C. ureter D. bladder

Answer: B. glomerulus. Explanation: Ultrafiltration occurs in the.

Q17. Choose the scientifically correct option: Ultrafiltration occurs in the.

A. glomerulus B. ureter C. bladder D. urethra

Answer: A. glomerulus. Explanation: Ultrafiltration occurs in the.

Q18. Which option correctly describes this fact: Ultrafiltration occurs in the?

A. ureter B. bladder C. urethra D. glomerulus

Answer: D. glomerulus. Explanation: Ultrafiltration occurs in the.

Q19. Select the best answer about this topic: Ultrafiltration occurs in the.

A. bladder B. urethra C. glomerulus D. ureter

Answer: C. glomerulus. Explanation: Ultrafiltration occurs in the.

Q20. Which of the following is true? Ultrafiltration occurs in the.

A. urethra B. glomerulus C. ureter D. bladder

Answer: B. glomerulus. Explanation: Ultrafiltration occurs in the.

Q21. Which statement is correct? ADH mainly influences.

A. water reabsorption B. oxygen binding C. blood clotting D. digestion of starch

Answer: A. water reabsorption. Explanation: ADH mainly influences.

Q22. Choose the scientifically correct option: ADH mainly influences.

A. oxygen binding B. blood clotting C. digestion of starch D. water reabsorption

Answer: D. water reabsorption. Explanation: ADH mainly influences.

Q23. Which option correctly describes this fact: ADH mainly influences?

A. blood clotting B. digestion of starch C. water reabsorption D. oxygen binding

Answer: C. water reabsorption. Explanation: ADH mainly influences.

Q24. Select the best answer about this topic: ADH mainly influences.

A. digestion of starch B. water reabsorption C. oxygen binding D. blood clotting

Answer: B. water reabsorption. Explanation: ADH mainly influences.

Q25. Which of the following is true? ADH mainly influences.

A. water reabsorption B. oxygen binding C. blood clotting D. digestion of starch

Answer: A. water reabsorption. Explanation: ADH mainly influences.

CHAPTER 28: Nervous System and Coordination

Biology — The nervous system detects stimuli, processes information and coordinates rapid responses.

Concept notes and exam points

1. The central nervous system consists of brain and spinal cord. The peripheral nervous system connects the CNS to the rest of the body.

2. A neuron is a specialised cell. Dendrites receive signals, the cell body integrates them and the axon conducts impulses away.

3. A synapse is a junction between neurons or between a neuron and an effector; chemical neurotransmitters commonly transmit signals across it.

4. The cerebrum is involved in conscious thought, memory, sensory processing and voluntary movement.

5. The cerebellum coordinates movement, posture and balance. The brainstem controls vital functions such as breathing and heart rate.

6. The spinal cord conducts signals and coordinates many reflexes. A reflex arc produces a rapid, automatic response.

7. The autonomic nervous system regulates involuntary functions; sympathetic activity supports fight-or-flight and parasympathetic activity supports rest-and-digest.

8. Endocrine glands release hormones into blood. Hormonal responses are often slower but longer-lasting than nerve impulses.

Formulae, units and quick facts

• reflex arc: receptor → sensory neuron → CNS → motor neuron → effector

Worked examples

Problem: Which brain region is strongly involved in balance and coordination?

Solution: Cerebellum. Cerebrum

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The central nervous system includes.

A. brain and spinal cord B. heart and lungs C. kidneys and liver D. nerves only

Answer: A. brain and spinal cord. Explanation: The central nervous system includes.

Q2. Choose the scientifically correct option: The central nervous system includes.

A. heart and lungs B. kidneys and liver C. nerves only D. brain and spinal cord

Answer: D. brain and spinal cord. Explanation: The central nervous system includes.

Q3. Which option correctly describes this fact: The central nervous system includes?

A. kidneys and liver B. nerves only C. brain and spinal cord D. heart and lungs

Answer: C. brain and spinal cord. Explanation: The central nervous system includes.

Q4. Select the best answer about this topic: The central nervous system includes.

A. nerves only B. brain and spinal cord C. heart and lungs D. kidneys and liver

Answer: B. brain and spinal cord. Explanation: The central nervous system includes.

Q5. Which of the following is true? The central nervous system includes.

A. brain and spinal cord B. heart and lungs C. kidneys and liver D. nerves only

Answer: A. brain and spinal cord. Explanation: The central nervous system includes.

Q6. Which statement is correct? The functional signalling cell is the.

A. nephron B. osteocyte C. erythrocyte D. neuron

Answer: D. neuron. Explanation: The functional signalling cell is the.

Q7. Choose the scientifically correct option: The functional signalling cell is the.

A. osteocyte B. erythrocyte C. neuron D. nephron

Answer: C. neuron. Explanation: The functional signalling cell is the.

Q8. Which option correctly describes this fact: The functional signalling cell is the?

A. erythrocyte B. neuron C. nephron D. osteocyte

Answer: B. neuron. Explanation: The functional signalling cell is the.

Q9. Select the best answer about this topic: The functional signalling cell is the.

A. neuron B. nephron C. osteocyte D. erythrocyte

Answer: A. neuron. Explanation: The functional signalling cell is the.

Q10. Which of the following is true? The functional signalling cell is the.

A. nephron B. osteocyte C. erythrocyte D. neuron

Answer: D. neuron. Explanation: The functional signalling cell is the.

Q11. Which statement is correct? Balance and coordination are strongly associated with the.

A. pancreas B. spleen C. cerebellum D. cerebrum only

Answer: C. cerebellum. Explanation: Balance and coordination are strongly associated with the.

Q12. Choose the scientifically correct option: Balance and coordination are strongly associated with the.

A. spleen B. cerebellum C. cerebrum only D. pancreas

Answer: B. cerebellum. Explanation: Balance and coordination are strongly associated with the.

Q13. Which option correctly describes this fact: Balance and coordination are strongly associated with the?

A. cerebellum B. cerebrum only C. pancreas D. spleen

Answer: A. cerebellum. Explanation: Balance and coordination are strongly associated with the.

Q14. Select the best answer about this topic: Balance and coordination are strongly associated with the.

A. cerebrum only B. pancreas C. spleen D. cerebellum

Answer: D. cerebellum. Explanation: Balance and coordination are strongly associated with the.

Q15. Which of the following is true? Balance and coordination are strongly associated with the.

A. pancreas B. spleen C. cerebellum D. cerebrum only

Answer: C. cerebellum. Explanation: Balance and coordination are strongly associated with the.

Q16. Which statement is correct? A rapid automatic response is a.

A. diffusion B. reflex action C. digestion D. secretion

Answer: B. reflex action. Explanation: A rapid automatic response is a.

Q17. Choose the scientifically correct option: A rapid automatic response is a.

A. reflex action B. digestion C. secretion D. diffusion

Answer: A. reflex action. Explanation: A rapid automatic response is a.

Q18. Which option correctly describes this fact: A rapid automatic response is a?

A. digestion B. secretion C. diffusion D. reflex action

Answer: D. reflex action. Explanation: A rapid automatic response is a.

Q19. Select the best answer about this topic: A rapid automatic response is a.

A. secretion B. diffusion C. reflex action D. digestion

Answer: C. reflex action. Explanation: A rapid automatic response is a.

Q20. Which of the following is true? A rapid automatic response is a.

A. diffusion B. reflex action C. digestion D. secretion

Answer: B. reflex action. Explanation: A rapid automatic response is a.

Q21. Which statement is correct? Hormones are transported mainly by.

A. blood B. bone C. air D. urine only

Answer: A. blood. Explanation: Hormones are transported mainly by.

Q22. Choose the scientifically correct option: Hormones are transported mainly by.

A. bone B. air C. urine only D. blood

Answer: D. blood. Explanation: Hormones are transported mainly by.

Q23. Which option correctly describes this fact: Hormones are transported mainly by?

A. air B. urine only C. blood D. bone

Answer: C. blood. Explanation: Hormones are transported mainly by.

Q24. Select the best answer about this topic: Hormones are transported mainly by.

A. urine only B. blood C. bone D. air

Answer: B. blood. Explanation: Hormones are transported mainly by.

Q25. Which of the following is true? Hormones are transported mainly by.

A. blood B. bone C. air D. urine only

Answer: A. blood. Explanation: Hormones are transported mainly by.

CHAPTER 29: Nutrition and Vitamins

Biology — Nutrition supplies energy and raw materials needed for growth, repair and regulation of body processes.

Concept notes and exam points

1. Carbohydrates are a major energy source; proteins supply amino acids for growth and repair; fats provide concentrated energy and help absorb fat-soluble vitamins.

2. Vitamins are organic micronutrients needed in small quantities. They do not directly provide energy but support metabolic functions.

3. Fat-soluble vitamins are A, D, E and K; water-soluble vitamins include B-complex and C. Excess fat-soluble vitamins can accumulate.

4. Vitamin A supports vision and epithelial health; deficiency can cause night blindness and, in severe cases, xerophthalmia.

5. Vitamin B■ (thiamine) deficiency can cause beriberi; vitamin C deficiency causes scurvy; vitamin D deficiency can cause rickets in children and osteomalacia in adults.

6. Vitamin K is important for blood clotting; vitamin E functions as an antioxidant. Vitamin B■■ is needed for red blood cell formation and nerve function.

7. Minerals include calcium for bones and teeth, iron for haemoglobin, iodine for thyroid hormones and sodium/potassium for fluid and nerve function.

8. A balanced diet varies with age, activity and health. Dietary fibre supports bowel function; water is essential for transport and temperature regulation.

Formulae, units and quick facts

• fat-soluble vitamins: A, D, E, K

• water-soluble: B group and C

Worked examples

Problem: Deficiency of vitamin C causes

Solution: scurvy. rickets

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Which group contains fat-soluble vitamins?.

A. A, D, E and K B. B and C only C. C and K only D. B■ and B■■ only

Answer: A. A, D, E and K. Explanation: Which group contains fat-soluble vitamins?.

Q2. Choose the scientifically correct option: Which group contains fat-soluble vitamins?.

A. B and C only B. C and K only C. B■ and B■■ only D. A, D, E and K

Answer: D. A, D, E and K. Explanation: Which group contains fat-soluble vitamins?.

Q3. Which option correctly describes this fact: Which group contains fat-soluble vitamins??

A. C and K only B. B■ and B■■ only C. A, D, E and K D. B and C only

Answer: C. A, D, E and K. Explanation: Which group contains fat-soluble vitamins?.

Q4. Select the best answer about this topic: Which group contains fat-soluble vitamins?.

A. B■ and B■■ only B. A, D, E and K C. B and C only D. C and K only

Answer: B. A, D, E and K. Explanation: Which group contains fat-soluble vitamins?.

Q5. Which of the following is true? Which group contains fat-soluble vitamins?.

A. A, D, E and K B. B and C only C. C and K only D. B■ and B■■ only

Answer: A. A, D, E and K. Explanation: Which group contains fat-soluble vitamins?.

Q6. Which statement is correct? Vitamin D deficiency in children can cause.

A. scurvy B. beriberi C. night blindness D. rickets

Answer: D. rickets. Explanation: Vitamin D deficiency in children can cause.

Q7. Choose the scientifically correct option: Vitamin D deficiency in children can cause.

A. beriberi B. night blindness C. rickets D. scurvy

Answer: C. rickets. Explanation: Vitamin D deficiency in children can cause.

Q8. Which option correctly describes this fact: Vitamin D deficiency in children can cause?

A. night blindness B. rickets C. scurvy D. beriberi

Answer: B. rickets. Explanation: Vitamin D deficiency in children can cause.

Q9. Select the best answer about this topic: Vitamin D deficiency in children can cause.

A. rickets B. scurvy C. beriberi D. night blindness

Answer: A. rickets. Explanation: Vitamin D deficiency in children can cause.

Q10. Which of the following is true? Vitamin D deficiency in children can cause.

A. scurvy B. beriberi C. night blindness D. rickets

Answer: D. rickets. Explanation: Vitamin D deficiency in children can cause.

Q11. Which statement is correct? Iron is required for formation of.

A. insulin B. bile salts C. haemoglobin D. chlorophyll in humans

Answer: C. haemoglobin. Explanation: Iron is required for formation of.

Q12. Choose the scientifically correct option: Iron is required for formation of.

A. bile salts B. haemoglobin C. chlorophyll in humans D. insulin

Answer: B. haemoglobin. Explanation: Iron is required for formation of.

Q13. Which option correctly describes this fact: Iron is required for formation of?

A. haemoglobin B. chlorophyll in humans C. insulin D. bile salts

Answer: A. haemoglobin. Explanation: Iron is required for formation of.

Q14. Select the best answer about this topic: Iron is required for formation of.

A. chlorophyll in humans B. insulin C. bile salts D. haemoglobin

Answer: D. haemoglobin. Explanation: Iron is required for formation of.

Q15. Which of the following is true? Iron is required for formation of.

A. insulin B. bile salts C. haemoglobin D. chlorophyll in humans

Answer: C. haemoglobin. Explanation: Iron is required for formation of.

Q16. Which statement is correct? Iodine is needed for production of.

A. collagen only B. thyroid hormones C. haemoglobin D. saliva

Answer: B. thyroid hormones. Explanation: Iodine is needed for production of.

Q17. Choose the scientifically correct option: Iodine is needed for production of.

A. thyroid hormones B. haemoglobin C. saliva D. collagen only

Answer: A. thyroid hormones. Explanation: Iodine is needed for production of.

Q18. Which option correctly describes this fact: Iodine is needed for production of?

A. haemoglobin B. saliva C. collagen only D. thyroid hormones

Answer: D. thyroid hormones. Explanation: Iodine is needed for production of.

Q19. Select the best answer about this topic: Iodine is needed for production of.

A. saliva B. collagen only C. thyroid hormones D. haemoglobin

Answer: C. thyroid hormones. Explanation: Iodine is needed for production of.

Q20. Which of the following is true? Iodine is needed for production of.

A. collagen only B. thyroid hormones C. haemoglobin D. saliva

Answer: B. thyroid hormones. Explanation: Iodine is needed for production of.

Q21. Which statement is correct? Vitamin K is important for.

A. blood clotting B. night vision C. calcium absorption only D. oxygen transport

Answer: A. blood clotting. Explanation: Vitamin K is important for.

Q22. Choose the scientifically correct option: Vitamin K is important for.

A. night vision B. calcium absorption only C. oxygen transport D. blood clotting

Answer: D. blood clotting. Explanation: Vitamin K is important for.

Q23. Which option correctly describes this fact: Vitamin K is important for?

A. calcium absorption only B. oxygen transport C. blood clotting D. night vision

Answer: C. blood clotting. Explanation: Vitamin K is important for.

Q24. Select the best answer about this topic: Vitamin K is important for.

A. oxygen transport B. blood clotting C. night vision D. calcium absorption only

Answer: B. blood clotting. Explanation: Vitamin K is important for.

Q25. Which of the following is true? Vitamin K is important for.

A. blood clotting B. night vision C. calcium absorption only D. oxygen transport

Answer: A. blood clotting. Explanation: Vitamin K is important for.

CHAPTER 30: Human Diseases and Immunity

Biology — Disease may result from infectious agents, genetic factors, nutritional deficiencies, environmental exposures or lifestyle factors.

Concept notes and exam points

1. Pathogens include bacteria, viruses, fungi, protozoa and parasitic worms. Transmission may occur through air, water, food, vectors, contact or body fluids.

2. Bacteria are cellular organisms; viruses are acellular infectious agents that replicate inside host cells. Antibiotics act against bacteria, not viruses.

3. Malaria is caused by Plasmodium and transmitted by female Anopheles mosquitoes. Dengue is transmitted mainly by Aedes mosquitoes.

4. Tuberculosis is caused by Mycobacterium tuberculosis and commonly affects lungs; it spreads through airborne particles from an infectious person.

5. Vaccination exposes the immune system to antigens safely or in a controlled form, helping build immune memory and reduce severe disease.

6. Innate immunity provides rapid, non-specific defence; adaptive immunity involves specific B and T lymphocytes and immunological memory.

7. Antibodies are proteins produced by B-cell-derived plasma cells that bind specific antigens. T cells coordinate immune responses or kill infected cells.

8. Prevention includes safe water, sanitation, vaccination, vector control, hand hygiene, ventilation and appropriate medical care.

Formulae, units and quick facts

• malaria: Plasmodium; vector: female Anopheles

• TB: Mycobacterium tuberculosis

• antibiotics do not treat viral infections

Worked examples

Problem: Malaria is caused by which organism?

Solution: Plasmodium. Aedes virus

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Malaria is transmitted by female.

A. Anopheles mosquito B. Aedes mosquito only C. housefly D. sandfly

Answer: A. Anopheles mosquito. Explanation: Malaria is transmitted by female.

Q2. Choose the scientifically correct option: Malaria is transmitted by female.

A. Aedes mosquito only B. housefly C. sandfly D. Anopheles mosquito

Answer: D. Anopheles mosquito. Explanation: Malaria is transmitted by female.

Q3. Which option correctly describes this fact: Malaria is transmitted by female?

A. housefly B. sandfly C. Anopheles mosquito D. Aedes mosquito only

Answer: C. Anopheles mosquito. Explanation: Malaria is transmitted by female.

Q4. Select the best answer about this topic: Malaria is transmitted by female.

A. sandfly B. Anopheles mosquito C. Aedes mosquito only D. housefly

Answer: B. Anopheles mosquito. Explanation: Malaria is transmitted by female.

Q5. Which of the following is true? Malaria is transmitted by female.

A. Anopheles mosquito B. Aedes mosquito only C. housefly D. sandfly

Answer: A. Anopheles mosquito. Explanation: Malaria is transmitted by female.

Q6. Which statement is correct? Tuberculosis is caused by.

A. Plasmodium B. HIV C. Vibrio cholerae D. Mycobacterium tuberculosis

Answer: D. Mycobacterium tuberculosis. Explanation: Tuberculosis is caused by.

Q7. Choose the scientifically correct option: Tuberculosis is caused by.

A. HIV B. Vibrio cholerae C. Mycobacterium tuberculosis D. Plasmodium

Answer: C. Mycobacterium tuberculosis. Explanation: Tuberculosis is caused by.

Q8. Which option correctly describes this fact: Tuberculosis is caused by?

A. Vibrio cholerae B. Mycobacterium tuberculosis C. Plasmodium D. HIV

Answer: B. Mycobacterium tuberculosis. Explanation: Tuberculosis is caused by.

Q9. Select the best answer about this topic: Tuberculosis is caused by.

A. Mycobacterium tuberculosis B. Plasmodium C. HIV D. Vibrio cholerae

Answer: A. Mycobacterium tuberculosis. Explanation: Tuberculosis is caused by.

Q10. Which of the following is true? Tuberculosis is caused by.

A. Plasmodium B. HIV C. Vibrio cholerae D. Mycobacterium tuberculosis

Answer: D. Mycobacterium tuberculosis. Explanation: Tuberculosis is caused by.

Q11. Which statement is correct? Antibiotics are primarily used against.

A. genetic disorders B. vitamin deficiencies C. bacterial infections D. all viral infections

Answer: C. bacterial infections. Explanation: Antibiotics are primarily used against.

Q12. Choose the scientifically correct option: Antibiotics are primarily used against.

A. vitamin deficiencies B. bacterial infections C. all viral infections D. genetic disorders

Answer: B. bacterial infections. Explanation: Antibiotics are primarily used against.

Q13. Which option correctly describes this fact: Antibiotics are primarily used against?

A. bacterial infections B. all viral infections C. genetic disorders D. vitamin deficiencies

Answer: A. bacterial infections. Explanation: Antibiotics are primarily used against.

Q14. Select the best answer about this topic: Antibiotics are primarily used against.

A. all viral infections B. genetic disorders C. vitamin deficiencies D. bacterial infections

Answer: D. bacterial infections. Explanation: Antibiotics are primarily used against.

Q15. Which of the following is true? Antibiotics are primarily used against.

A. genetic disorders B. vitamin deficiencies C. bacterial infections D. all viral infections

Answer: C. bacterial infections. Explanation: Antibiotics are primarily used against.

Q16. Which statement is correct? Vaccination helps develop.

A. digestive enzymes B. immune memory C. bone fractures D. blood glucose

Answer: B. immune memory. Explanation: Vaccination helps develop.

Q17. Choose the scientifically correct option: Vaccination helps develop.

A. immune memory B. bone fractures C. blood glucose D. digestive enzymes

Answer: A. immune memory. Explanation: Vaccination helps develop.

Q18. Which option correctly describes this fact: Vaccination helps develop?

A. bone fractures B. blood glucose C. digestive enzymes D. immune memory

Answer: D. immune memory. Explanation: Vaccination helps develop.

Q19. Select the best answer about this topic: Vaccination helps develop.

A. blood glucose B. digestive enzymes C. immune memory D. bone fractures

Answer: C. immune memory. Explanation: Vaccination helps develop.

Q20. Which of the following is true? Vaccination helps develop.

A. digestive enzymes B. immune memory C. bone fractures D. blood glucose

Answer: B. immune memory. Explanation: Vaccination helps develop.

Q21. Which statement is correct? Antibodies are produced by.

A. plasma cells B. red blood cells C. platelets D. neurons

Answer: A. plasma cells. Explanation: Antibodies are produced by.

Q22. Choose the scientifically correct option: Antibodies are produced by.

A. red blood cells B. platelets C. neurons D. plasma cells

Answer: D. plasma cells. Explanation: Antibodies are produced by.

Q23. Which option correctly describes this fact: Antibodies are produced by?

A. platelets B. neurons C. plasma cells D. red blood cells

Answer: C. plasma cells. Explanation: Antibodies are produced by.

Q24. Select the best answer about this topic: Antibodies are produced by.

A. neurons B. plasma cells C. red blood cells D. platelets

Answer: B. plasma cells. Explanation: Antibodies are produced by.

Q25. Which of the following is true? Antibodies are produced by.

A. plasma cells B. red blood cells C. platelets D. neurons

Answer: A. plasma cells. Explanation: Antibodies are produced by.

CHAPTER 31: Plant Biology

Biology — Plants are multicellular eukaryotes; most green plants make organic food by photosynthesis.

Concept notes and exam points

1. Roots anchor plants and absorb water and mineral ions. Root hairs increase the surface area for absorption.

2. Stems support leaves and transport materials. Xylem carries water and minerals mainly upward; phloem transports sugars and other organic substances.

3. Leaves are major photosynthetic organs. Stomata allow gas exchange; guard cells regulate their opening and closing.

4. Photosynthesis uses light energy, chlorophyll, carbon dioxide and water to form carbohydrates and release oxygen.

5. Transpiration is loss of water vapour from aerial plant parts, mainly through stomata. It helps pull water through xylem and cools leaves.

6. Pollination is transfer of pollen from anther to stigma; fertilisation is fusion of male and female gametes to form a zygote.

7. A seed contains an embryo and stored food or cotyledons; germination requires suitable water, oxygen and temperature.

8. Plant hormones include auxins (growth and tropisms), gibberellins (stem growth and germination), cytokinins (cell division), abscisic acid (stress and dormancy) and ethylene (fruit ripening).

Formulae, units and quick facts

• photosynthesis: 6CO■ + 6H■O → C■H■■O■ + 6O■

• xylem: water/minerals; phloem: food

Worked examples

Problem: Which tissue transports water in plants?

Solution: Xylem. Phloem

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? Water is transported mainly by.

A. xylem B. phloem C. cortex only D. stomata

Answer: A. xylem. Explanation: Water is transported mainly by.

Q2. Choose the scientifically correct option: Water is transported mainly by.

A. phloem B. cortex only C. stomata D. xylem

Answer: D. xylem. Explanation: Water is transported mainly by.

Q3. Which option correctly describes this fact: Water is transported mainly by?

A. cortex only B. stomata C. xylem D. phloem

Answer: C. xylem. Explanation: Water is transported mainly by.

Q4. Select the best answer about this topic: Water is transported mainly by.

A. stomata B. xylem C. phloem D. cortex only

Answer: B. xylem. Explanation: Water is transported mainly by.

Q5. Which of the following is true? Water is transported mainly by.

A. xylem B. phloem C. cortex only D. stomata

Answer: A. xylem. Explanation: Water is transported mainly by.

Q6. Which statement is correct? Food is transported through.

A. xylem B. root cap C. cuticle D. phloem

Answer: D. phloem. Explanation: Food is transported through.

Q7. Choose the scientifically correct option: Food is transported through.

A. root cap B. cuticle C. phloem D. xylem

Answer: C. phloem. Explanation: Food is transported through.

Q8. Which option correctly describes this fact: Food is transported through?

A. cuticle B. phloem C. xylem D. root cap

Answer: B. phloem. Explanation: Food is transported through.

Q9. Select the best answer about this topic: Food is transported through.

A. phloem B. xylem C. root cap D. cuticle

Answer: A. phloem. Explanation: Food is transported through.

Q10. Which of the following is true? Food is transported through.

A. xylem B. root cap C. cuticle D. phloem

Answer: D. phloem. Explanation: Food is transported through.

Q11. Which statement is correct? Gas exchange in leaves occurs mainly through.

A. petals B. seeds C. stomata D. root hairs

Answer: C. stomata. Explanation: Gas exchange in leaves occurs mainly through.

Q12. Choose the scientifically correct option: Gas exchange in leaves occurs mainly through.

A. seeds B. stomata C. root hairs D. petals

Answer: B. stomata. Explanation: Gas exchange in leaves occurs mainly through.

Q13. Which option correctly describes this fact: Gas exchange in leaves occurs mainly through?

A. stomata B. root hairs C. petals D. seeds

Answer: A. stomata. Explanation: Gas exchange in leaves occurs mainly through.

Q14. Select the best answer about this topic: Gas exchange in leaves occurs mainly through.

A. root hairs B. petals C. seeds D. stomata

Answer: D. stomata. Explanation: Gas exchange in leaves occurs mainly through.

Q15. Which of the following is true? Gas exchange in leaves occurs mainly through.

A. petals B. seeds C. stomata D. root hairs

Answer: C. stomata. Explanation: Gas exchange in leaves occurs mainly through.

Q16. Which statement is correct? Loss of water vapour from aerial parts is.

A. fertilisation B. transpiration C. translocation D. pollination

Answer: B. transpiration. Explanation: Loss of water vapour from aerial parts is.

Q17. Choose the scientifically correct option: Loss of water vapour from aerial parts is.

A. transpiration B. translocation C. pollination D. fertilisation

Answer: A. transpiration. Explanation: Loss of water vapour from aerial parts is.

Q18. Which option correctly describes this fact: Loss of water vapour from aerial parts is?

A. translocation B. pollination C. fertilisation D. transpiration

Answer: D. transpiration. Explanation: Loss of water vapour from aerial parts is.

Q19. Select the best answer about this topic: Loss of water vapour from aerial parts is.

A. pollination B. fertilisation C. transpiration D. translocation

Answer: C. transpiration. Explanation: Loss of water vapour from aerial parts is.

Q20. Which of the following is true? Loss of water vapour from aerial parts is.

A. fertilisation B. transpiration C. translocation D. pollination

Answer: B. transpiration. Explanation: Loss of water vapour from aerial parts is.

Q21. Which statement is correct? Transfer of pollen from anther to stigma is.

A. pollination B. fertilisation C. germination D. respiration

Answer: A. pollination. Explanation: Transfer of pollen from anther to stigma is.

Q22. Choose the scientifically correct option: Transfer of pollen from anther to stigma is.

A. fertilisation B. germination C. respiration D. pollination

Answer: D. pollination. Explanation: Transfer of pollen from anther to stigma is.

Q23. Which option correctly describes this fact: Transfer of pollen from anther to stigma is?

A. germination B. respiration C. pollination D. fertilisation

Answer: C. pollination. Explanation: Transfer of pollen from anther to stigma is.

Q24. Select the best answer about this topic: Transfer of pollen from anther to stigma is.

A. respiration B. pollination C. fertilisation D. germination

Answer: B. pollination. Explanation: Transfer of pollen from anther to stigma is.

Q25. Which of the following is true? Transfer of pollen from anther to stigma is.

A. pollination B. fertilisation C. germination D. respiration

Answer: A. pollination. Explanation: Transfer of pollen from anther to stigma is.

CHAPTER 32: Reproduction and Genetics

Biology — Reproduction produces offspring; heredity passes genetic information from parents to offspring.

Concept notes and exam points

1. Asexual reproduction involves one parent and no fusion of gametes; offspring are often genetically similar to the parent, except for mutations.

2. Sexual reproduction involves formation and fusion of gametes. In humans, fertilisation usually occurs in the oviduct and the embryo implants in the uterus.

3. In humans, sperm are produced in testes and ova in ovaries. The placenta enables exchange of nutrients, gases and wastes between maternal and fetal blood without normally mixing them directly.

4. DNA stores hereditary information. A gene is a DNA sequence that contributes to a functional product; chromosomes contain DNA and proteins.

5. Mitosis produces two genetically similar daughter cells and supports growth and repair. Meiosis produces haploid cells and contributes to genetic variation.

6. In humans, body cells typically have 46 chromosomes (23 pairs); gametes have 23. At fertilisation, the diploid number is restored.

7. Gregor Mendel's pea plant experiments established basic principles of inheritance, including segregation of alleles.

8. Sex determination in humans usually involves XX chromosomes in females and XY in males; the sperm contributes either X or Y.

Formulae, units and quick facts

• human somatic cells: 46 chromosomes

• human gametes: 23 chromosomes

• mitosis: 2 similar cells; meiosis: 4 haploid cells (typical outcome)

Worked examples

Problem: How many chromosomes are normally present in a human gamete?

Solution: 23. 46

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The basic unit of heredity is the.

A. gene B. tissue C. organ D. enzyme

Answer: A. gene. Explanation: The basic unit of heredity is the.

Q2. Choose the scientifically correct option: The basic unit of heredity is the.

A. tissue B. organ C. enzyme D. gene

Answer: D. gene. Explanation: The basic unit of heredity is the.

Q3. Which option correctly describes this fact: The basic unit of heredity is the?

A. organ B. enzyme C. gene D. tissue

Answer: C. gene. Explanation: The basic unit of heredity is the.

Q4. Select the best answer about this topic: The basic unit of heredity is the.

A. enzyme B. gene C. tissue D. organ

Answer: B. gene. Explanation: The basic unit of heredity is the.

Q5. Which of the following is true? The basic unit of heredity is the.

A. gene B. tissue C. organ D. enzyme

Answer: A. gene. Explanation: The basic unit of heredity is the.

Q6. Which statement is correct? Mitosis usually produces.

A. four haploid cells B. one gamete only C. eight nuclei D. two similar daughter cells

Answer: D. two similar daughter cells. Explanation: Mitosis usually produces.

Q7. Choose the scientifically correct option: Mitosis usually produces.

A. one gamete only B. eight nuclei C. two similar daughter cells D. four haploid cells

Answer: C. two similar daughter cells. Explanation: Mitosis usually produces.

Q8. Which option correctly describes this fact: Mitosis usually produces?

A. eight nuclei B. two similar daughter cells C. four haploid cells D. one gamete only

Answer: B. two similar daughter cells. Explanation: Mitosis usually produces.

Q9. Select the best answer about this topic: Mitosis usually produces.

A. two similar daughter cells B. four haploid cells C. one gamete only D. eight nuclei

Answer: A. two similar daughter cells. Explanation: Mitosis usually produces.

Q10. Which of the following is true? Mitosis usually produces.

A. four haploid cells B. one gamete only C. eight nuclei D. two similar daughter cells

Answer: D. two similar daughter cells. Explanation: Mitosis usually produces.

Q11. Which statement is correct? Human gametes contain.

A. 23 pairs B. 44 chromosomes C. 23 chromosomes D. 46 chromosomes

Answer: C. 23 chromosomes. Explanation: Human gametes contain.

Q12. Choose the scientifically correct option: Human gametes contain.

A. 44 chromosomes B. 23 chromosomes C. 46 chromosomes D. 23 pairs

Answer: B. 23 chromosomes. Explanation: Human gametes contain.

Q13. Which option correctly describes this fact: Human gametes contain?

A. 23 chromosomes B. 46 chromosomes C. 23 pairs D. 44 chromosomes

Answer: A. 23 chromosomes. Explanation: Human gametes contain.

Q14. Select the best answer about this topic: Human gametes contain.

A. 46 chromosomes B. 23 pairs C. 44 chromosomes D. 23 chromosomes

Answer: D. 23 chromosomes. Explanation: Human gametes contain.

Q15. Which of the following is true? Human gametes contain.

A. 23 pairs B. 44 chromosomes C. 23 chromosomes D. 46 chromosomes

Answer: C. 23 chromosomes. Explanation: Human gametes contain.

Q16. Which statement is correct? Fertilisation is fusion of.

A. two embryos B. male and female gametes C. two body cells D. two chromosomes only

Answer: B. male and female gametes. Explanation: Fertilisation is fusion of.

Q17. Choose the scientifically correct option: Fertilisation is fusion of.

A. male and female gametes B. two body cells C. two chromosomes only D. two embryos

Answer: A. male and female gametes. Explanation: Fertilisation is fusion of.

Q18. Which option correctly describes this fact: Fertilisation is fusion of?

A. two body cells B. two chromosomes only C. two embryos D. male and female gametes

Answer: D. male and female gametes. Explanation: Fertilisation is fusion of.

Q19. Select the best answer about this topic: Fertilisation is fusion of.

A. two chromosomes only B. two embryos C. male and female gametes D. two body cells

Answer: C. male and female gametes. Explanation: Fertilisation is fusion of.

Q20. Which of the following is true? Fertilisation is fusion of.

A. two embryos B. male and female gametes C. two body cells D. two chromosomes only

Answer: B. male and female gametes. Explanation: Fertilisation is fusion of.

Q21. Which statement is correct? Meiosis is important because it.

A. reduces chromosome number in gametes B. doubles body chromosome number C. stops variation D. forms identical skin cells only

Answer: A. reduces chromosome number in gametes. Explanation: Meiosis is important because it.

Q22. Choose the scientifically correct option: Meiosis is important because it.

A. doubles body chromosome number B. stops variation C. forms identical skin cells only D. reduces chromosome number in gametes

Answer: D. reduces chromosome number in gametes. Explanation: Meiosis is important because it.

Q23. Which option correctly describes this fact: Meiosis is important because it?

A. stops variation B. forms identical skin cells only C. reduces chromosome number in gametes D. doubles body chromosome number

Answer: C. reduces chromosome number in gametes. Explanation: Meiosis is important because it.

Q24. Select the best answer about this topic: Meiosis is important because it.

A. forms identical skin cells only B. reduces chromosome number in gametes C. doubles body chromosome number D. stops variation

Answer: B. reduces chromosome number in gametes. Explanation: Meiosis is important because it.

Q25. Which of the following is true? Meiosis is important because it.

A. reduces chromosome number in gametes B. doubles body chromosome number C. stops variation D. forms identical skin cells only

Answer: A. reduces chromosome number in gametes. Explanation: Meiosis is important because it.

CHAPTER 33: Ecology and Environment

Biology — Ecology studies relationships among organisms and between organisms and their physical environment.

Concept notes and exam points

1. An ecosystem includes a community of organisms and the abiotic environment interacting as a system.

2. Producers such as green plants capture energy; consumers obtain energy by eating organisms; decomposers break down dead material and recycle nutrients.

3. A food chain shows a single pathway of energy transfer; food webs show interconnected feeding relationships.

4. Energy transfer between trophic levels is inefficient; much energy is lost as heat and through metabolism. The 10% rule is a rough ecological approximation.

5. Biotic factors include living organisms; abiotic factors include light, temperature, water, soil and air.

6. Biodiversity is variety of life at genetic, species and ecosystem levels. Habitat loss, invasive species, pollution, overexploitation and climate change threaten it.

7. The carbon cycle moves carbon among atmosphere, organisms, oceans and rocks through photosynthesis, respiration, decomposition and combustion.

8. Conservation includes protected areas, habitat restoration, sustainable use, pollution reduction and community participation.

Formulae, units and quick facts

• energy transfer: roughly 10% passed to next trophic level (approximation)

• producers → consumers → decomposers

Worked examples

Problem: Organisms that break down dead organic matter are

Solution: decomposers. producers

Practice MCQs — 25 questions

Choose one best answer. Each question includes its answer and a short explanation.

Q1. Which statement is correct? The living components of an ecosystem are.

A. biotic factors B. abiotic factors C. minerals only D. climate only

Answer: A. biotic factors. Explanation: The living components of an ecosystem are.

Q2. Choose the scientifically correct option: The living components of an ecosystem are.

A. abiotic factors B. minerals only C. climate only D. biotic factors

Answer: D. biotic factors. Explanation: The living components of an ecosystem are.

Q3. Which option correctly describes this fact: The living components of an ecosystem are?

A. minerals only B. climate only C. biotic factors D. abiotic factors

Answer: C. biotic factors. Explanation: The living components of an ecosystem are.

Q4. Select the best answer about this topic: The living components of an ecosystem are.

A. climate only B. biotic factors C. abiotic factors D. minerals only

Answer: B. biotic factors. Explanation: The living components of an ecosystem are.

Q5. Which of the following is true? The living components of an ecosystem are.

A. biotic factors B. abiotic factors C. minerals only D. climate only

Answer: A. biotic factors. Explanation: The living components of an ecosystem are.

Q6. Which statement is correct? Green plants are usually.

A. decomposers B. top carnivores C. parasites only D. producers

Answer: D. producers. Explanation: Green plants are usually.

Q7. Choose the scientifically correct option: Green plants are usually.

A. top carnivores B. parasites only C. producers D. decomposers

Answer: C. producers. Explanation: Green plants are usually.

Q8. Which option correctly describes this fact: Green plants are usually?

A. parasites only B. producers C. decomposers D. top carnivores

Answer: B. producers. Explanation: Green plants are usually.

Q9. Select the best answer about this topic: Green plants are usually.

A. producers B. decomposers C. top carnivores D. parasites only

Answer: A. producers. Explanation: Green plants are usually.

Q10. Which of the following is true? Green plants are usually.

A. decomposers B. top carnivores C. parasites only D. producers

Answer: D. producers. Explanation: Green plants are usually.

Q11. Which statement is correct? A food web consists of.

A. only abiotic factors B. a single straight pathway C. interconnected food chains D. one organism only

Answer: C. interconnected food chains. Explanation: A food web consists of.

Q12. Choose the scientifically correct option: A food web consists of.

A. a single straight pathway B. interconnected food chains C. one organism only D. only abiotic factors

Answer: B. interconnected food chains. Explanation: A food web consists of.

Q13. Which option correctly describes this fact: A food web consists of?

A. interconnected food chains B. one organism only C. only abiotic factors D. a single straight pathway

Answer: A. interconnected food chains. Explanation: A food web consists of.

Q14. Select the best answer about this topic: A food web consists of.

A. one organism only B. only abiotic factors C. a single straight pathway D. interconnected food chains

Answer: D. interconnected food chains. Explanation: A food web consists of.

Q15. Which of the following is true? A food web consists of.

A. only abiotic factors B. a single straight pathway C. interconnected food chains D. one organism only

Answer: C. interconnected food chains. Explanation: A food web consists of.

Q16. Which statement is correct? The approximate 10% rule relates to.

A. soil pH B. energy transfer between trophic levels C. rainfall D. population age

Answer: B. energy transfer between trophic levels. Explanation: The approximate 10% rule relates to.

Q17. Choose the scientifically correct option: The approximate 10% rule relates to.

A. energy transfer between trophic levels B. rainfall C. population age D. soil pH

Answer: A. energy transfer between trophic levels. Explanation: The approximate 10% rule relates to.

Q18. Which option correctly describes this fact: The approximate 10% rule relates to?

A. rainfall B. population age C. soil pH D. energy transfer between trophic levels

Answer: D. energy transfer between trophic levels. Explanation: The approximate 10% rule relates to.

Q19. Select the best answer about this topic: The approximate 10% rule relates to.

A. population age B. soil pH C. energy transfer between trophic levels D. rainfall

Answer: C. energy transfer between trophic levels. Explanation: The approximate 10% rule relates to.

Q20. Which of the following is true? The approximate 10% rule relates to.

A. soil pH B. energy transfer between trophic levels C. rainfall D. population age

Answer: B. energy transfer between trophic levels. Explanation: The approximate 10% rule relates to.

Q21. Which statement is correct? A major threat to biodiversity is.

A. habitat loss B. pollination C. photosynthesis D. nutrient recycling

Answer: A. habitat loss. Explanation: A major threat to biodiversity is.

Q22. Choose the scientifically correct option: A major threat to biodiversity is.

A. pollination B. photosynthesis C. nutrient recycling D. habitat loss

Answer: D. habitat loss. Explanation: A major threat to biodiversity is.

Q23. Which option correctly describes this fact: A major threat to biodiversity is?

A. photosynthesis B. nutrient recycling C. habitat loss D. pollination

Answer: C. habitat loss. Explanation: A major threat to biodiversity is.

Q24. Select the best answer about this topic: A major threat to biodiversity is.

A. nutrient recycling B. habitat loss C. pollination D. photosynthesis

Answer: B. habitat loss. Explanation: A major threat to biodiversity is.

Q25. Which of the following is true? A major threat to biodiversity is.

A. habitat loss B. pollination C. photosynthesis D. nutrient recycling

Answer: A. habitat loss. Explanation: A major threat to biodiversity is.